Cu 29 — 2 8 18 (1)

Chapter 8
Modern electronics

Materials

Conductors (Metals)
Valency < 4

Na 11 — 2 8 (1)

Semi-conductors (Metaloids)
Valency = 4

Si_{14} — 2 8 (4)

Ge_{32} — 2 8 18 (4)

Insulators (Non-Metals)
Valency > 4

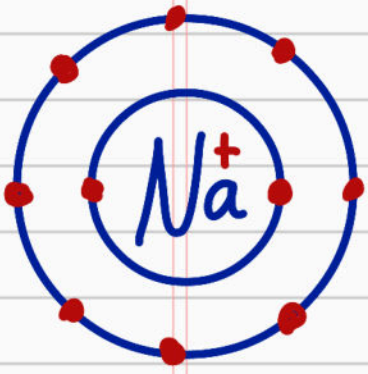
Cl 17 — 2 8 (7)

$\sigma_{conductors} > \sigma_{semi} > \sigma_{Insulators} = 0$

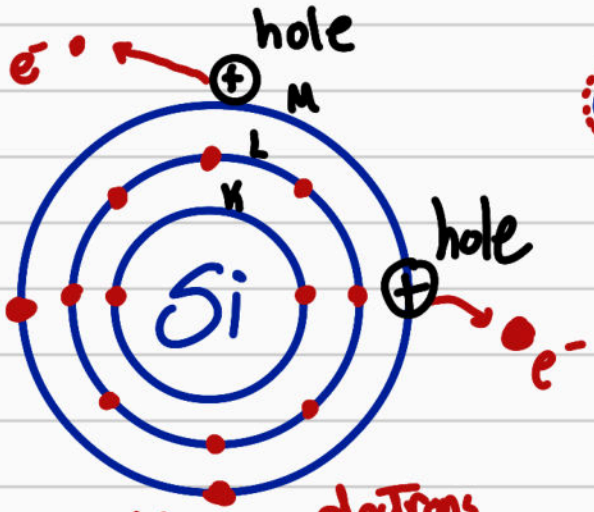
Charge carriers → free electrons

Charge carriers → free electrons $[n^-]$ + holes $[p^+]$

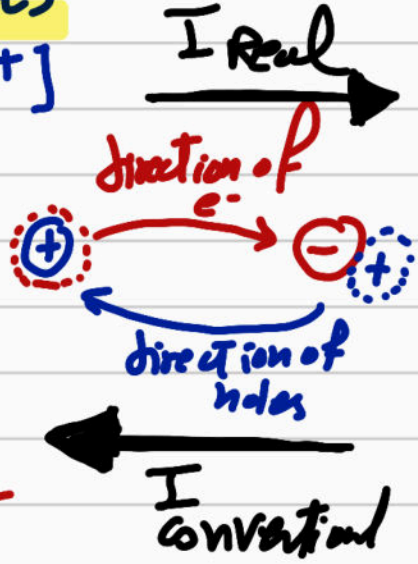
No charge carriers

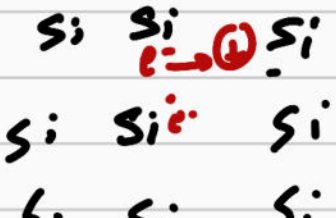


e^- free electron



Valency electrons





Chapter ① → conductors

Chapter ② → semi-cond.

How To increase the Conductivity of semi-conductor?

inc. charge Carriers → $\sigma \uparrow\uparrow$

→ $p^+ \uparrow$ holes ↑

→ $n^- \uparrow$ free electrons ↑

1 Intrinsic [Internally]: $p^+ = n^- = n_i$ Pure semi Cond.

Theory (absolute Zero) 0K = -273 c°

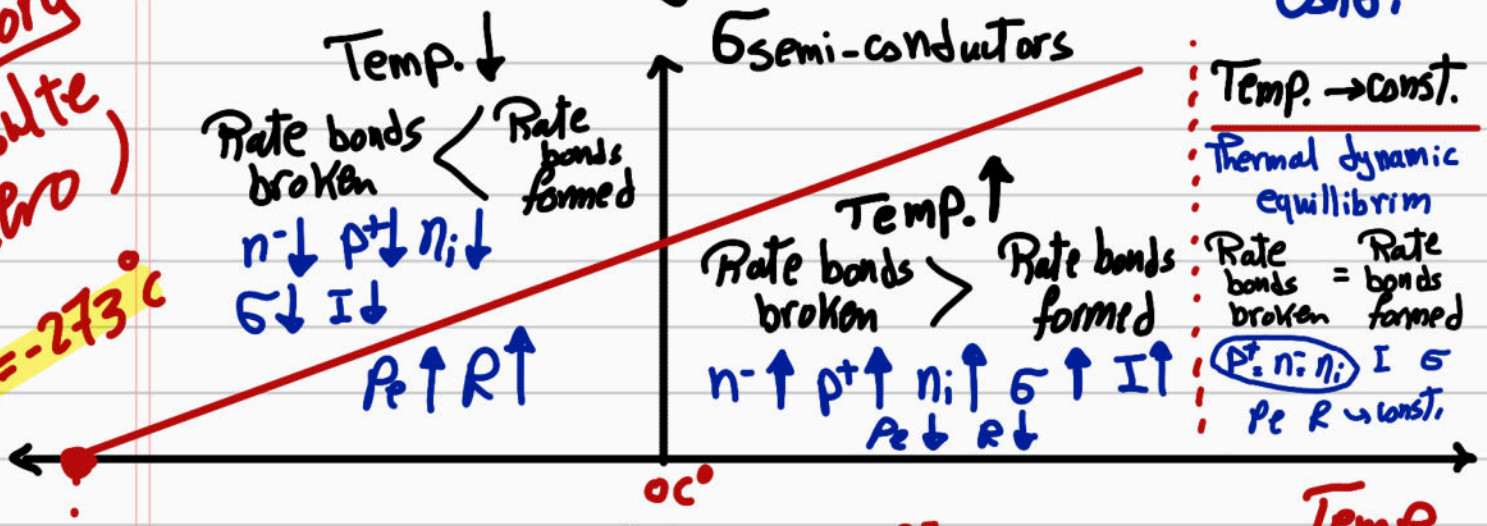
Temp. ↓

Rate bonds broken < Rate bonds formed

$n^- \downarrow$ $p^+ \downarrow$ $n_i \downarrow$

$\sigma \downarrow$ $I \downarrow$

$\rho_e \uparrow$ $R \uparrow$



Temp. → const.

Thermal dynamic equillibrim

Rate bonds broken = Rate bonds formed

$p^+ = n^- = n_i$

I σ

ρ_e R → const.

Temp. ↑

Rate bonds broken > Rate bonds formed

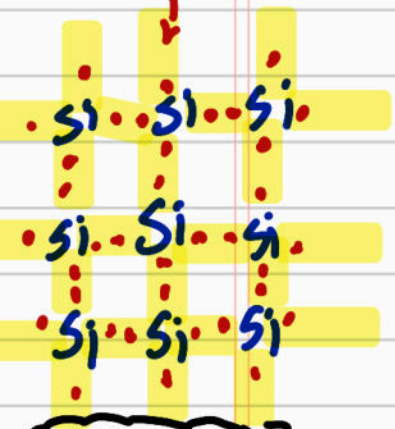
$n^- \uparrow$ $p^+ \uparrow$ $n_i \uparrow$ $\sigma \uparrow$ $I \uparrow$

$\rho_e \downarrow$ $R \downarrow$

p^+ → conc. of holes [Cm^{-3}]

n^- → conc. of free electrons [Cm^{-3}]

n_i → conc. of free electrons or holes in any Pure Semi-conductor [Cm^{-3}]



$n^- = 0$

$p^+ = 0$

$n_i = 0$

$\sigma = 0$ → Perfect Insulator

$I = 0$ → all the bonds are formed

$R = \infty$ $\rho_e = \infty$ No free e^- No holes

$Cm \xrightarrow{\times 10^{-2}} m$

$Cm^{-3} \xrightarrow{\times 10^{6}} m^{-3}$

n_i → Pure

2 Extrinsic "Externally" Doping حقن / تطعيم

σ ↑ impure σ ↑

P-Type [$p^+ > n^-$]

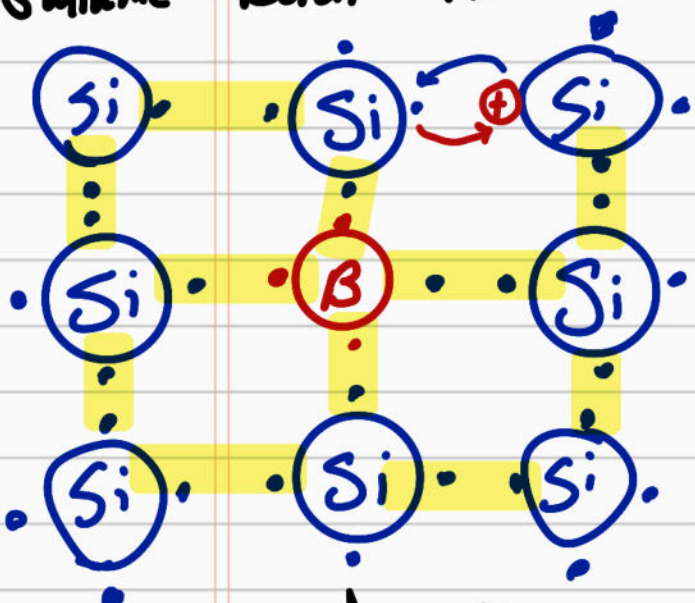
Trivalent "Acceptors"

Valency = 3

جبل

Ga, B, AL

Galiume Boron ALminume



Crystal neutral

$p^+ = n^- + N_A^-$

holes — free e^- — Acceptors

Ga B AL

$N_A^- >>>>> n^-$

$p^+ = N_A^-$ #

$p^+ \times n^- = n_i^2$ (Pure)

$n^- = \frac{n_i^2}{p^+} \approx \frac{n_i^2}{N_A^-}$

N-Type [$p^+ < n^-$]

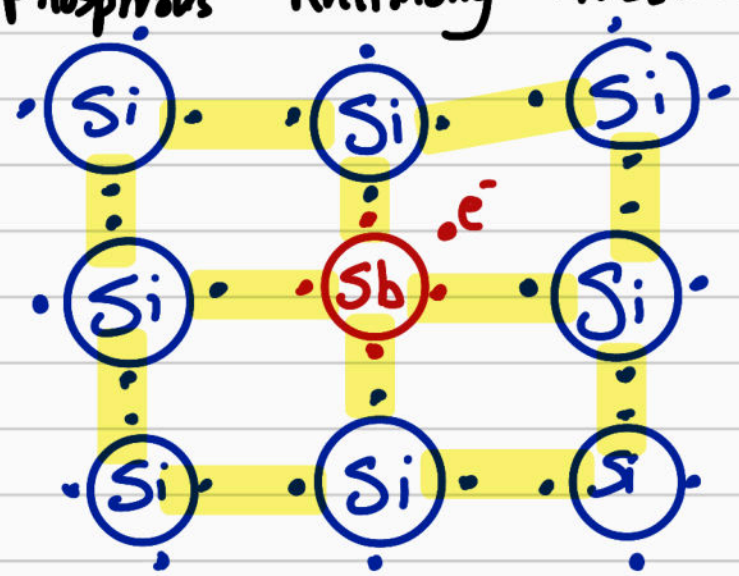
Pentavalent "Donors"

Valency = 5

بس باس

P, sb, As

Phosphrous Antimony Arsenic



Crystal neutral

$n^- = p^+ + N_D^+$

Donors

P sb As

$N_D^+ >>>>> p^+$

$n^- = N_D^+$ #

$p^+ \times n^- = n_i^2$ (Pure)

$p^+ = \frac{n_i^2}{n^-} = \frac{n_i^2}{N_D^+}$

Law of mass action

Pure
impure
P-Type
N-Type

* Pure → $p^+ = n^- = n_i$ ← Pure

$$p^+ \times n^- = n_i \times n_i = n_i^2$$

$$p^+ \times n^- = n_i^2$$

Pure

$$n_i = \sqrt{p^+ \times n^-}$$

P-Type
N-Type

Conc. of free electrons or holes

Pure only

Conc. of holes

Pure impure

$p^+ = n^-$

P-Type N-Type

$p^+ > n^-$ $p^+ < n^-$

Ga B AL P Sb As

conc. of free electron

Pure impure

Summary

Semi-conductor

Silicon - Germanium

Si_{14} Ge_{32}

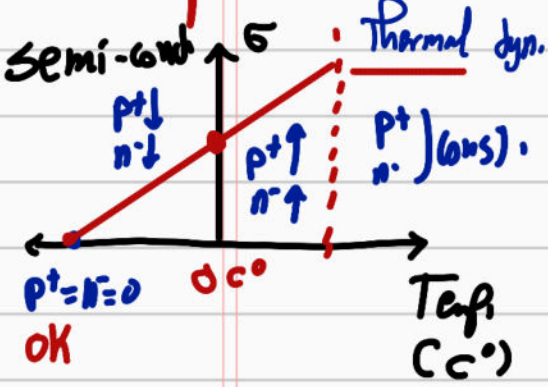
Law of mass Action

$$p^+ \times n^- = n_i^2$$

$$n_i = \sqrt{p^+ \times n^-}$$

Pure InTernsic

Temp. ↑



OK

n_i → concentration of electrons or holes in Pure semi-conductor

impure ExTernsic

Doping

P-Type

Ga/B/AL

$p^+ = N_A^-$

$n^- = \frac{n_i^2}{p^+} = \frac{n_i^2}{N_A^-}$

N_A^- → conc. of Acceptors (اجبل)

N-Type

P/Sb/As

$n^- = N_D^+$

$p^+ = \frac{n_i^2}{n^-} = \frac{n_i^2}{N_D^+}$

N_D^+ → conc. of Donors (ربس باس)

Questions on Lesson One

- The Semiconductor Crystal.
- The pn Junction.

Part (1) الأصل
Part (2) الأربعاء

Interactive test

The questions signed by ✳ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

Important guidelines

Law of mass action: $np = n_i^2$

(Where: (n_i) is the concentration of free electrons or holes in the pure (intrinsic) semiconductor crystal).

Semiconductor

- **Intrinsic crystals** — $n = p = n_i$
- **Extrinsic crystals (doped)**

Extrinsic crystals (doped)	n-type	p-type
Impurity element valency	Pentavalent (doners)	Trivalent (acceptors)
Free electrons concentration	$n = N_D^+ + P$ (Electrically neutral) $n \approx N_D^+$	$n = \frac{n_i^2}{N_A^-}$
Holes concentration	$p = \frac{n_i^2}{N_D^+}$	$p = N_A^- + n$ (Electrically neutral) $p = N_A^-$
The relation between the two types of charge carriers	$n > p$	$p > n$
Crystal returns back to its initial state (n = p) when	$N_A^- = N_D^+$	$N_A^- = N_D^+$

First Multiple choice questions

The semiconductor crystal

1 Pure silicon or germanium crystals become perfectly insulating to electricity at a temperature of ..... .

(a) 0°C (b) 273°C (c) – 273°C (d) 273 K

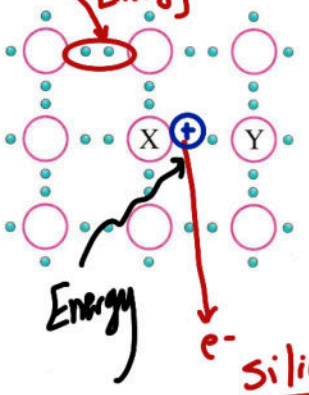
2 The opposite figure shows a section of pure silicon crystal at room temperature in which the covalent bond between atom (X) and atom (Y) in the crystal is incomplete and that's because one of the electrons of the bond ..... .

(a) has repelled the other electron in the bond

(b) has gained enough energy from the surrounding environment to get free

(c) has moved into an inner energy level in atom (X)

(d) has been captured by atom (Y)



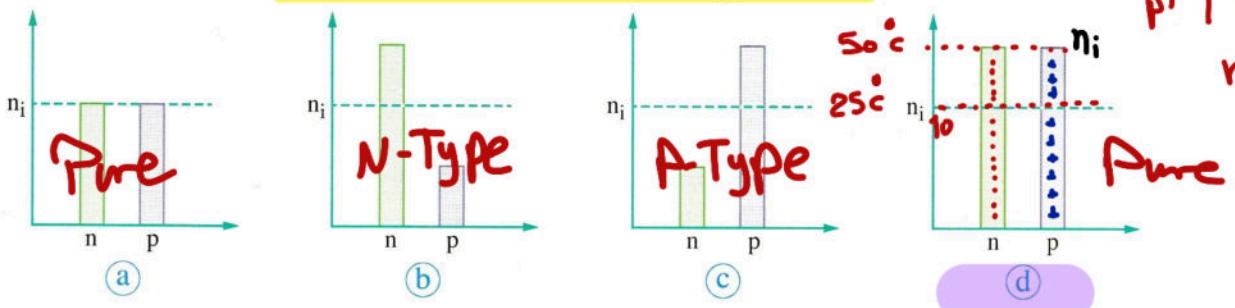
3 If a copper chip and a germanium crystal are cooled down from 300 K to 200 K, the electrical resistance ..... .

(a) increases for each of them (b) decreases for each of them

(c) increases for the copper chip while it decreases for the germanium crystal

(d) decreases for the copper chip while it increases for the germanium crystal

4 In a pure semiconductor crystal at 25°C, the concentration of free electrons = The concentration of holes = n_i, so which of the following graphs represents the concentration of free electrons (n) and that of holes (p) at 50°C?



(a) (b) (c) (d)

5 Two intrinsic (pure) silicon crystals A and B are put at temperatures t_1 and t_2 respectively. If $t_2 > t_1$, the mathematical relation that represents the concentration of free electrons (n) and the concentration of holes (p) in both crystals is ..... .

(a) $n_A + n_B = p_A + p_B$

(b) $n_A + p_A = n_B + p_B$

(c) $n_A \times p_A = n_B \times p_B$

(d) $n_B - n_A > p_B - p_A$

6 Three intrinsic silicon crystals x, y, z where crystal (x) is in a state of thermodynamic equilibrium, crystal (y) temperature is increased, crystal (z) temperature is decreased, which of the three crystals:

(i) Contains positive charge carriers?

(a) Crystal (x). (b) Crystal (y). (c) Crystal (z). (d) The three crystals.

(ii) The charge carriers' concentration in it increases?

(a) Crystal (x). (b) Crystal (y). (c) Crystal (z). (d) The three crystals.

(iii) Its resistivity increases?

(a) Crystal (x). (b) Crystal (y). (c) Crystal (z). (d) The three crystals.

7 The element with which a silicon crystal is doped to form an n-type semiconductor is ..... .

(a) boron (trivalent) (b) antimony (pentavalent)

(c) nickel (divalent) (d) aluminum (trivalent)

8 When increasing the percentage of trivalent impurities in a p-type semiconductor crystal, the electrical conductivity of the crystal will .....

(a) increase (b) decrease

(c) doesn't change (d) be indeterminable

9 A p-type semiconductor crystal in which the concentration of holes equals X, so when adding pentavalent impurities to the crystal by a concentration of $\frac{1}{2}$X, its electric conductivity .....

(a) decreases (b) remains unchanged

(c) increases (d) can't be determined

10 The dominant (majority) charge carriers in p-type crystal are .....

(a) free electrons (b) holes (c) positive ions (d) negative ions

11 A silicon crystal is doped with atoms of a pentavalent element, so at equilibrium the ratio between the concentration of holes to that of free electrons is .....

(a) less than one

(b) greater than one

(c) equal to one

(d) the information is insufficient for determining the answer

12 The n-type semiconductor crystal is ..... .

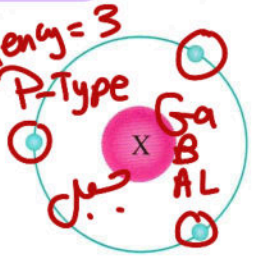
(a) negatively charged (b) electrically neutral

(c) positively charged (d) electrically insulator

13 In an impure semiconductor crystal, if the concentrations of free electrons and holes are n and p respectively, certainly .....

(a) $n > p$ (b) $n < p$ (c) $n = p$ (d) $n \neq p$

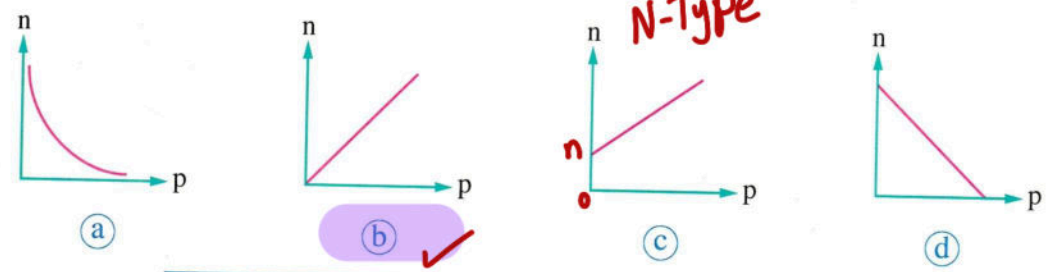
14 The opposite diagram illustrates the electrons of the outermost shell for element (49)X, if a pure semiconductor crystal is doped with atoms of this element, the type and the charge of the resulted crystal respectively will be ..... .



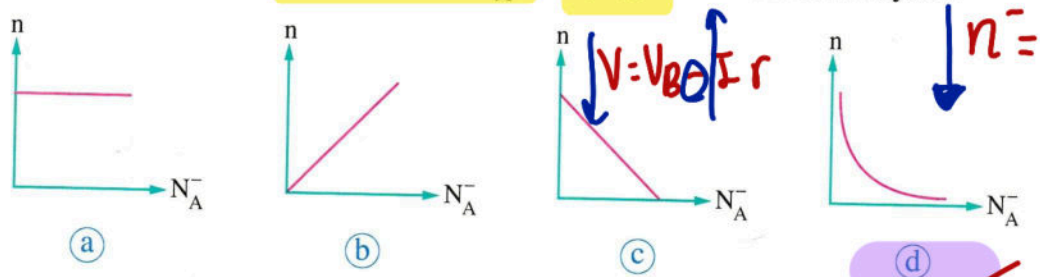
(a) n, neutral (b) p, positive

(c) p, neutral (d) n, negative

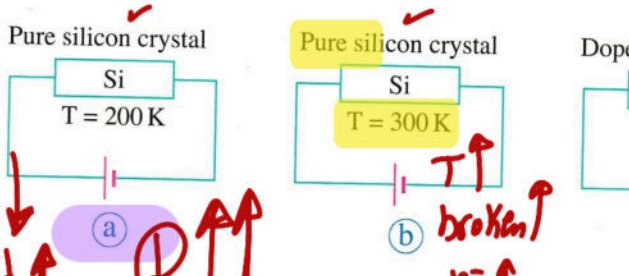
15 The graph that represents the relation between the concentration of free electrons (n) and the concentration of holes (p) in a pure silicon crystal when increasing its temperature is ..... .



16 Which of the following graphs represents the relation between the concentration of free electrons (n) and that of aluminum ions (N_A^-) in a p-type semiconductor crystal?

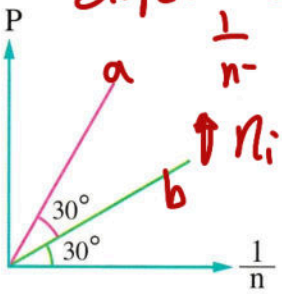


17 Which of the following circuits has the greatest electrical resistance to the flow of electric current?



$\frac{P^+_a}{P^+_b} = \frac{n_{i_a}}{n_{i_b}} = \sqrt{\frac{\text{Slope a}}{\text{Slope b}}} = \sqrt{\frac{\tan 60}{\tan 30}}$

18 Two intrinsic silicon crystals a, b at different temperatures where $(t_a > t_b)$ were doped with phosphorus atoms, the opposite graph represents the relation between the holes concentration (P) in the doped crystal and the inverse of the free electron concentration $\left(\frac{1}{n}\right)$ in it, then the ratio of the holes concentrations in the two pure crystals before doping $\left(\frac{P_a}{P_b}\right)$ is ..... .



$\text{Slope} = \frac{P^+}{\frac{1}{n^-}} = P^+ \times n^- = n_i^2$

$\uparrow n_i = \sqrt{\text{slope}\uparrow}$

$t_a > t_b$

$n_{i_a} > n_{i_b}$

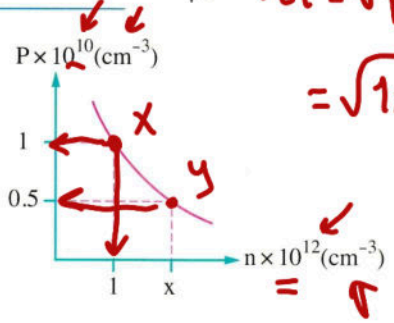
n_i

(a) 3 (b) $\sqrt{3}$ ✓ (c) $\frac{1}{3}$ (d) $\frac{1}{\sqrt{3}}$

$n_i = \sqrt{P^+ \times n^-}$

$= \sqrt{1 \times 10^{10} \times 1 \times 10^{12}}$

19 A pure semiconductor crystal is doped with phosphorus atoms, and the opposite graph represents the relation between the holes concentration (P) and the concentration of free electrons (n) in the crystal, then:



(i) The concentration of free electrons in the intrinsic semiconductor crystal is equal to ..... .

Pure → n_i

(a) 10^{10} cm^{-3} (b) 10^{11} cm^{-3}

$n_i = 10^{11}$

(c) 10^{12} cm^{-3} (d) 10^{22} cm^{-3}

(ii) The value of (x) on the graph is equal to ..... .

$n_i^2 = (10^{11})^2$

(a) 1.5×10^{12} cm^{-3} (b) 2×10^{12} cm^{-3}

(c) 2.5×10^{12} cm^{-3} (d) 3×10^{12} cm^{-3}

$P^+ \times n^- = n_i^2$

$0.5 \times 10^{10} \times n^- = (10^{11})^2$

20 Three identical intrinsic silicon crystals (a), (b), and (c), crystal (b) was doped with aluminum atoms at a concentration of (X) cm^{-3} and crystal (c) was doped with phosphorus atoms at the same concentration of (X) cm^{-3}. Assuming all of the three crystals are at the same temperature, the correct order of the crystals based on the concentration of free electrons in each is ..... .

P-Type

N-Type

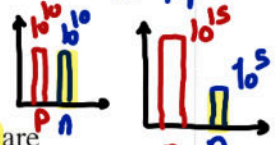
(a) a = b = c (b) a < b < c ✓ (c) c = b > a (d) b < a < c ✓

a → Pure $n^- = P^+$

b → P-Type $n^- < P^+$

c → N-Type

$n^- > P^+$



$P \times n = P \times n$

$10^{10} \times 10^{10} = 10^{15} \times 10^{5}$

21 If the concentrations of free electrons and holes in a silicon crystal doped with arsenic are 10^{13} cm^{-3} and 10^{9} cm^{-3} respectively, the concentration of each of free electrons and holes in a pure silicon crystal equals ..... .

n^- P^+

As N-Type

$n_i = \sqrt{P^+ \times n^-} = \sqrt{10^9 \times 10^{13}}$

(a) 10^{9} cm^{-3} (b) 10^{10} cm^{-3} (c) 10^{11} cm^{-3} ✓ (d) 10^{13} cm^{-3}

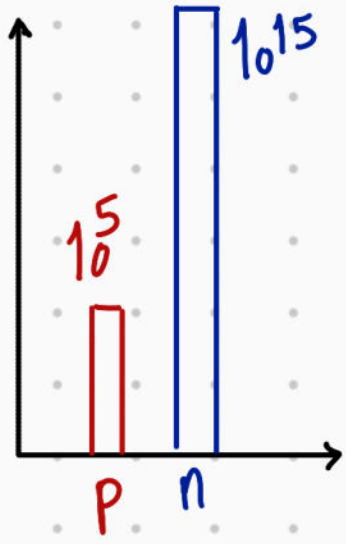
n^-

22 In a semiconductor crystal, the concentration of free electrons is 5×10^{13} cm^{-3} and the concentration of holes is 8×10^{12} cm^{-3}, so this semiconductor crystal is ..... .

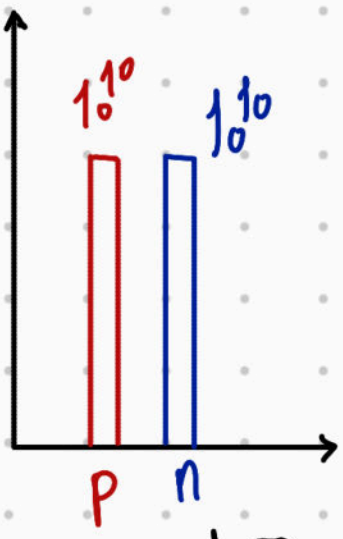
(a) n-type ✓ (b) p-type P^+ (c) positively charged (d) pure

$n^- > P^+$

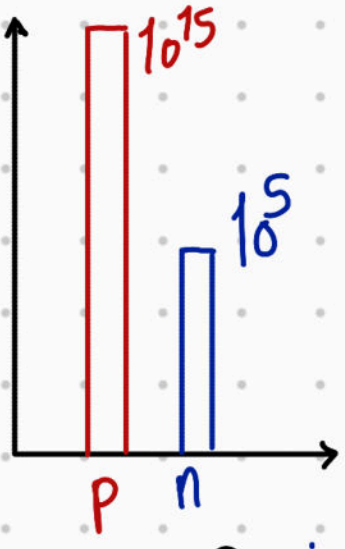
$50 \times 10^{12} \quad 8 \times 10^{12}$



Crystal ©
impure
N-type



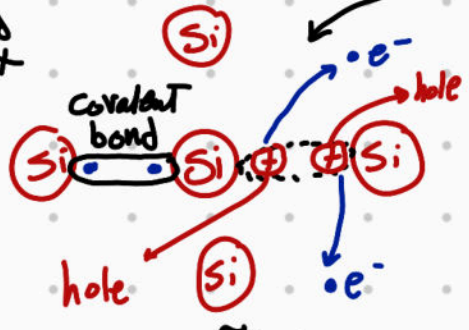
Crystal ⓐ
Pure



Crystal ⓑ
impure
P-Type

Free electrons
b < a < c

electron free
bond
+ + +



Semi-conductor
Valency = 4

Si_{14} 2 8 4

Ge_{32} 2 8 18 4

Responsible for Conducting (I) / (σ)

Charge Carriers

Free electrons (n⁻) — semi cond. — holes (A⁺)

Intrinsic (Pure)

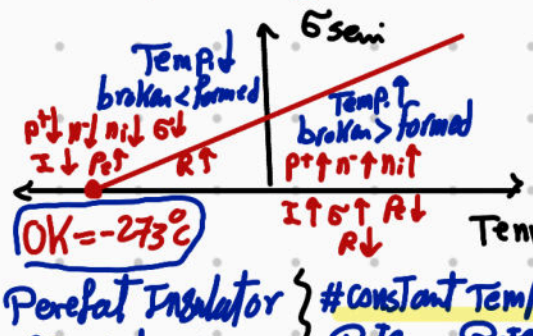
$p^+ = n^- = n_i$

Extrinsic (impure)

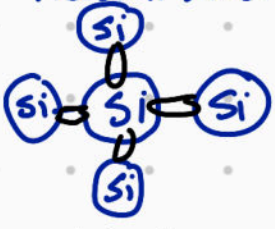
$p^+ \neq n^-$

$p^+ > n^-$
P-Type
Trivalent
Acceptors
(N_A^-) → Ga/B/Al

$p^+ < n^-$
N-Type
Pentavalent
Donors
(N_D^+) → P/Sb/As



Perfect Insulator
Absolute Zero



all bonds are formed

$p^+ = n^- = n_i = \sigma = I = Zero$

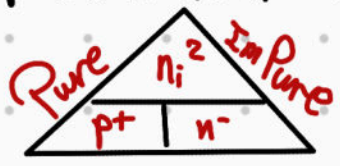
$Pe = R = \infty$

constant Temp.
Rate broken = Rate formed
* Thermal dynamic equilibrium
p^+ n^- n_i I σ
R Pe → constant

Law of mass action

$p^+ = n^- = n_i$ → Pure

$p^+ \times n^- = n_i \times n_i = n_i^2$



$p^+ \times n^- = n_i^2 \rightarrow n_i = 10^{10}$

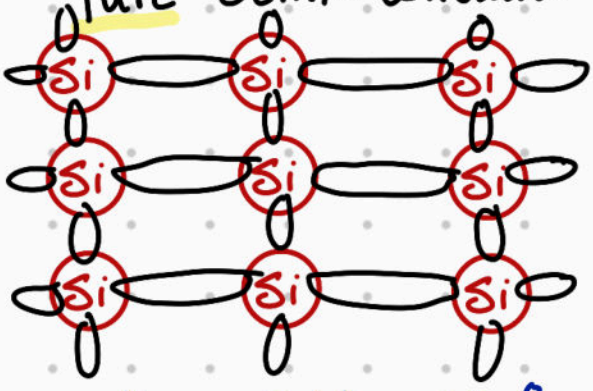
$10^{10} \times 10^{10} = 10^{20}$

P-Type | neutral | N-Type

$p^+ = N_A^- + n^-$
$N_A^- >>>> n^-$
$p^+ = N_A^-$

$n^- = N_D^+ + p^+$
$N_D^+ >>>> p^+$
$n^- = N_D^+$

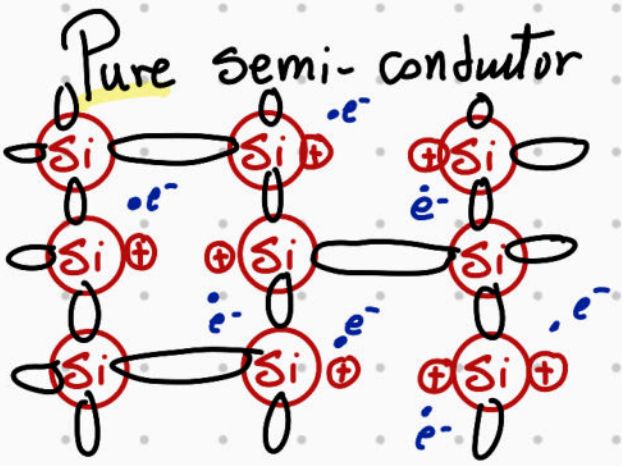
Pure semi-conductor



at Zero Kelvin = -273°c

$p^+ = n^- = n_i = Zero$

Pure semi-conductor



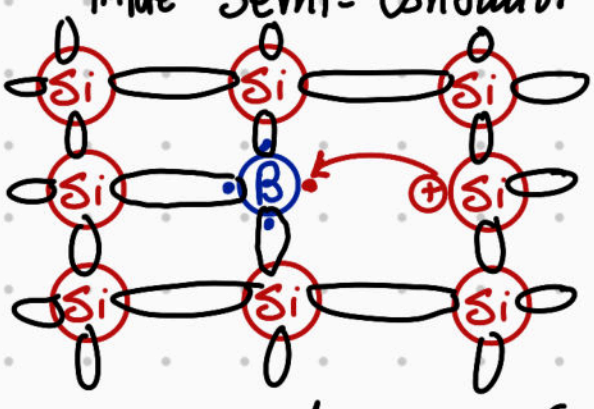
at Room Temp. 27°c = 300K

$p^+ = n^- = n_i = 7$ 25°c = 298K

↳ intrinsic no. (Pure)

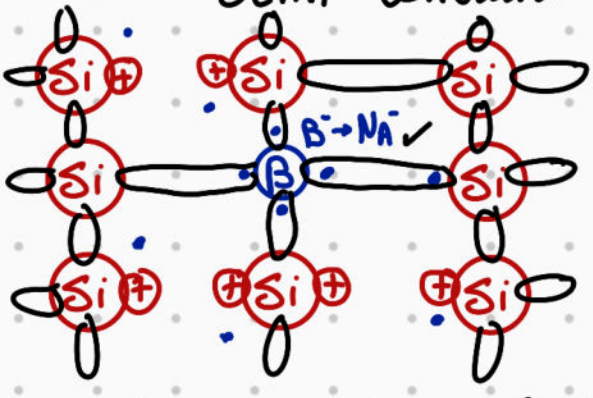
n_i → conc. of free electrons or holes in Pure semi-conductor. (intrinsic)

Impure semi-conductor



at Zero kelvin = -273°c

Impure semi-conductor



at Room Temp 27c° = 300K

$p^+ = 6 > n^- = 5$

P-Type → GaIBIAl

$p^+ = n^- + N_A^-$ → neutral

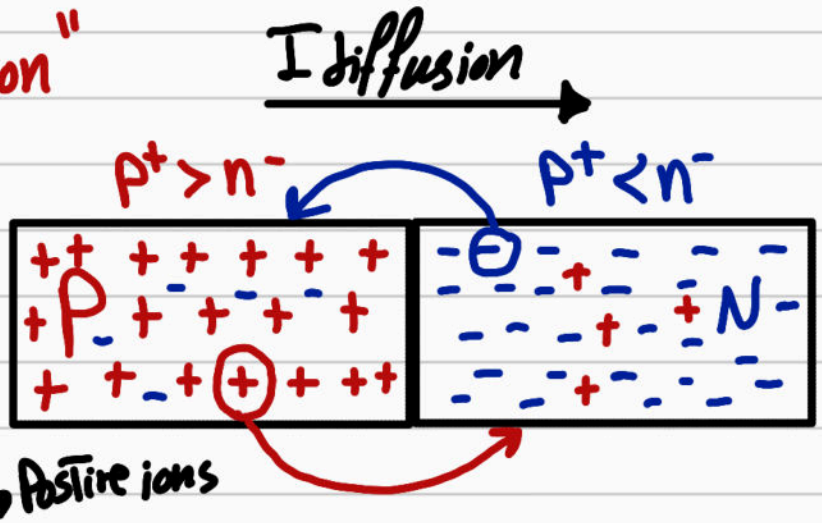
L.E.D

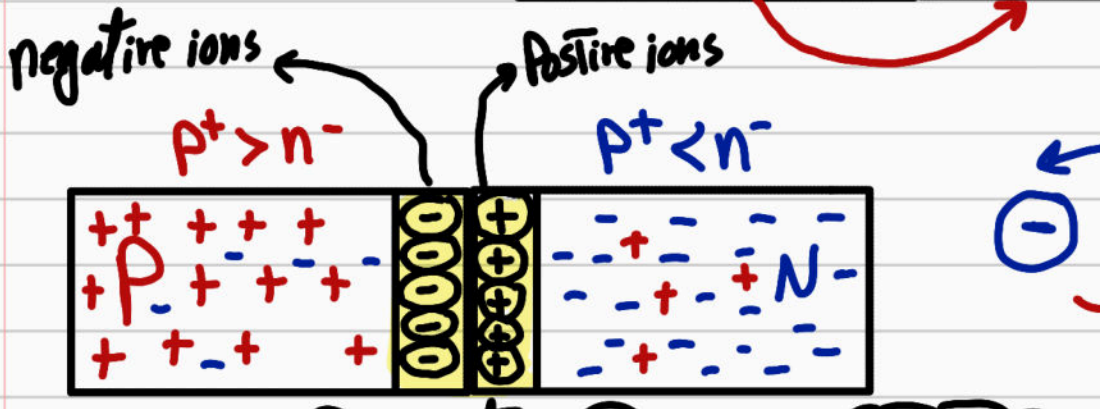
Light Emitting Diode

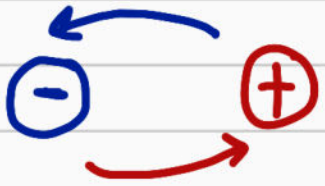
*Diode "PN-Junction"

* diffusion :

high Conc. Area → low conc. Area







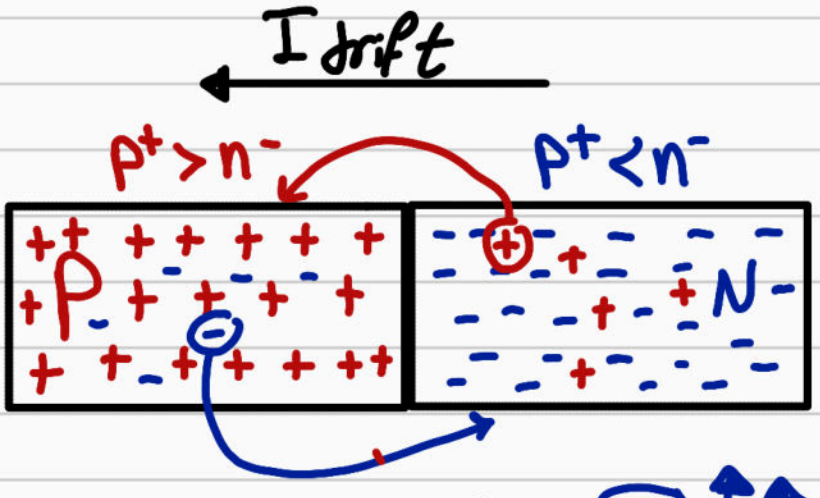
Depletion Region "Transition"

R_{Diode}

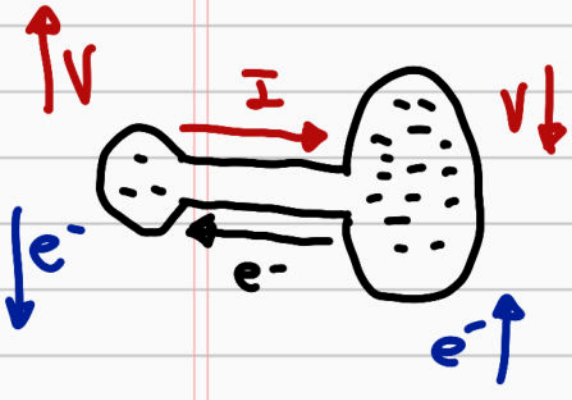
No free electrons or holes

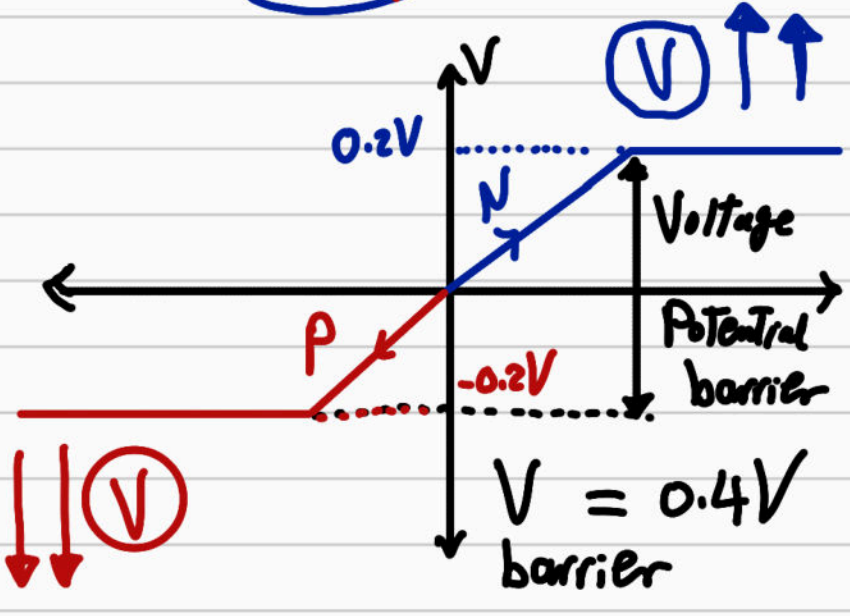
* drift

low conc. → high conc.



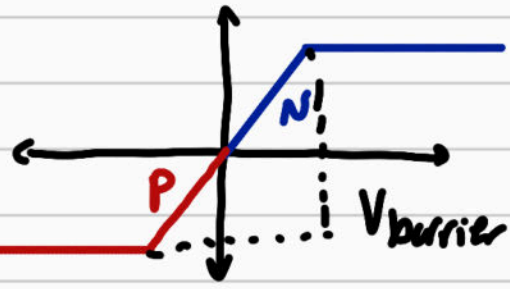
$I_{diffusion} = I_{drift}$

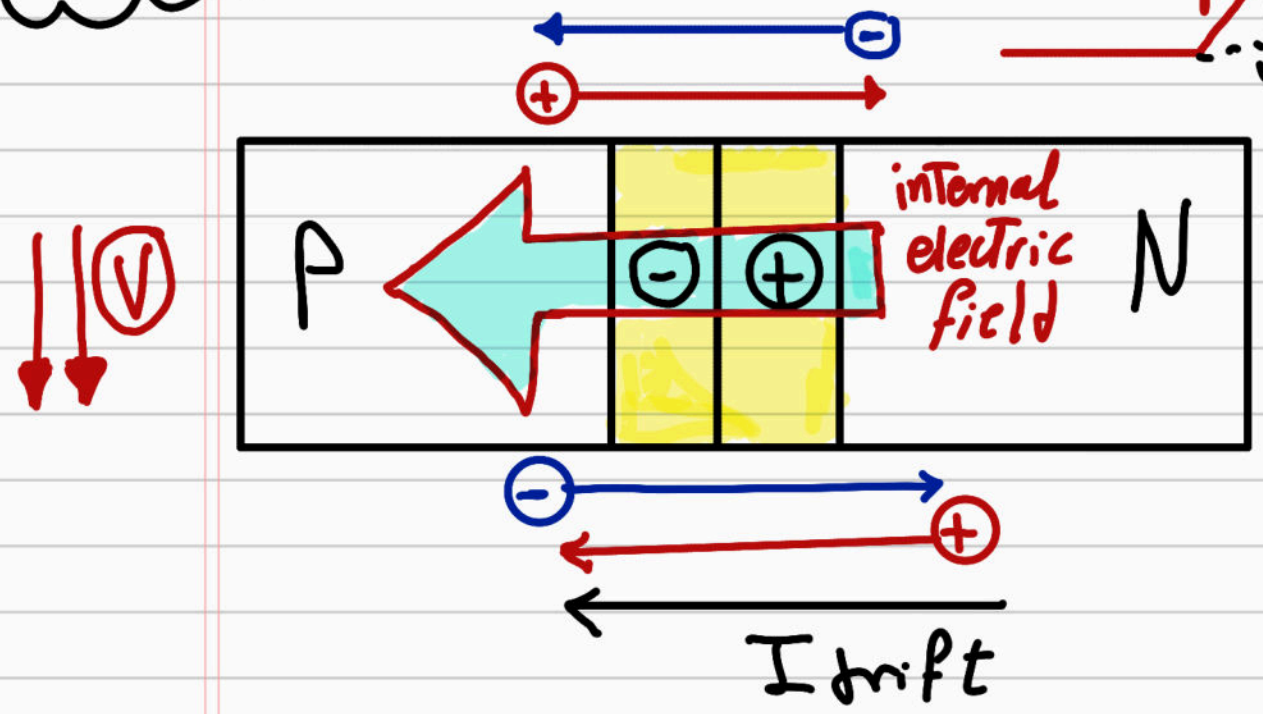




$I_{diffusion} = I_{drift}$

$I_{diffusion}$

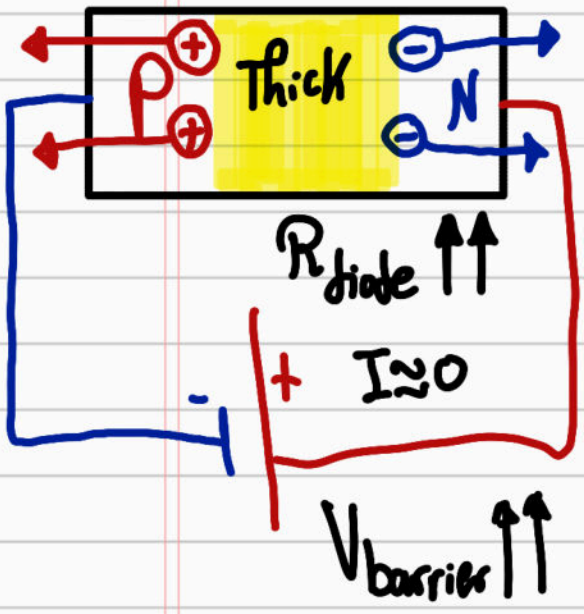




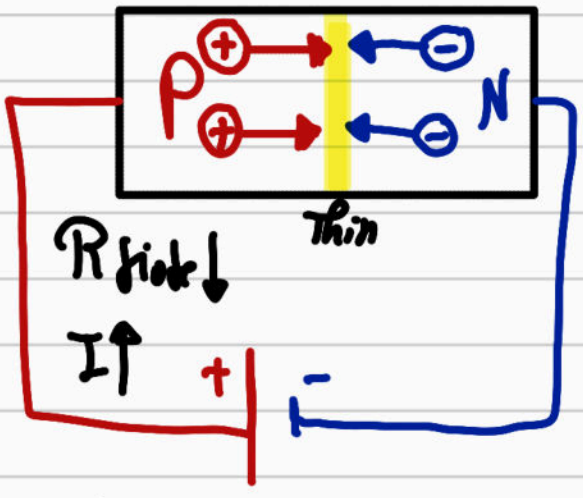
I_{drift}

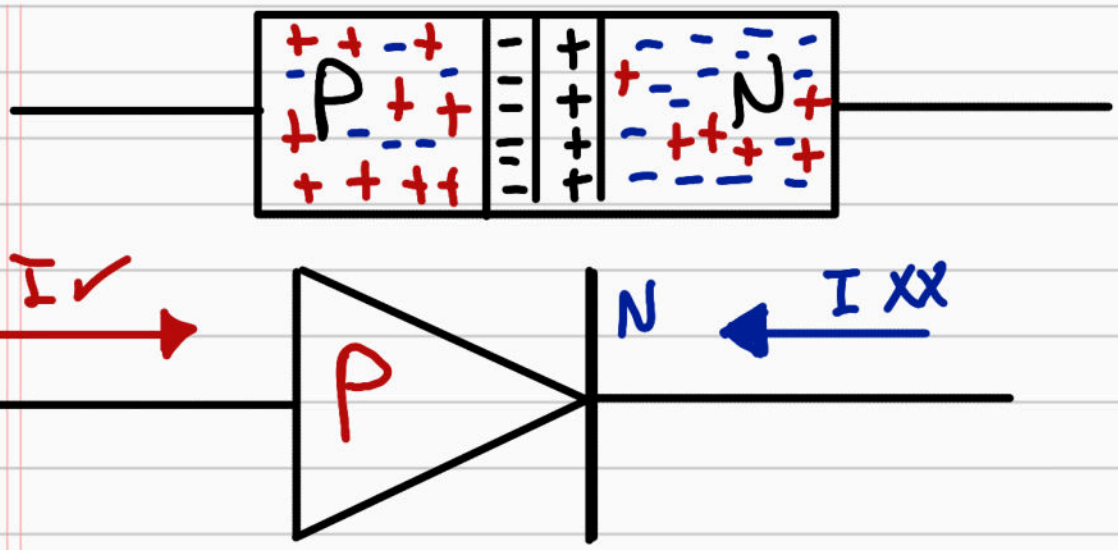
Connection

Reverse "Backward"

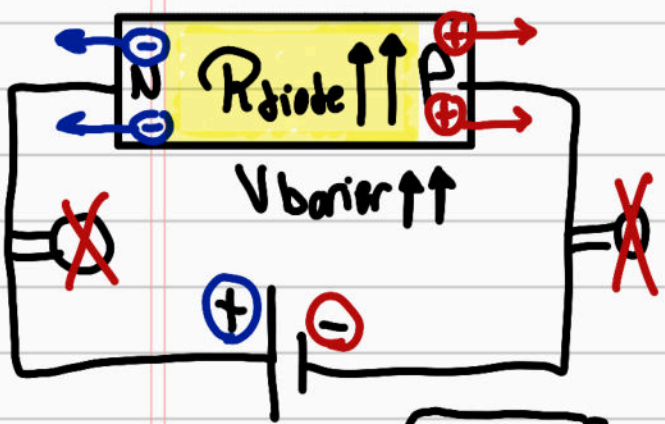


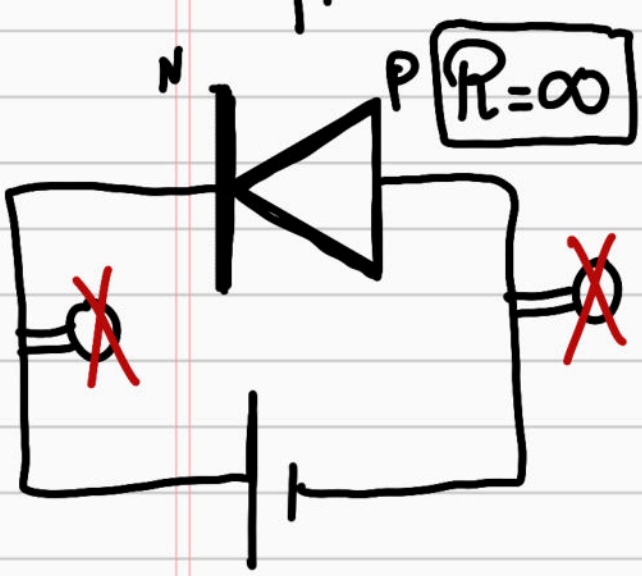
forward





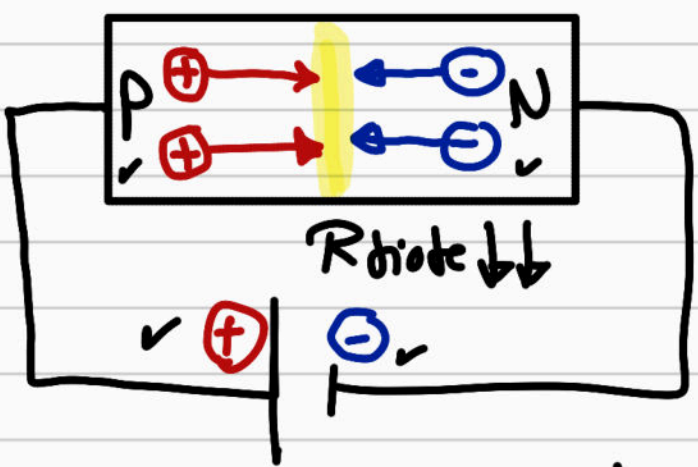
Reverse/BackWard

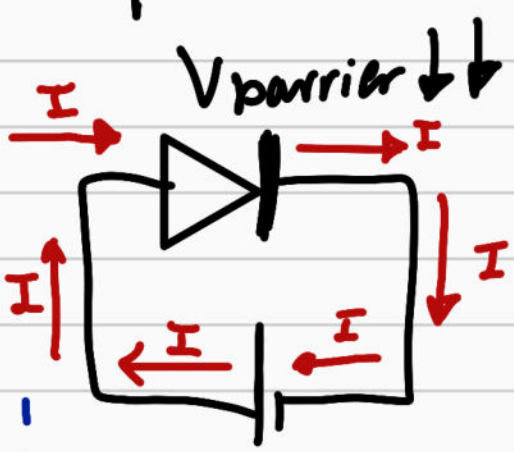




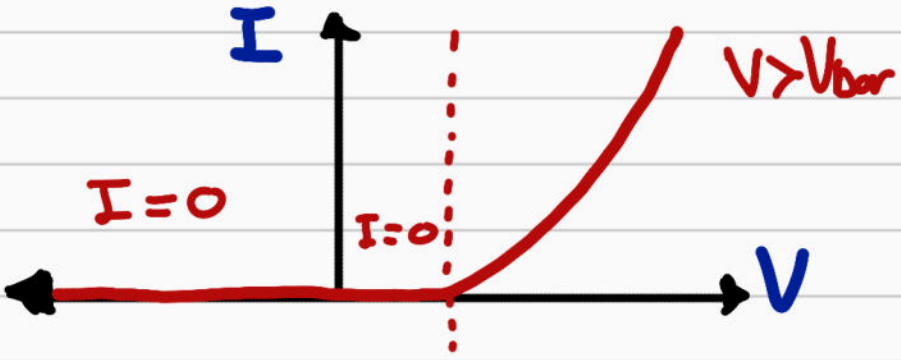
Reverse

forWard

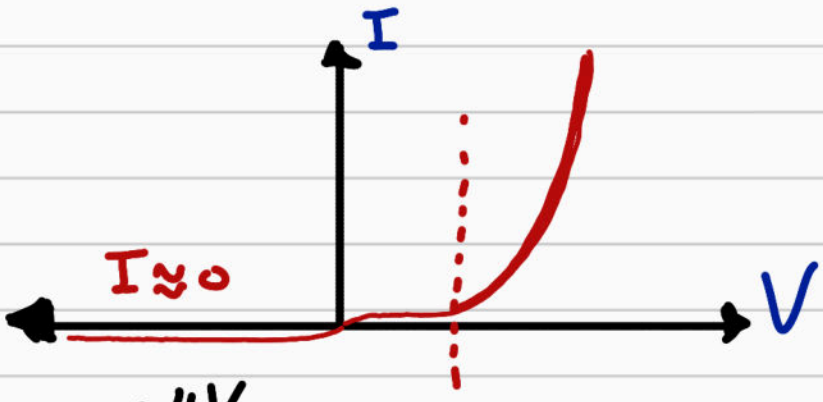


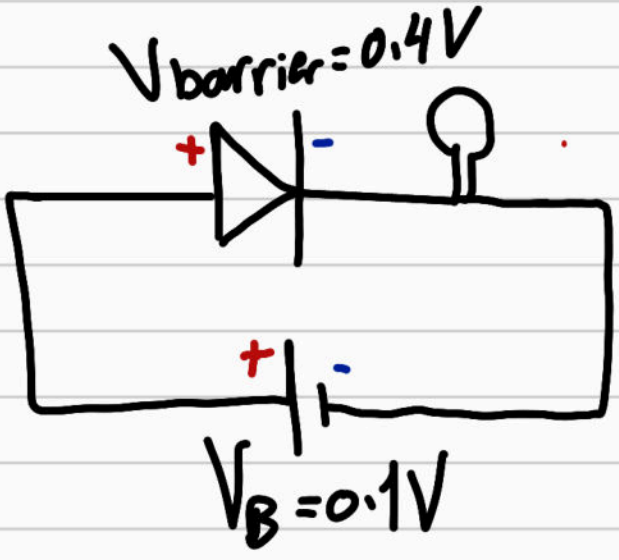


Ideal



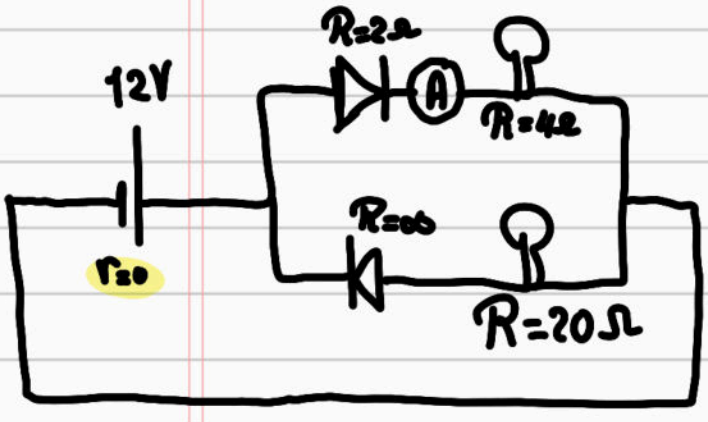
non-Ideal

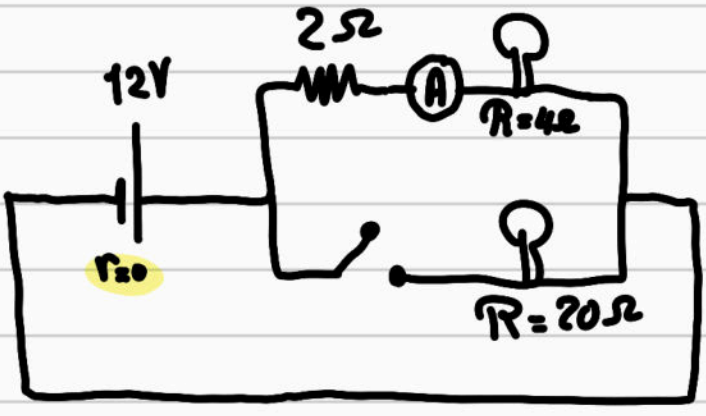




$V_B < V_{barrier}$

$I = 0$





$$I = \frac{12}{2+4} = 2A$$

23 The concentration of either free electrons or holes in each of two identical intrinsic (pure) silicon crystals x and y is 10^{12} cm^{-3}, so if crystal x gets doped with antimony atoms in a concentration of 10^{14} cm^{-3} while crystal y gets doped with aluminum atoms in a concentration of 10^{14} cm^{-3}, the ratio between the concentrations of holes in the two crystals $\left(\frac{p_x}{p_y}\right)$ will be ..... .

(a) 10^{-2}
(b) 10^{-4}
(c) 10^{-6}
(d) 10^{-8}

24 * The concentration of free electrons or that of holes in a pure silicon crystal is 10^{8} cm^{-3}, if the crystal is doped with aluminum atoms with a concentration of 10^{10} cm^{-3}, hence the concentration of holes and that of free electrons in this case at the fully ionization of the atoms of impurities respectively will be ..... .

(a) 10^{10} cm^{-3}, 10^{8} cm^{-3}
(b) 10^{8} cm^{-3}, 10^{10} cm^{-3}
(c) 10^{6} cm^{-3}, 10^{10} cm^{-3}
(d) 10^{10} cm^{-3}, 10^{6} cm^{-3}

The pn junction

25 When connecting a diode in reverse direction to a battery, ...

(a) the potential barrier increases and the resistance increases
(b) the potential barrier decreases and the resistance decreases
(c) the potential barrier increases and the resistance decreases
(d) neither the potential barrier nor the resistance changes

26 The opposite graph represents the relation between the intensity of the current (I) through a semiconductor diode and the potential difference (V) across its terminals, so the potential barrier for this diode is ..... .

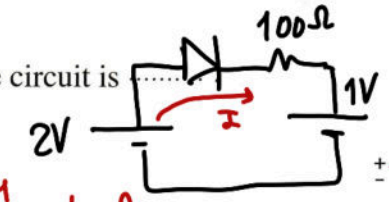
(a) 1.2 V
(b) 0.8 V
(c) 0.7 V
(d) zero

27 The depletion region of the pn junction has a high electrical resistance because of ..... .

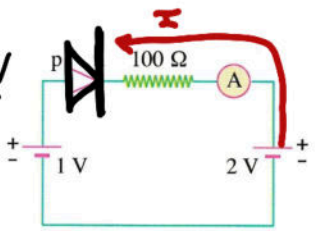
(a) not containing charge carriers
(b) containing a large number of charge carriers
(c) containing only free electrons
(d) containing only holes

28 The ammeter reading in the opposite circuit is .....

(a) zero ✓
(b) 1 mA
(c) 10 mA
(d) 30 mA

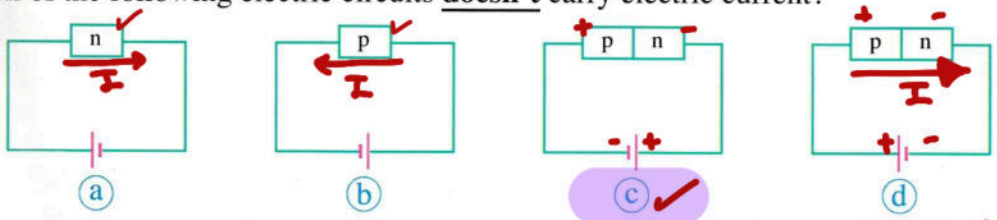


$I = \frac{2-1}{100} = \frac{1}{100} = 10mA$



Reverse

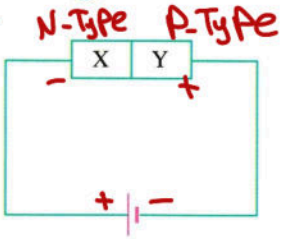
29 Which of the following electric circuits **doesn't** carry electric current?



(a) (b) (c) ✓ (d)

$V_B > V_{barrier}$ forward

30 The opposite figure represents a diode made of silicon in which region X contains arsenic atoms and region Y contains boron atoms, hence ..... .

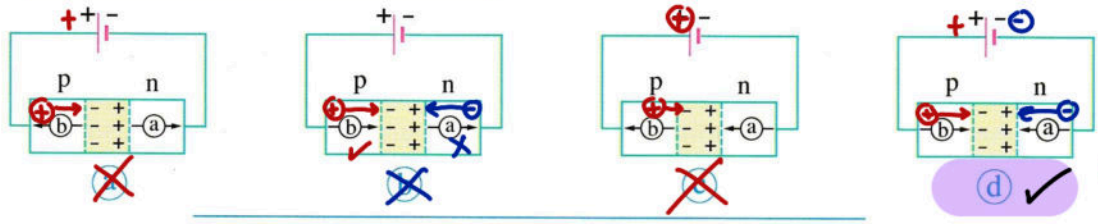


	Type of region X	Type of region Y	Bias
(a)	p	n	forward
(b)	n ✓	p ✓	forward
(c)	p	n	reverse
(d) ✓	n ✓	p ✓	reverse ✓

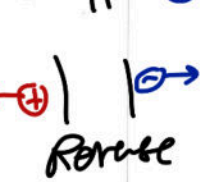
N-Type $p^+ < n^-$ dominant
(a) → free electrons

31 If a and b are referring to the dominant charge carriers in the two regions (n, p) of a diode respectively, which of the following figures represents the directions of their motions in the case of forward bias connection?

P-Type
(b) → holes

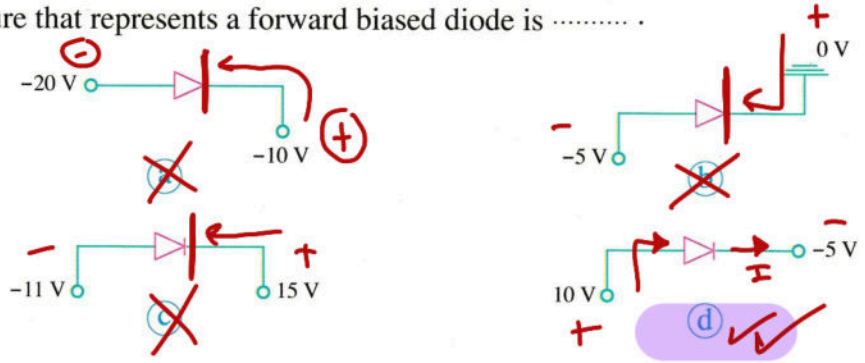


forward



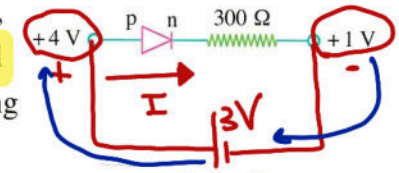
Reverse

32 The figure that represents a forward biased diode is ..... .



$I = \frac{V}{R}$ → Potential difference $\quad I = \frac{4-1}{300}$

33 The opposite figure represents a part of an electric circuit, if the resistance of the pn junction is negligible in forward bias and infinite in reverse bias, the intensity of the passing current is ..... .



(a) 0 (b) 10^{-2} A 0.01A (c) 10^{2} A (d) 10^{-3} A

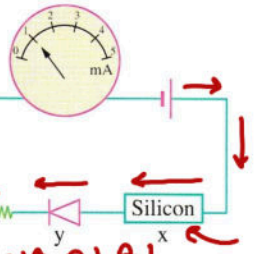
34 In the opposite circuit, which of the following actions leads to an increase in the reading of the milliammeter? I↑

(a) Reversing the polarity of the battery ✗ (Reverse) I=0

(b) Reversing the direction of component y ✗ (Reverse) I=0

(c) ✓ Increasing the temperature of component x Temp.↑ broken > found

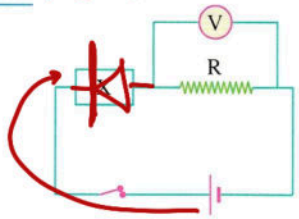
(d) Increasing the temperature of component z



p⁺↑ n⁻↑ n_i↑ I↑ or R↓ Rel↓

↑R ∝ temp.↑ ∝ Pel↑ I↓

35 In the opposite electric circuit, if the reading of the voltmeter is approximately equal to zero, then component X is ..... .



(a) n (b) p (c) (d) ✓

36 When a certain electrical component was connected between the terminals of an ohmmeter, the pointer position was as in figure (1) and when reversing the connection of the component between the ohmmeter terminals, the pointer position became as in figure (2), then this component is ..... .

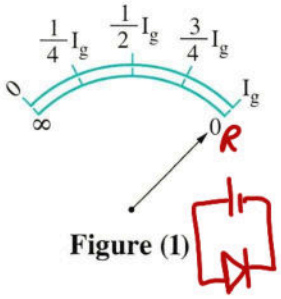


Figure (1)

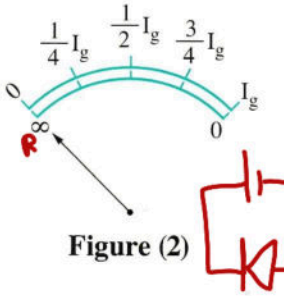
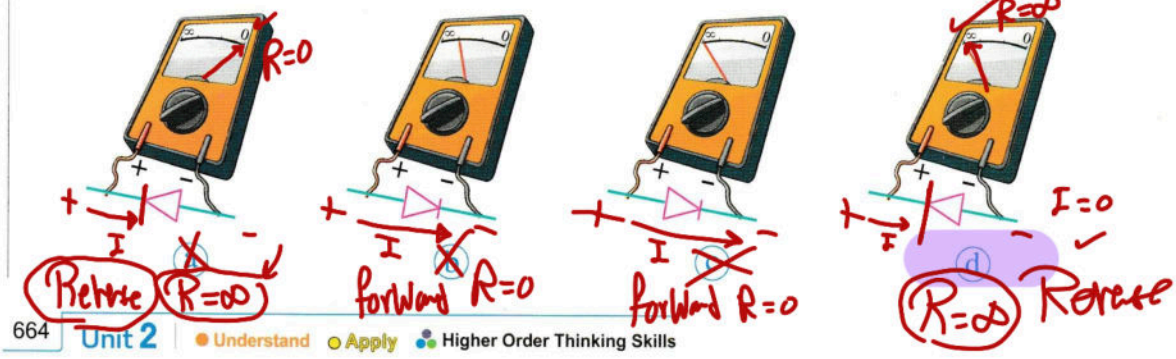


Figure (2)

(a) a pure silicon crystal (b) an n-type silicon crystal

(c) a p-type silicon crystal (d) ✓ a diode

37 An ohmmeter gives zero reading when its terminals are connected together, so if a pn junction has negligible resistance in forward bias and infinite resistance in reverse bias, which of the following figures illustrates the correct deflection of the ohmmeter pointer when the pn junction is connected between its terminals?

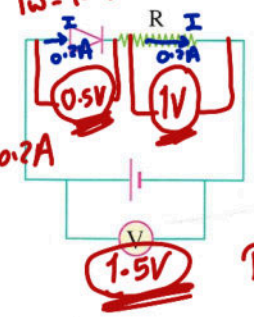


38 ✱ A diode with a forward bias resistance of 100 Ω and a reverse bias resistance of infinity is connected to a source of potential difference 5 V such that the p-side of the diode gets connected with the positive potential at a time then reversed so that the p-side gets connected to the negative potential at another time, so the current intensities through the diode in the two cases respectively are ..... .

(a) 0, 0 (b) 0.05 A, 0 (c) 0, 0.05 A (d) 0.05 A, 0.05 A

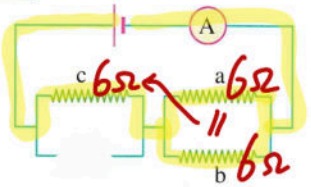
39 ✱ In the opposite electric circuit, the voltmeter reading is 1.5 V, so if the potential difference across the diode terminals is 0.5 V and the electric power consumed through it equals 100 mW, so the resistance of R is .....

(a) 2.5 Ω (b) 5 Ω (c) 7.5 Ω (d) 10 Ω



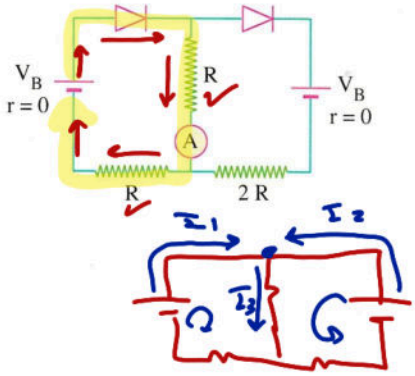
40 ✱ The electric circuit in the opposite figure consists of a dry cell of emf V_B whose internal resistance is negligible, three identical ohmic resistances (a, b, c) and a diode with a forward bias resistance equal to the value of one of the three ohmic resistances and an infinite resistance in reverse bias, hence the ratio between the readings of the ammeter before and after reversing the diode connection respectively equals ..... .

(a) $\frac{1}{2}$ (b) $\frac{2}{1}$ (c) $\frac{2}{3}$ (d) $\frac{3}{2}$



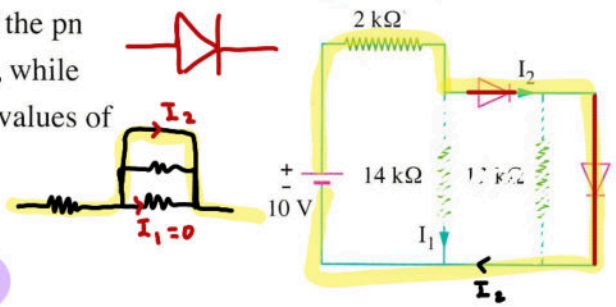
41 In the shown electric circuit, if the resistance of the diode is negligible in forward bias and infinite in reverse bias, the reading of the ammeter is .....

(a) $\frac{V_B}{3R}$ (b) $\frac{V_B}{2R}$ (c) $\frac{2V_B}{3R}$ (d) $\frac{3V_B}{2R}$

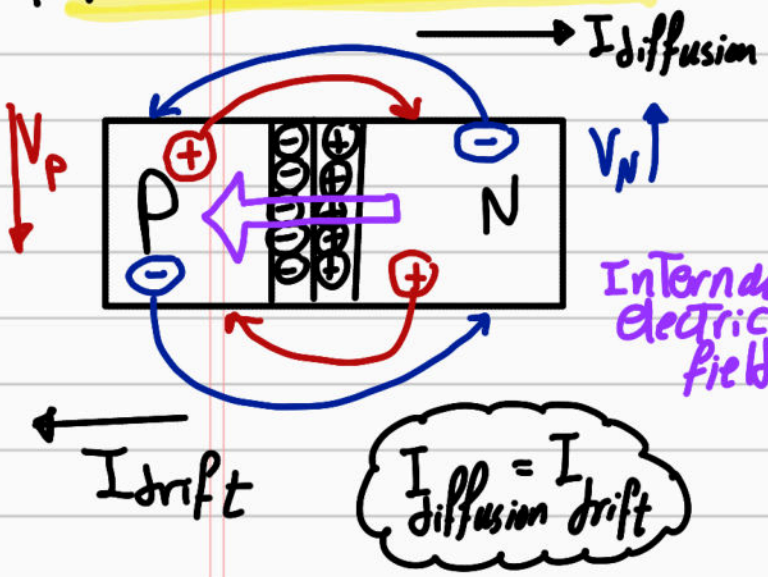


42 In the opposite circuit, if the resistance of the pn junction can be neglected in forward bias, while considered infinite in reverse bias, so the values of I_1 and I_2 respectively are ..... .

(a) 0, 0 (b) 5 mA, 5 mA (c) 5 mA, 0 (d) 0, 5 mA

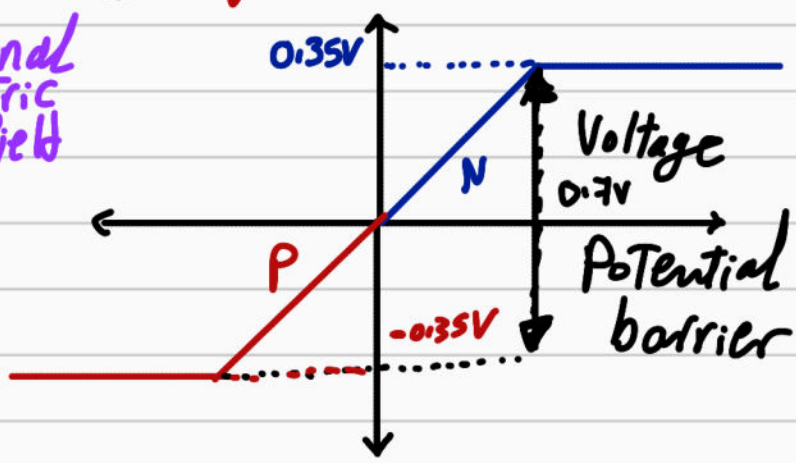


* PN-Junction "Diode"



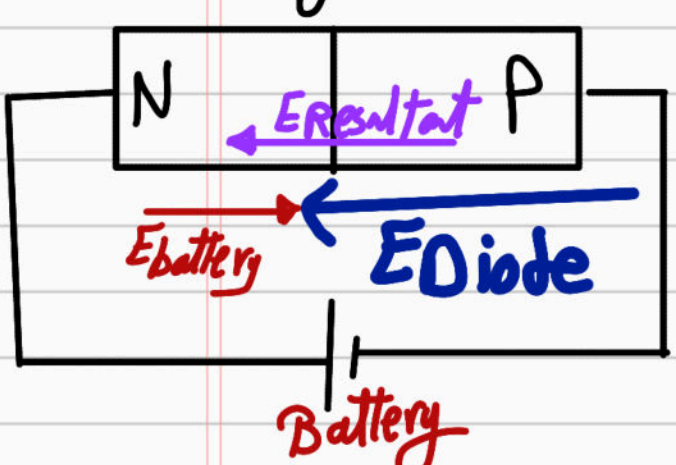
- Diffusion I
- Drift I
- Depletion Region
- Voltage Pot. barrier

نظري



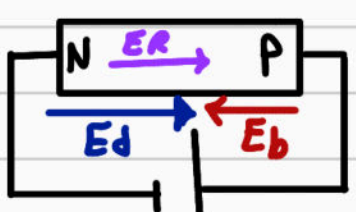
Reverse (Backward)

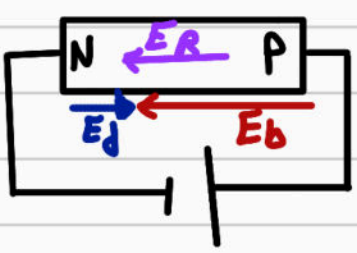
Voltage barrier ↑↑



$I \simeq 0$

for Ward

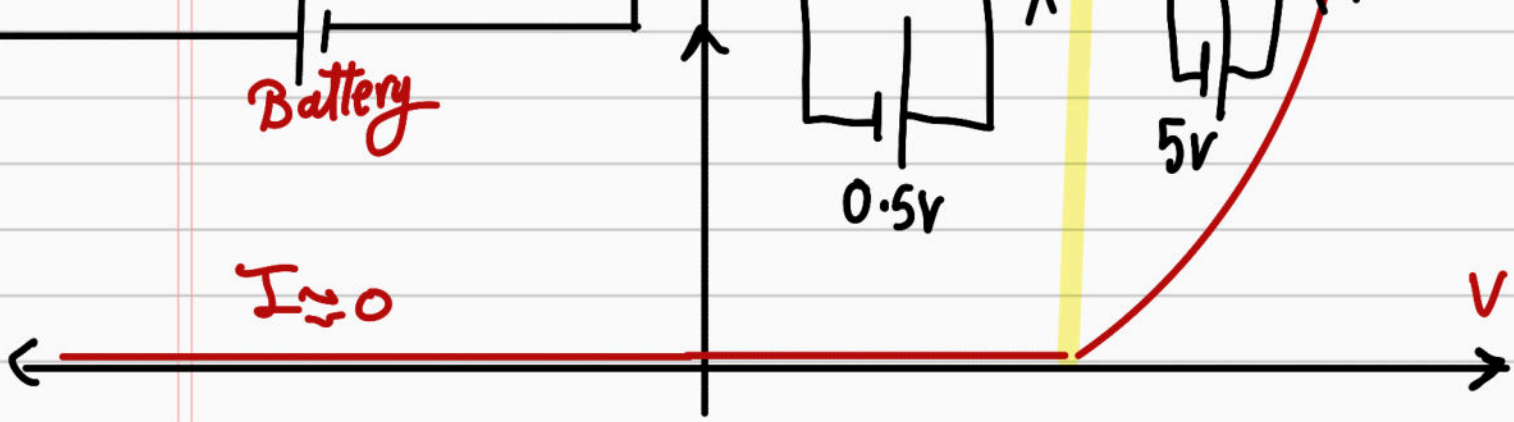




E_{diode} $E_{battery}$

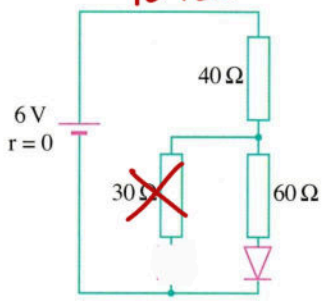
$V < V_{barrier}$

$V > V_{barrier}$



$I = \frac{6}{40+60} = 0.06A$

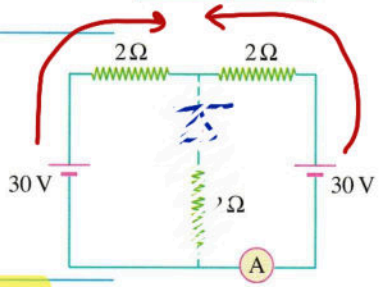
43 ✱ In the opposite diagram, considering that the internal resistance of the electric source is negligible and supposing that the resistances of the diodes in forward connection are negligible while in reverse connection are infinite, the intensity of the current in the 40 Ω resistor is ..... .



(a) 0 (b) 0.06 A
(c) 0.1 A (d) 0.12 A

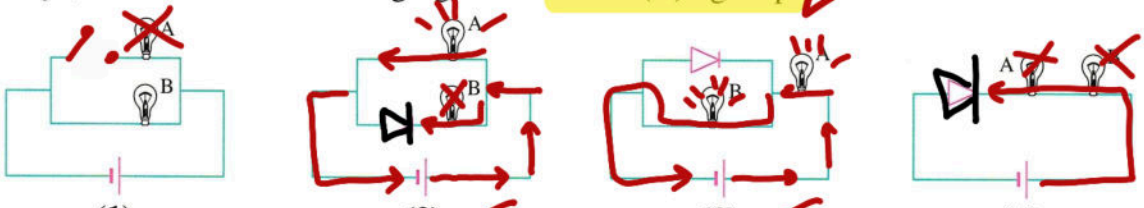
44 What happens to the reading of the ammeter (A) when reversing the connection of the diode in the circuit shown in the opposite figure?

$I = \frac{30-30}{2+2}$



(a) Increases. (b) Decreases.
(c) Becomes zero. (d) Doesn't change.

45 Two identical lightbulbs (A, B) are connected with a pn junction to a battery in several ways, in which of the following figures does bulb (A) light up?

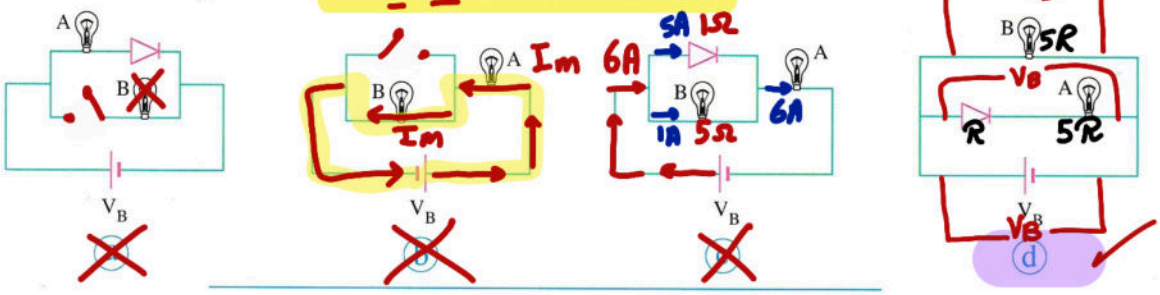


(1) (2) (3) (4)

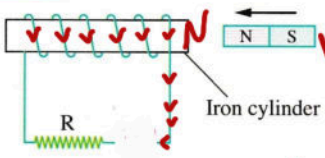
(a) (1), (3). (b) (2), (3). (c) (2), (4). (d) (3), (4).

$R_{lamp} = 5\Omega$ $R_{diode} = 1\Omega$

46 In the following electric circuits, given that the two lightbulbs have the same resistance which is five times as large as the resistance of the pn junction in forward bias, so in which circuit is the brightness of bulb A less than that of bulb B?



47 A coil is wound around an iron cylinder and connected to an ohmic resistance (R) and a diode, as shown in the opposite figure. When a magnet approaches the coil, ...



Motion ⇒ Variation ϕ_m ⇒ Induction emf

	In the coil	In the resistance
(a)	An emf is induced	An electric current flows
(b)	An emf is induced	No electric current flows
(c)	No emf is induced	An electric current flows
(d)	No emf is induced	No electric current flows

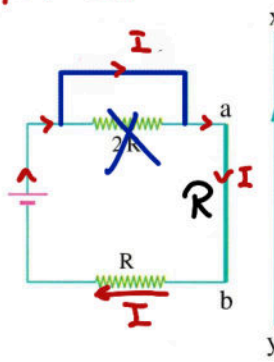
Open circuit: emf ✓, $I = 0$, $F = 0$
Closed circuit: emf ✓, I ✓, F ✓

open switch $R=\infty$ — Closed switch $R=0$

$R_{eq}=2R+R+R=4R \rightarrow I$

$R_{eq}=R+R=2R \rightarrow 2I$

48 A long straight wire (xy) carries an electric current and is placed parallel to another straight wire (ab) with resistance (R) connected to an electric circuit as shown in the opposite figure, if the diode in the circuit is assumed to have negligible resistance in forward bias and in reverse bias, what happens to the magnitude of the mutual magnetic force between the two wires (xy) and (ab) when the switch (K) is closed?



$F=\frac{\mu I_1 I_2 \ell}{2\pi d}$

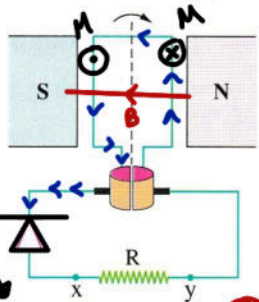
(a) It decreases to half. (b) It doubles. ✓

(c) It vanishes. (d) It doesn't change.

49 The opposite figure shows a dynamo connected to an ohmic resistance (R) and a diode, then the resistance (R) .....

Induction → Fleming Right

Force + B → I induced



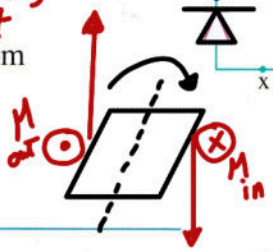
(a) has an electric current always passing through it from point x to y

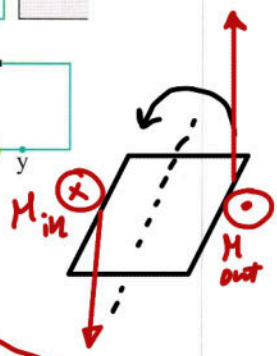
I + B → Force → Fleming left

(b) has an electric current always passing through it from point y to x

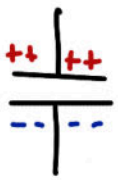
(c) has an alternating current passing through it

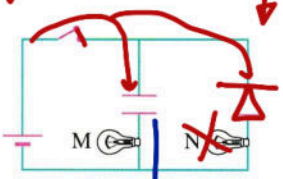
(d) doesn't have electric current ✓





50 Two identical lamps (N, M) are connected to a battery, a capacitor and a pn junction as shown in the opposite electric circuit, which of the two lamps flashes at the moment of closing the switch?





Uncharged

(a) Only M. ✓ during charging the capacitor then goes off.

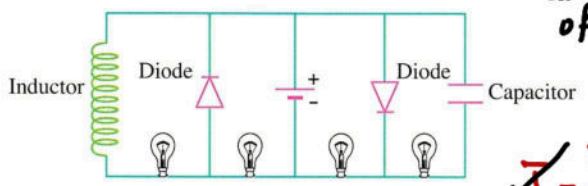
(b) Only N.

(c) M and N.

(d) No one will glow.

at the moment of closing the switch

51 The opposite electric circuit contains multiple electric components and multiple identical lightbulbs, so the number of lightbulbs that are glowing in the circuit is .....



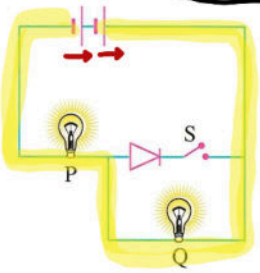
at the moment of closing

$I=\frac{V_B-\varepsilon f}{R}=0$

$V_B=\varepsilon mf_{back}$

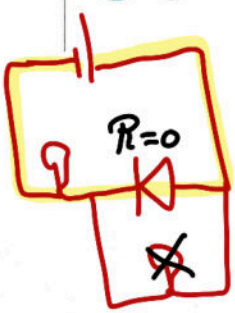
(a) 1 (b) 2 (c) 3 ✓ (d) 4

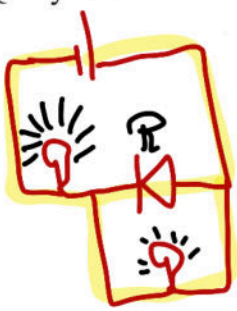
52 Two identical lamps P, Q are connected in an electric circuit with a pn junction as shown in the opposite figure, what happens to the lights of the two lamps when closing switch S?



(a) Both go off. (b) P stays on, Q goes off.

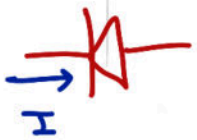
(c) P goes off, Q stays on. (d) Both stay on. ✓





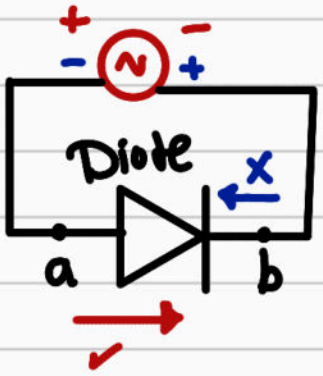
$R_{diode}=??$ forward

$R_{diode}=\infty$ Reverse

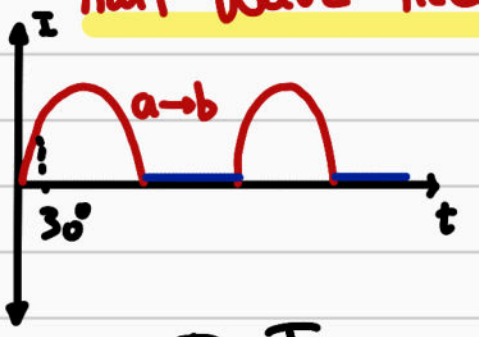


Alternating Current

Input



half Wave Rectification



output

$$P_{w_{AC}} = V_{eff} \times I_{eff}$$

$$V_{eff} = V_{max} \sin(45)$$

$$\frac{1}{2} \times \frac{V_{max}\ I_{max}}{2}$$

$$\frac{V_{max} \times I_{max}}{2 \times 2}$$

$$\frac{V_{max}}{2} \times \frac{I_{max}}{2}$$

$$P_w = \frac{1}{2} P_{w_{AC}}$$

$$V_{eff}\ I_{eff} = \frac{1}{2} \times V_{eff} \times I_{eff}$$

half Wave — full Wave

$$V_{eff} \times I_{eff} = \frac{1}{2} \times \frac{V_{max}}{\sqrt{2}} \times \frac{I_{max}}{\sqrt{2}}$$

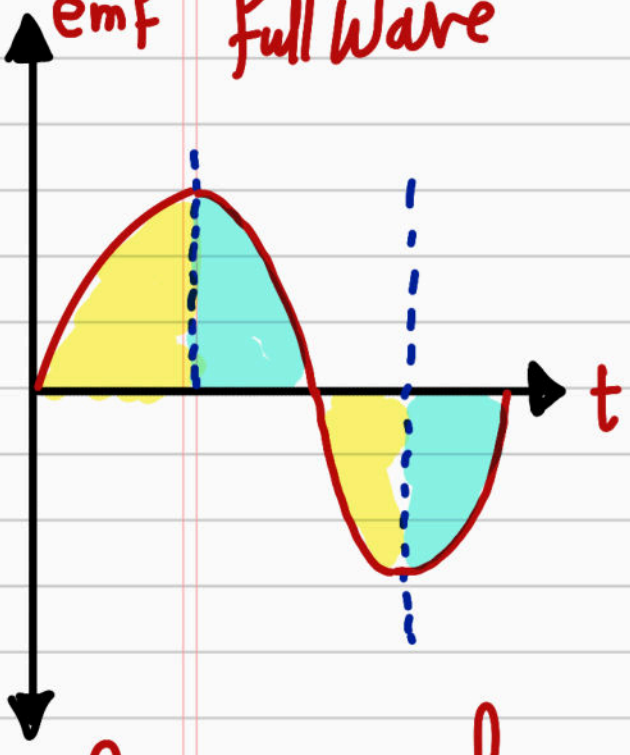
$$V_{eff} \times I_{eff} = \frac{1}{2} \times \frac{V_{max}\ I_{max}}{2}$$

$$V_{eff} \times I_{eff} = \frac{V_{max}}{2} \times \frac{I_{max}}{2}$$

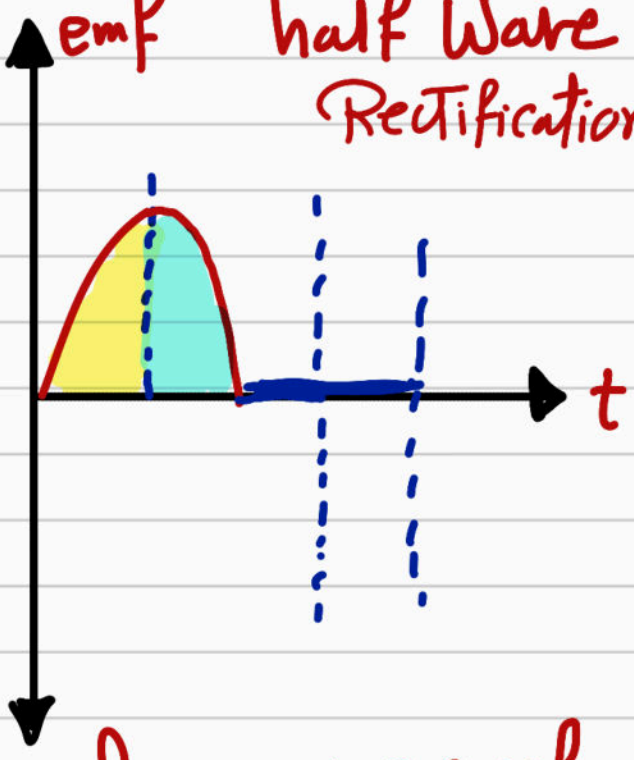
$$V_{eff} = \frac{V_{max}}{2} \qquad I_{eff} = \frac{I_{max}}{2}$$

$$V_{eff} = V_{max} \sin(30)$$

emf full Wave



emf half Wave Rectification



$$emf_{aw\ 1/4} = \frac{4BANf}{1}$$

$$emf_{aw\ 1/2} = \frac{4BANf + 4BANf}{2} = 4BANf$$

$$emf_{aw\ 3/4} = \frac{4BANf + 4BANf - 4BANf}{3} = \frac{4}{3}BANf$$

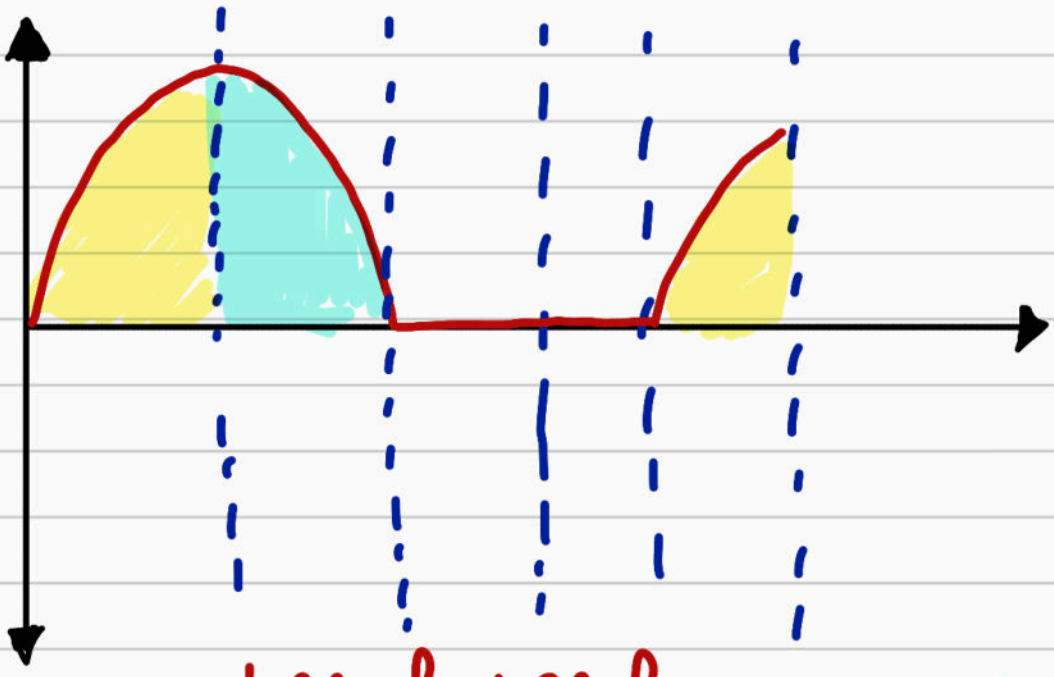
$$emf_{aw\ 1} = \frac{4BANf + 4BANf - 4BANf - 4BANf}{4} = Zero$$

$$emf_{aw\ 1/4} = \frac{4BANf}{1}$$

$$emf_{aw\ 1/2} = \frac{4BANf + 4BANf}{2} = 4BANf$$

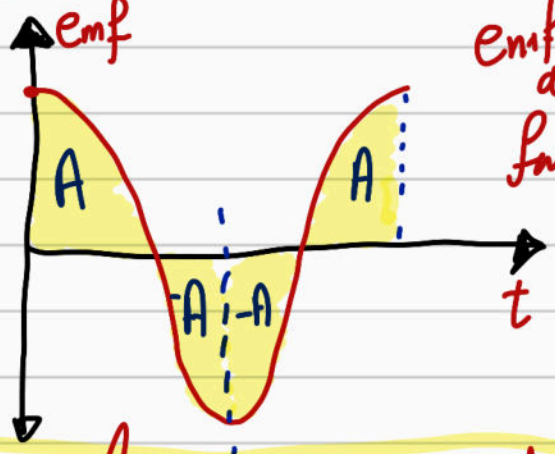
$$emf_{aw\ 3/4} = \frac{4BANf + 4BANf}{3} = \frac{8}{3}BANf$$

$$emf_{aw\ 1} = \frac{4BANf + 4BANf}{4} = 2BANf$$



$$emf_{av} = \frac{4BANf + 4BANf + 0 + 0 + 4BANf}{1 + 1 + 1 + 1 + 1}$$

Alternating Current AC

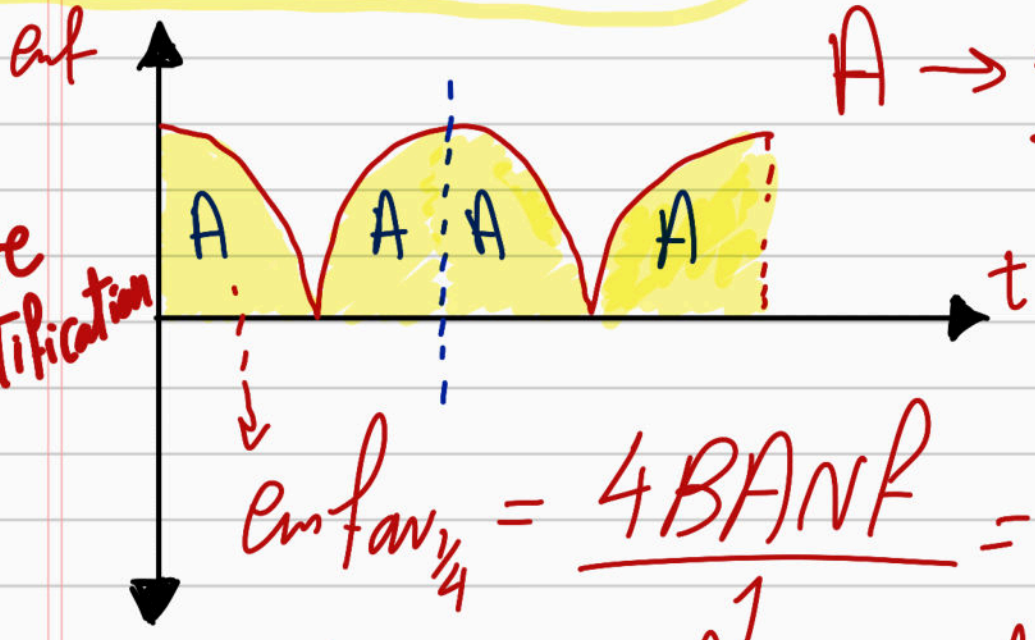


$$\text{emf}_{\text{average full cycle}} = \frac{\text{Sum Area}}{\text{Total}}$$

$$= \frac{A+(-A)+(-A)+A}{4}$$

$$= \frac{0}{4} = \text{Zero}$$

unidirectional Directional



Full Wave Rectification

$$A \rightarrow 4BANf$$

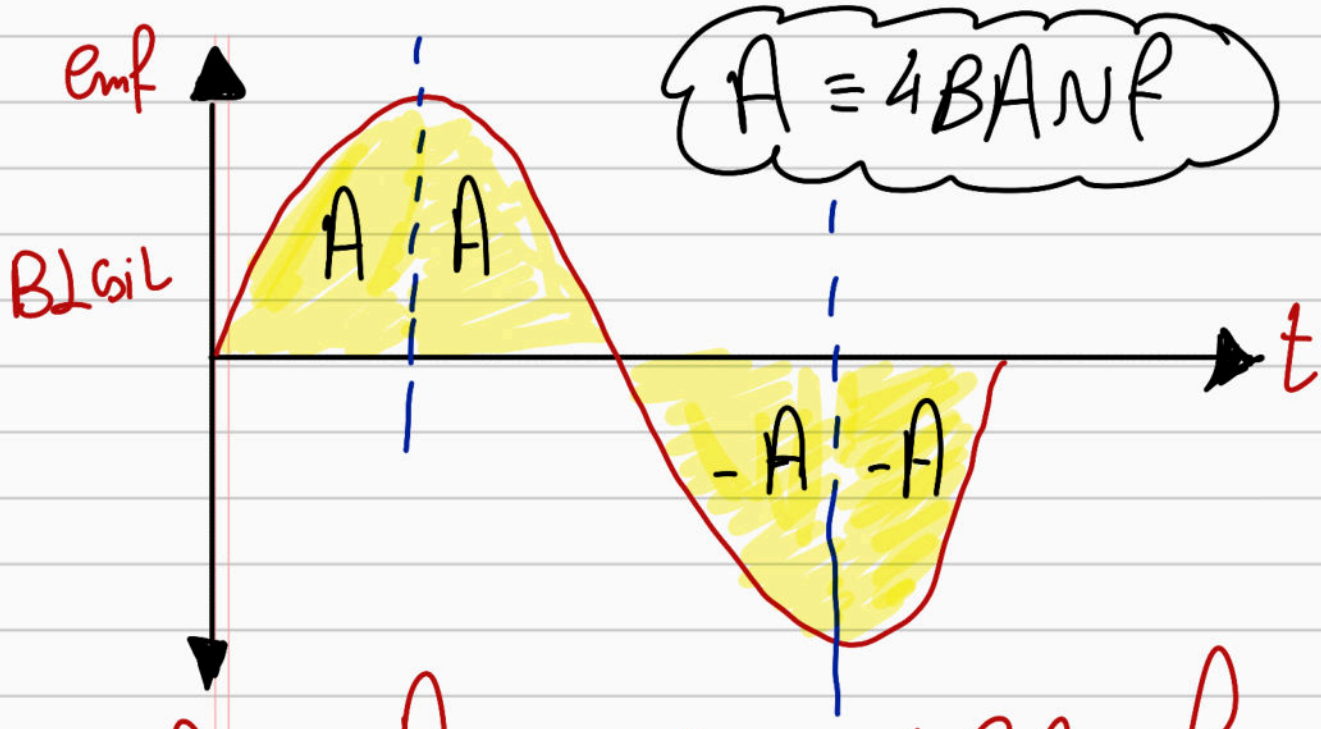
$$\text{emf}_{av\,1/4} = \frac{4BANf}{1} = 4BANf$$

$$\text{emf}_{av\,1/2} = \frac{4BANf + 4BANf}{2} = 4BANf$$

$$\text{emf}_{av\,3/4} = \frac{4BANf + 4BANf + 4BANf}{3} = 4BANf$$

$$\text{emf}_{av\,1} = \frac{A+A+A+A}{4} = A = 4BANf$$

$$\text{emf}_{av\,1/4} = \text{emf}_{av\,1/2} = \text{emf}_{av\,3/4} = \text{emf}_{av\,1} = 4BANf$$

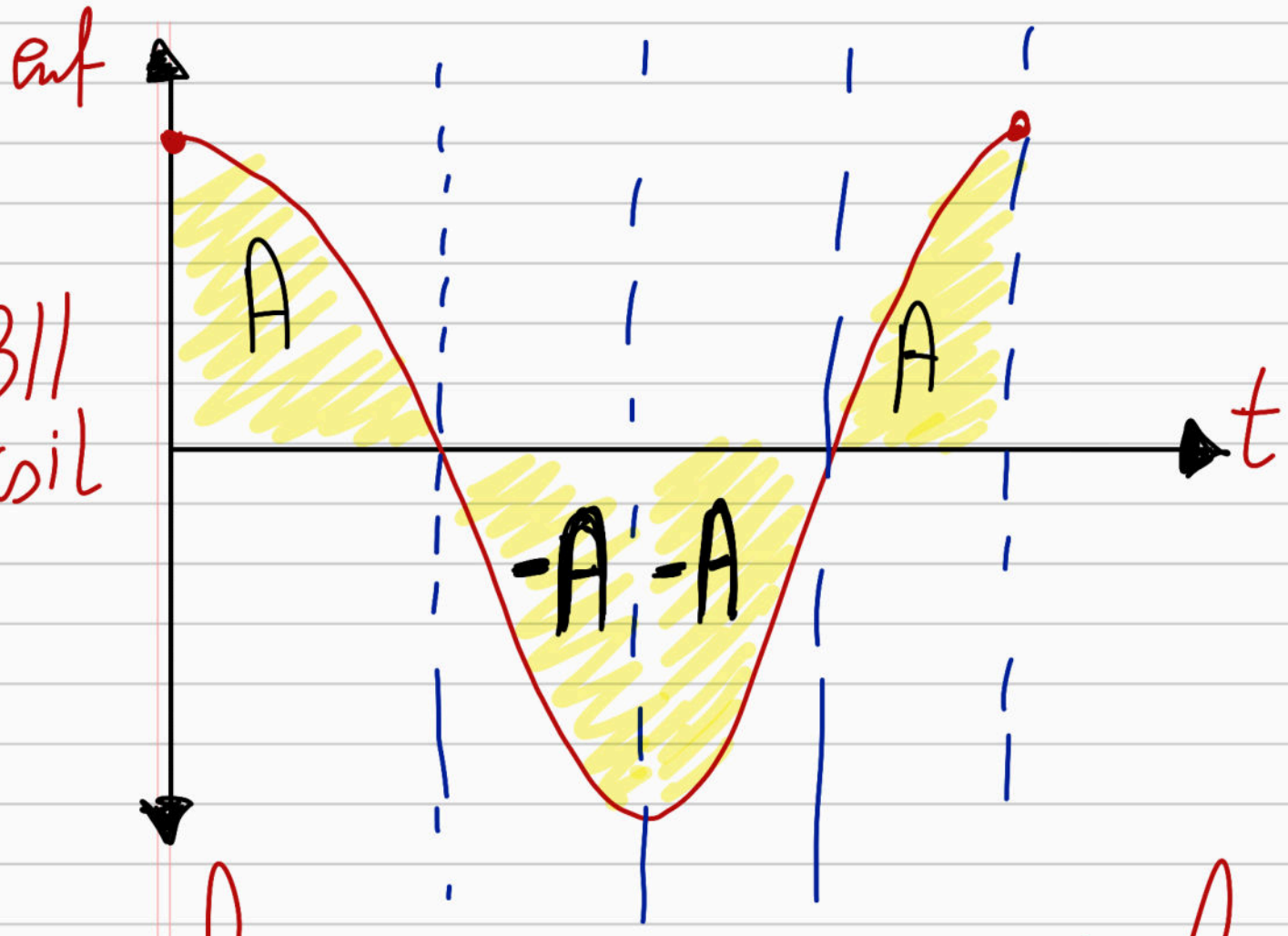


$$emf_{av\,1/4} = \frac{A}{1} = A = 4BANf$$

$$emf_{av\,1/2} = \frac{A+A}{2} = A = 4BANf$$

$$emf_{av\,3/4} = \frac{A+A+(-A)}{3} = \frac{4BANf}{3}$$

$$emf_{av\,1} = \frac{A+A+(-A)+(-A)}{4} = Zero$$



$$emf_{av\,1/4} = \frac{A}{1} \simeq A = 4BANf$$

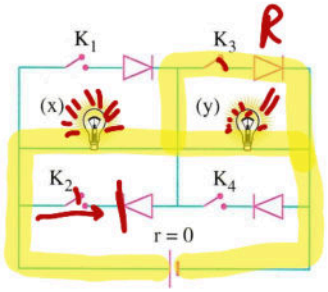
$$emf_{av\,1/2} = \frac{A + (-A)}{2} = Zero$$

$$emf_{av\,3/4} = \frac{\cancel{A} + (-\cancel{A}) + (-A)}{3} = \cancel{A}\,\frac{4}{3}BANf$$

direction

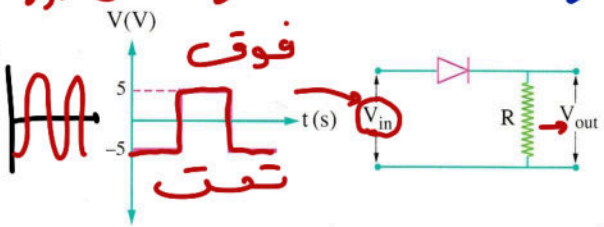
$$emf_{av\,1} = \frac{A + (-A) + (-A) + A}{4} = Zero$$

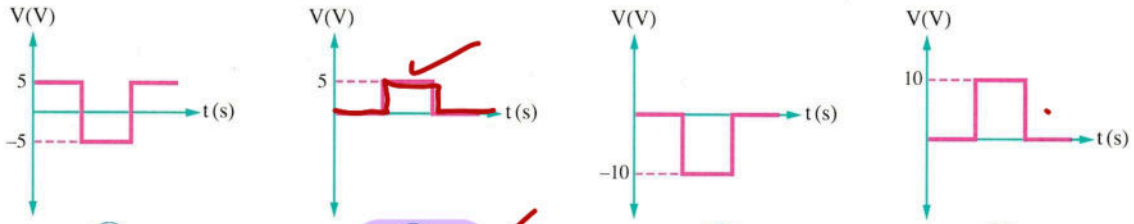
53 Two identical light bulbs x and y, are connected with several diodes through four switches as shown in the opposite electrical circuit. Assuming that the resistance of each diode is R when forward-biased and infinite when reverse-biased, then which two switches, when closed, will increase the brightness of bulb (x) and decrease the brightness of bulb (y)?



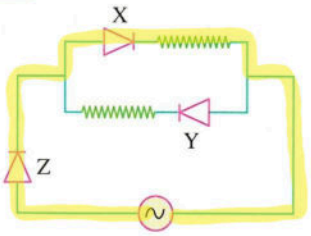
(a) Switches (K_1) and (K_3).
(b) Switches (K_2) and (K_4).
(c) Switches (K_1) and (K_4).
(d) Switches (K_2) and (K_3)

54 In the opposite figure, an electric signal passes through an ideal diode, so which of the following graphs represents the output signal (V_{out}) across resistor R?



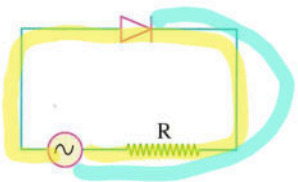


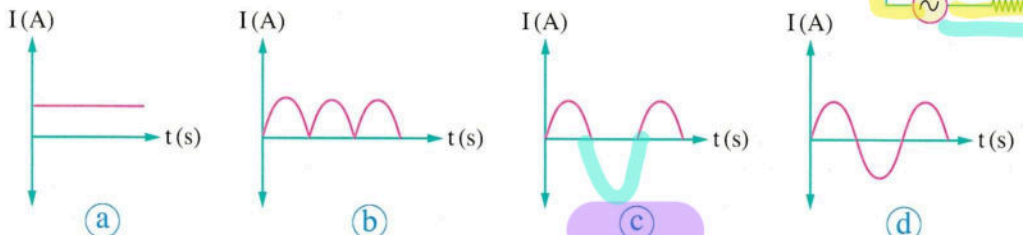
55 Some types of diodes are designed to emit light when connected in forward bias only. These diodes are called LEDs (light-emitting diodes). If three of these LEDs are connected to a low frequency electric AC source as shown in the opposite circuit, which of the following choices is correct?



(a) LED X lights up only when LED Z lights up.
(b) LED Z lights up only when LED X goes off.
(c) LED Y lights up only when LED X goes off.
(d) The three LEDs will always light up.

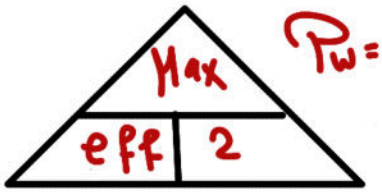
56 From the opposite circuit, the graph that represents the relation between the electric current intensity (I) through resistor R and time (t) is ..... .





Half Wave Rectification

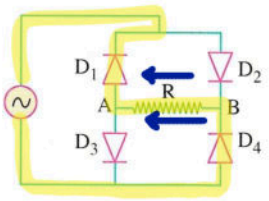
$P_w = \frac{1}{2} P_{w\,AC}$

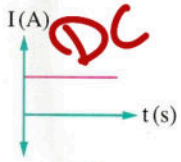


eff → θ = 30°

eff → θ = 60°

57 The opposite figure shows four diodes connected in an electric circuit, hence the graph that represents the variation of the current intensity (I) through resistor R versus time (t) is ..... .

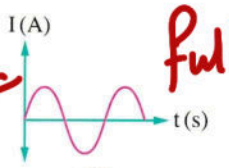




a

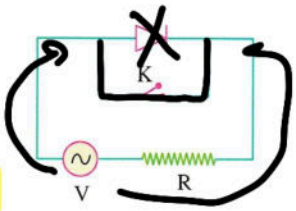
b

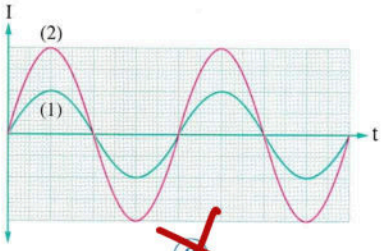
c



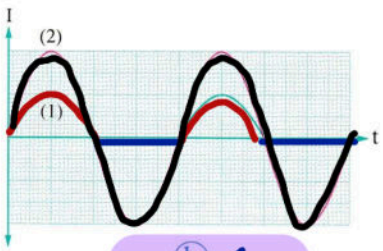
d

58 In the opposite circuit diagram, given that the resistance of the diode in forward bias equals R and in reverse bias is infinite, so the graph in which curve (1) and curve (2) represent the change of current intensity (I) versus time (t) through resistor R before and after closing switch K respectively is ..... .

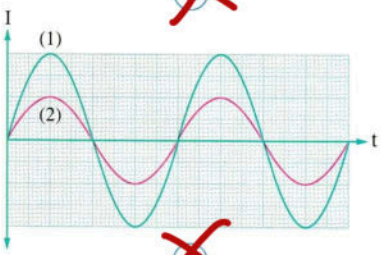




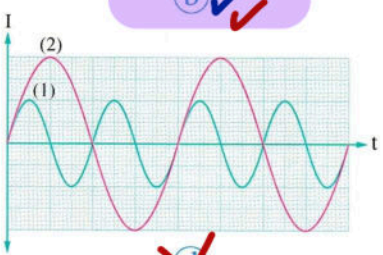
a



b

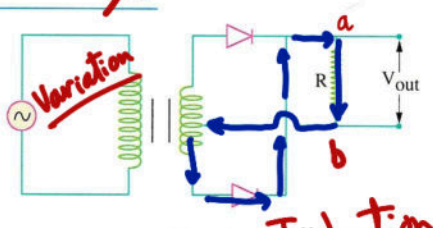


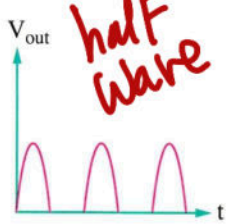
c



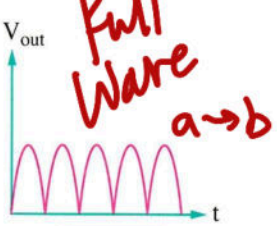
d

59 The graph that expresses the relation between the output voltage (V_{out}) and the time (t) for the opposite electric circuit is ..... .

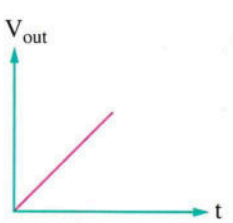




a



b



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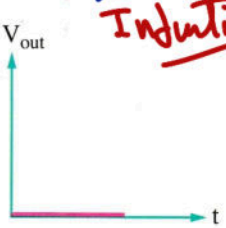
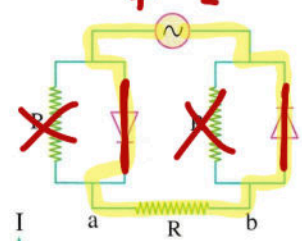
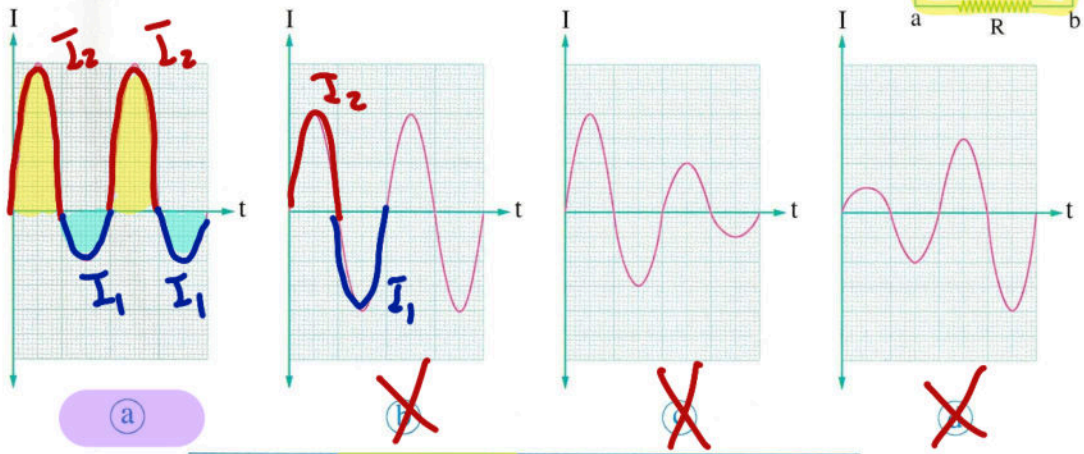


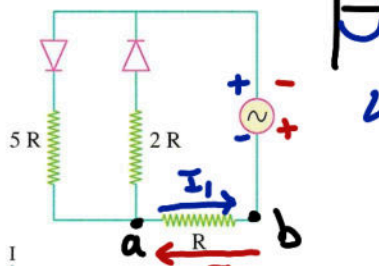
d

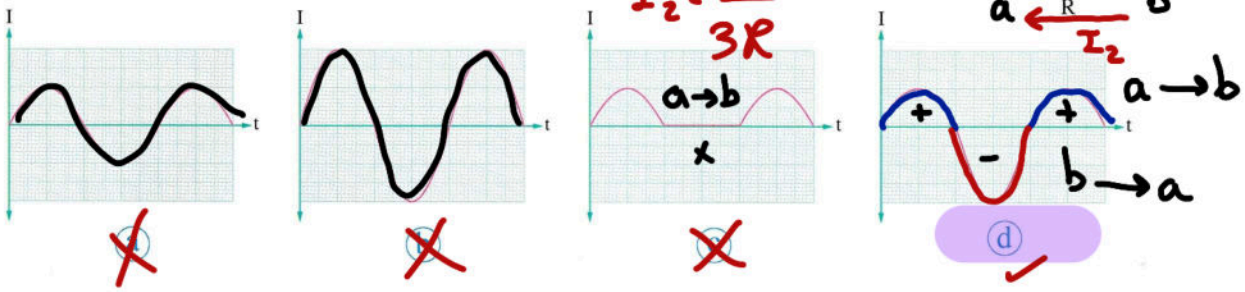
60 In the opposite circuit, if the resistances of the diodes are negligible in forward bias and infinite in reverse bias, the graph which represents the electric current through the resistance between the two points a and b is ..... .





61 In the opposite figure; if the diodes have negligible resistances in forward bias and infinite resistance in reverse bias, which of the following graphs represents the change of the current intensity (I) in resistor R with time (t)?





Second Essay questions

Semiconductor crystal

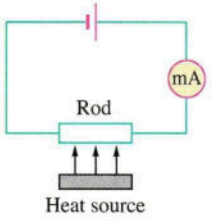
1 **Why** is pure silicon a poor conductor of electric current at low temperatures?

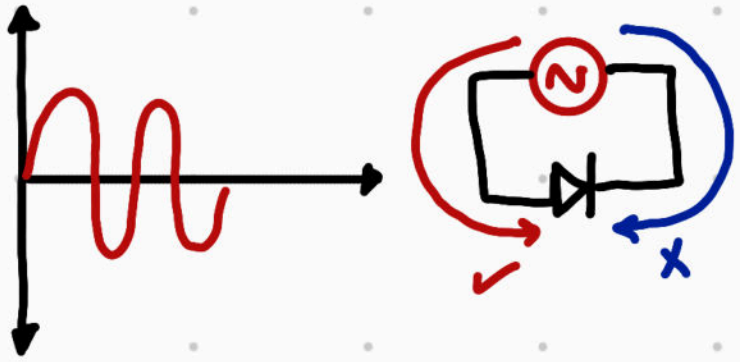
2 **Give reasons for:** It is not preferable to heat the pure semiconductor for a high temperature to increase its conductivity to electric current.

3 **Using the opposite figure, explain what happens to the reading of the milliammeter in the two following cases:**

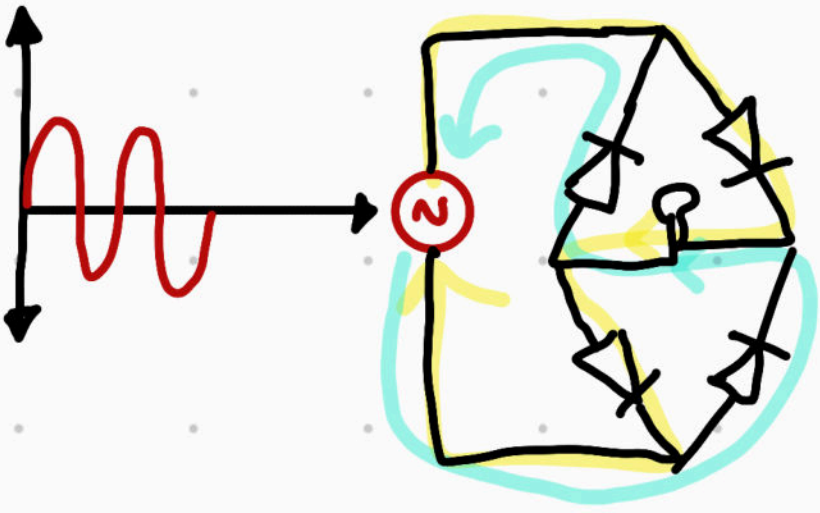
(a) If the rod is made of copper.

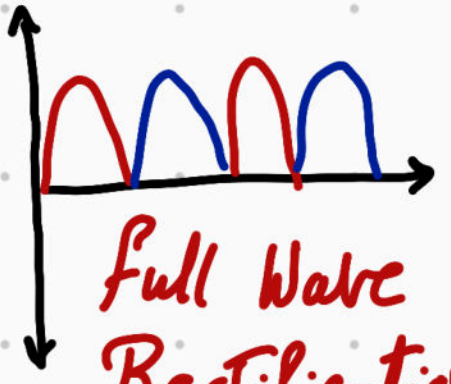
(b) If the rod is made of silicon.



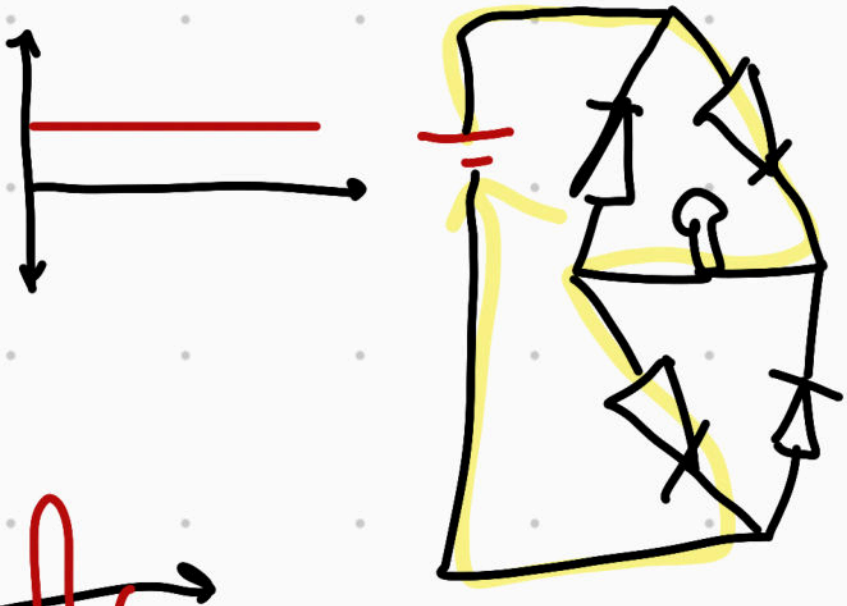


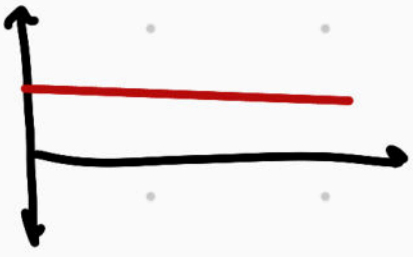
half Wave Rectification

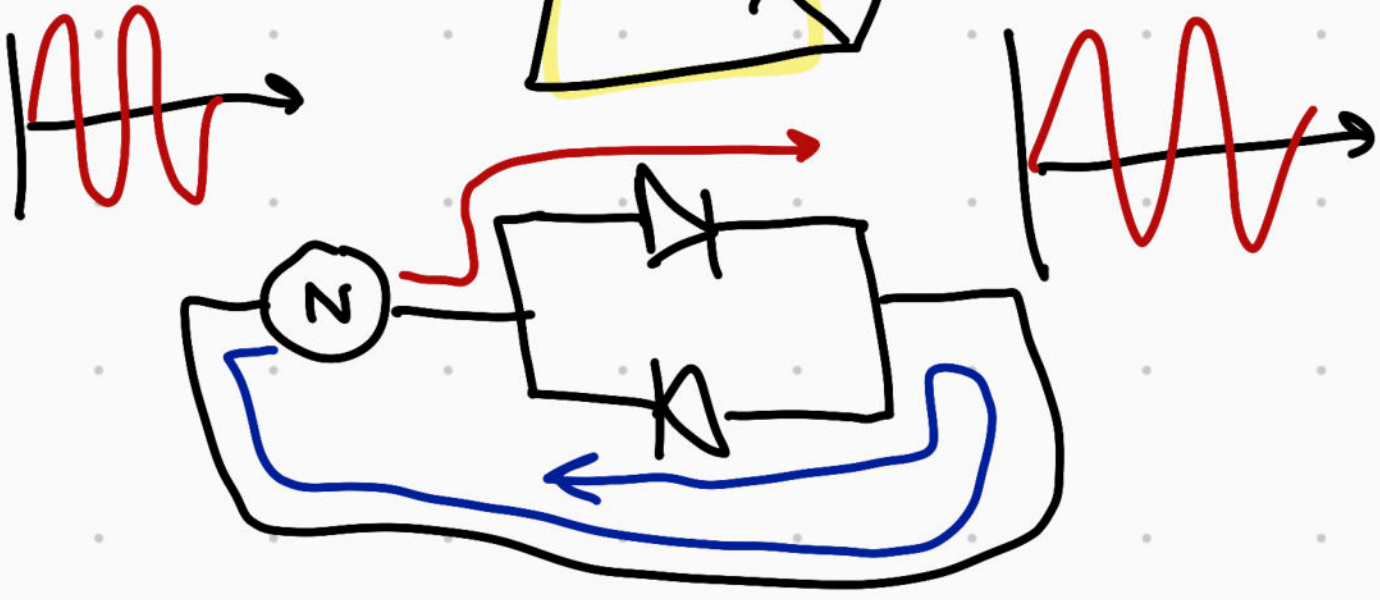




full Wave Rectification







4 **Explain why**, when a pure silicon crystal is doped with phosphorus atoms, the electrical conductivity of the crystal increases even though its total electric charge does not change.

5 **In a semiconductor crystal of p-type:**

(a) **How many** valence electrons should be in an atom of the impurity substance?

(b) **Explain**, is doping the pure crystal with atoms of impurity substance makes it positively charged.

(c) **What is the type of** the dominant charge carriers in the p-type semiconductor?

6 If the concentration of free electrons or that of holes in a pure silicon crystal is 10^{10} cm^{-3}, then phosphorus atoms are added to the crystal with a concentration of 10^{12} cm^{-3}, **so calculate:**

(a) The concentrations of free electrons and that of holes in this case.

(b) The concentration of aluminum atoms that are required to be added to the silicon crystal to return to be as if it is a pure crystal.

7 Three crystals of silicon (x, y, z) are of the same size at the same temperature, crystal (x) is pure, crystal (y) is doped with impurities of phosphorus atoms with a concentration of 2×10^{10} cm^{-3} and crystal (z) is doped with impurities of boron atoms with a concentration of 10^{10} cm^{-3}, **explain why** the conductivity of crystal (z) is greater than that of crystal (x) and less than that of the crystal (y).

8 A silicon crystal is doped with phosphorus atoms at a concentration of 2×10^{10} cm^{-3}, **explain what happens to its conductivity when adding:**

(a) Phosphorus atoms with a concentration of 10^{10} cm^{-3}.

(b) Aluminum atoms with a concentration of 10^{10} cm^{-3}.

The pn junction

9 **You have a pn junction and a battery:**

(a) **Explain how they can be connected so that the electric field of the battery works:**

(1) In the same direction as the electric field inside the junction.

(2) In a direction opposite to the electric field inside the junction.

(b) **Compare the results of the method of connection in each case on:**

(1) Potential barrier.

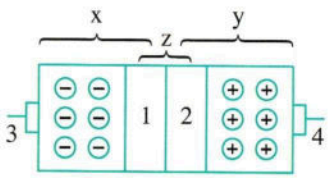
(2) The thickness of the depletion region.

(3) The flow of current through the junction.

10 The opposite diagram represents a pn junction:

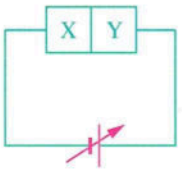
(a) **What** is the type of the semiconductor crystal denoted by (x) and that denoted by (y)?

(b) **What** is the name of region (z)?



(c) In the case of forward connection, **which** pole of the battery should be connected to terminal (3)?

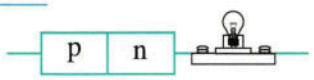
11 The opposite circuit contains a diode in which region X is doped with antimony and region Y is doped with boron, and then the junction is connected to a battery as shown in the figure:



(a) **Is** the junction forward biased or reverse biased?

(b) **Plot the graph** that represents the relation between the current intensity (I) and the potential difference (V) across the junction.

12 The opposite figure represents a diode connected in series to a small lamp:



(a) **Complete the diagram** of the electric circuit by connecting a DC source to light up the lamp **with explanation**.

(b) **What will happen if** the connection of the circuit terminals is reversed with respect to the DC source?

(c) **Determine with explanation** the type of the current that passes through the lamp, if the DC source is replaced by an alternating current source.

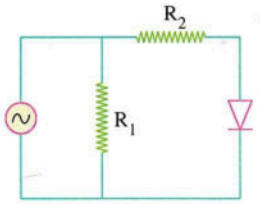
13 **Compare between each of the following:**

(a) Diffusion current and drift current in the diode (in terms of: the direction of free electrons - the cause of occurrence of each of them).

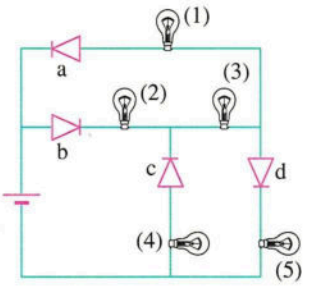
(b) The pn junction and the ohmic electric resistance (in terms of: charge carriers - the direction of electric current - the effect of heating).

14 **Show how** the polarity of pn junction can be determined.

15 In the opposite electric circuit, **why** are the types of electric currents passing in resistors R_1 and R_2 different?



16 In the opposite circuit, a number of identical pn junctions and identical lamps are connected to an electric cell:



(a) **Which** lamp lights up? **And why**?

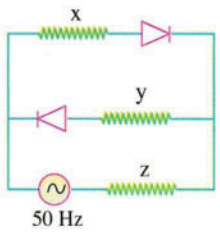
(b) **Which** of the four pn junctions if its connection method is reversed, no lamp lights up? **Explain your answer.**

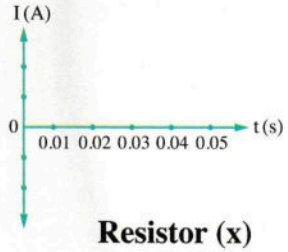
(c) **What lamps are lit with equal brightness when reversing the method of connecting:**

(i) The electric cell? **(ii)** Both pn junctions (a) and (c)?

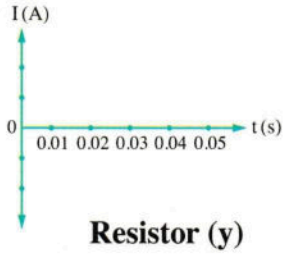
Explain your answer.

17 Three identical resistors (x, y, z) and two identical diodes are connected to an alternating electrical source of frequency 50 Hz as shown in the opposite figure, **use the following graphs to graphically represent the current passing through each of the three resistors during 0.05 s.**

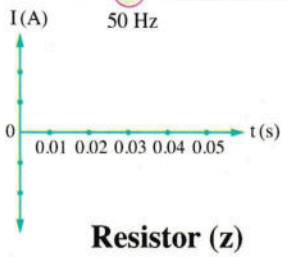




Resistor (x)

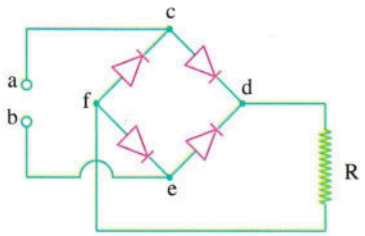


Resistor (y)



Resistor (z)

18 The opposite figure represents an electronic circuit containing four pn junctions, ohmic resistor R, and an electric source (ab):



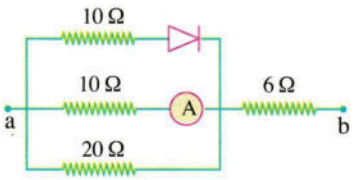
(a) Trace with letters the path of electric current through the circuit when the electric potential of:

(1) terminal a is positive.

(2) terminal b is positive.

(b) According to your answer in part (1), could the current passing in resistor R be used in recharging a car battery? **And why?**

19 The electrical circuit in the opposite figure shows several resistors, an ammeter, and a diode connected together. If a battery of electromotive force 5 volts that has a negligible internal resistance is connected between the two points a, b. Considering that the resistance of the pn junction is negligible in the case of forward connection and infinite in the case of reverse connection, **Calculate the reading of the ammeter when:**



(a) Potential V_b < Potential V_a

(b) Potential V_b > Potential V_a

Questions of the Previous Years Exams on Lesson 1

The semiconductor crystal

1 When the temperature of a pure germanium crystal (Ge) is lowered to zero degree Celsius (0°C), its electrical conductivity ..... . *(Exp. June 21)*

(a) increases (b) decreases (c) vanishes (d) does not change

2 Given that, the concentration of the free electrons in a pure germanium crystal in the case of dynamic equilibrium equals 2×10^8 cm^{-3}, hence the expected concentration of holes is ..... . *(1st session 21)*

(a) equal to 2×10^8 cm^{-3}
(b) greater than 2×10^8 cm^{-3}
(c) less than 2×10^8 cm^{-3}
(d) equal to zero

3 Assuming that the temperature of both pure silicon crystal (Si) and a copper wire is dropped to the absolute zero (0 K), so the electrical conductivity ..... . *(2nd session 21)*

(a) vanishes for Si but increases for copper
(b) vanishes for both Si and copper
(c) increases for both Si and copper
(d) increases for Si but vanishes for copper

4 The table illustrates four samples of the same pure semiconductor material at different temperatures.

Sample	Temperature	Charge carriers concentration
W	T_W	1.6×10^{16} m^{-3}
X	T_X	1.5×10^{11} cm^{-3}
Y	T_Y	1.6×10^{15} m^{-3}
Z	T_Z	1.5×10^{10} cm^{-3}

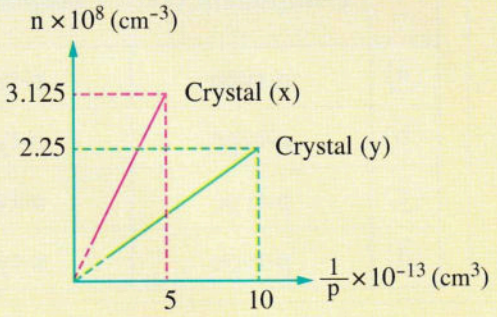
Which choice is the correct arrangement for the crystal samples temperatures?

(a) $T_W > T_Y > T_X > T_Z$
(b) $T_X > T_W > T_Z > T_Y$
(c) $T_Z > T_X > T_Y > T_W$
(d) $T_Y > T_Z > T_W > T_X$ *(1st session 22)*

5 The graph illustrates the relation between the free electrons concentration (n) and the reciprocal of the holes concentration $\left(\frac{1}{p}\right)$ for two extrinsic semiconductor crystals (x) and (y), so the ratio between:

$$\frac{\text{Free electrons concentration in the pure crystal x } [n_{ix}]}{\text{Holes concentration in the pure crystal y } [n_{iy}]}$$

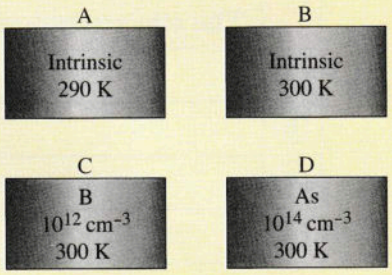
= ..... .



(2nd session 22)

(a) $\frac{25}{9}$ (b) $\frac{25}{36}$ (c) $\frac{5}{6}$ (d) $\frac{5}{3}$

6 The figure represents four identical silicon sheets having equal dimensions (the temperature of each one, the type of impurity and its concentration if exists is recorded), so the arrangement of them according to their electric conductivity from higher to lower is ..... .

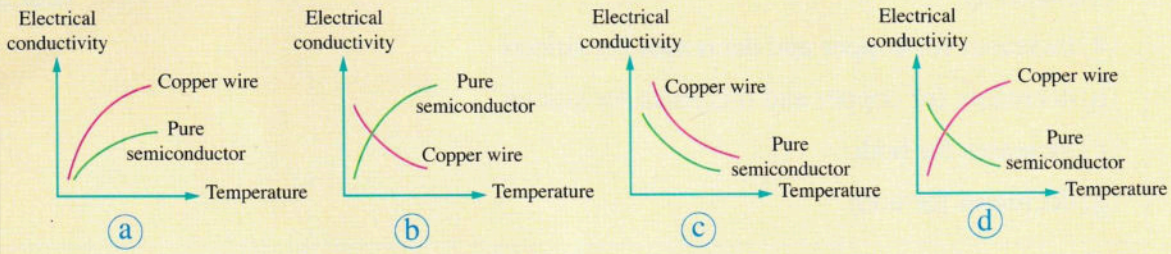


(a) A > B > C > D (b) D > C > B > A

(c) B = C = D > A (d) C = D > B > A *(Exp. 23)*

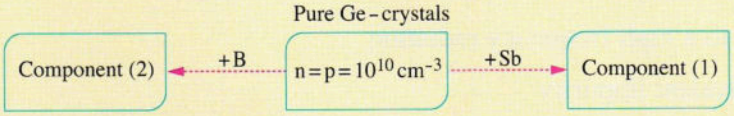
7 Which graph from the choices illustrates the relation between the electrical conductivity for a pure semiconductor crystal and a copper wire as the absolute temperature changes?

(1st session 23)



8 The following diagram shows the doping processes for pure Ge-crystal to increase the conductivity.

(1st session 24)



If the concentration of the impurities added in each case is 10^{12} cm^{-3}, so ..... .

	Component (1)	Component (2)	Ratio $\frac{p_1}{p_2}$	Ratio $\frac{n_1}{n_2}$
(a)	n-type	p-type	10^4	10^{-4}
(b)	n-type	p-type	10^{-4}	10^4
(c)	p-type	n-type	10^4	10^{-4}
(d)	p-type	n-type	10^{-4}	10^4

9 If the concentration of holes in pure semi-conductor crystal was 10^{11} cm^{-3}, then it doped by adding impurity atoms till the concentration of the holes became 10^9 cm^{-3}, which choice is correct?

(2^{nd} session 24)

	The concentration of electrons in the doped crystal	The impurity atoms
(a)	10^2 cm^{-3}	Phosphorus
(b)	10^2 cm^{-3}	Aluminum
(c)	10^{13} cm^{-3}	Boron
(d)	10^{13} cm^{-3}	Antimony

10 The element that does not produce a p-type semiconductor when doped into a silicon crystal is ..... .

(Exp. 25)

(a) boron (B^{+3})
(b) antimony (Sb^{+5})
(c) nickel (Ni^{+2})
(d) aluminum (Al^{+3})

11 When the temperature of a copper wire and a silicon crystal is raised, the electrical conductivity ..... .

(Exp. 25)

(a) increases for copper and decreases for silicon
(b) decreases for copper and increases for silicon
(c) increases for both
(d) decreases for both

12 The resistivity of intrinsic germanium crystal can be increased by ..... .

(Exp. 25)

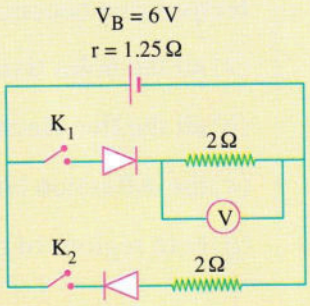
(a) adding a pentavalent impurity
(b) decreasing its temperature
(c) exposing it to a high frequency radiation
(d) adding a trivalent impurity

13 Two identical intrinsic semiconductor crystals A, B their temperatures are T_1 and T_2 respectively, they are placed in a medium of temperature T_o where $T_1 > T_o > T_2$.
At thermal equilibrium with the medium for each of the crystals, the conductivity of crystal A ..... and the resistivity of crystal B ..... . *(Exp. 25)*

(a) decreases, decreases
(b) increases, increases
(c) increases, decreases
(d) decreases, increases

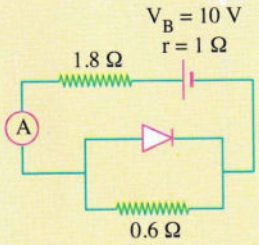
The pn junction

14 In the shown electric circuit. On closing the two keys (K_1), (K_2), the voltmeter reads ..... .
(where the resistance of each diode in forward bias connection = 0.75 Ω and equals (∞) at reverse bias)



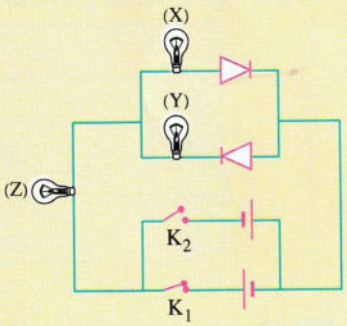
(a) 3 V
(b) 0 V
(c) 6 V
(d) 4 V *(1^{st} session 22)*

15 In the shown electric circuit, by considering the resistance of the diode in forward bias connection is 0.3 Ω and its resistance in reverse bias connection is infinite (∞), so, the reading of the ammeter equals ..... . *(2^{nd} session 22)*



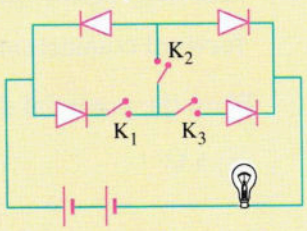
(a) 2.94 A
(b) 3.33 A
(c) 2.71 A
(d) 3.57 A

16 The figure illustrates an electric circuit that contains three bulbs X, Y and Z connected as in the figure, when K_1 is opened and K_2 is closed. Which of the following choices represents the correct change in the lighting of bulbs? *(Exp. 23)*



(a) Bulb (Y) lights and bulb (X) remains lighted.
(b) Bulb (X) is switched off and bulb (Z) is switched off too.
(c) Bulb (Y) doesn't light and bulb (Z) is switched off.
(d) Bulb (X) is switched off and bulb (Z) remains lighted.

17 According to the given figure:
The resistance of the diode in case of forward connection is 2 Ω and its resistance in backward (reverse) connection is infinite.
Which of the following choices is the correct answer to obtain the largest consumed power in the electric lamp?



	K_1	K_2	K_3
(a)	Closed	Closed	Closed
(b)	Closed	Opened	Opened
(c)	Closed	Closed	Opened
(d)	Closed	Opened	Closed

(1st session 23)

18 The value of the intensity of the electric field established inside the diode at a certain temperature becomes constant when ..... . *(2nd session 23)*

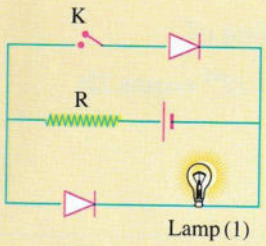
(a) all the holes transfer from the p-region to the n-region

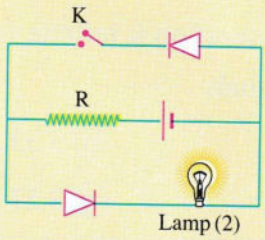
(b) all the free electrons transfer from the n-region to the p-region

(c) the diffussion current equals the drift current

(d) both regions of the diode become electrically neutral

19 If the resistance of the diode in forward bias is neglected and in reverse bias is infinite. *(1st session 24)*

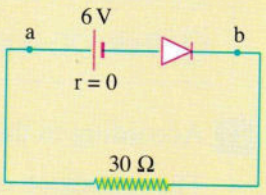




So when the key is closed in the two circuits, ...

	Lamp (1)	Lamp (2)
(a)	goes off	does not change
(b)	glows brighter	goes off
(c)	dims	glows brighter
(d)	does not change	dims

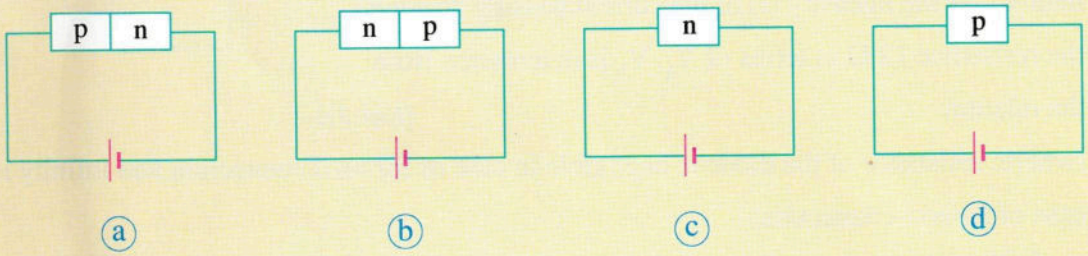
20 The opposite figure shows a battery (of negligible internal resistance) connected in series with a diode and an ohmic resistance, (knowing that the resistance of the diode is negligible in forward connection and infinite in case of backward connection).



Then the potential difference between the two points a and b is ..... . *(2nd session 24)*

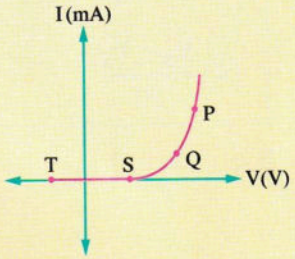
(a) 3 V (b) 0 V (c) 2 V (d) 6 V

21 Which of the following electrical circuits has the greatest electrical resistance to the flow of electric current? *(Exp. 25)*



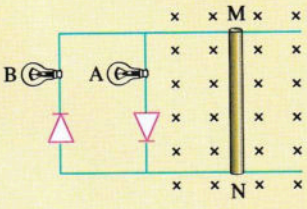
(a) (b) (c) (d)

22 The graph illustrates the (I,V) characteristics curve of a diode, so at which of the points shown on the graph is the diode's resistance at its highest value? *(Exp. 25)*



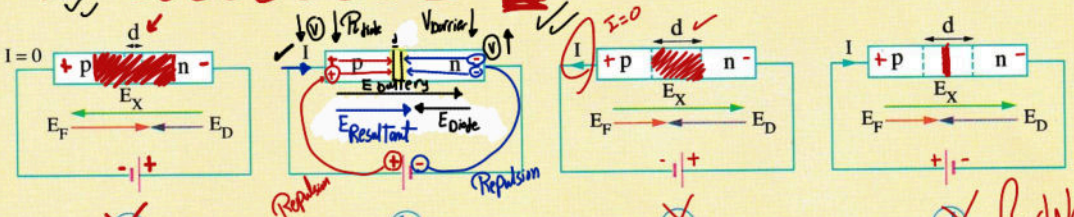
(a) S
(b) P
(c) Q
(d) T

23 In the opposite figure, the electric bulb (A) lights when the wire MN moves ..... . *(Exp. 25)*



(a) to the right of the page
(b) to the left of the page
(c) perpendicular into the page
(d) perpendicular out of the page

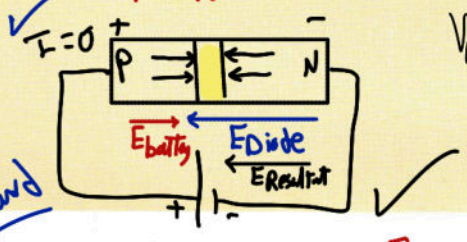
24 A diode is connected to a DC source, which of the following choices shows correctly the direction of the conventional current (I), thickness of the depletion layer (d), the electric field strength and its directions due to the battery (E_X), the internal electric field (E_D) and the resultant of the produced electric field (E_F) which is acting on the p-n junction? *(Exp. 25)*



(a) (b) (c) (d)

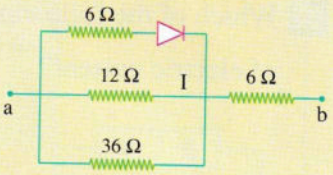
$E_{Resultant} = E_{battary} - E_{Diode}$

$V_{Battary} > V_{Barrier}$



Forward
I=0
$V_B < V_{Barrier}$
$E_{Battary} < E_{Diode}$

25 The opposite figure shows a part of an electric circuit, if the electric potential at a is V_a and at b is V_b , where V_b is greater than V_a, so the intensity (I) of the current passing through the resistance 12 Ω in terms of V_a, V_b is determined from the relation ..... .



(Exp. 25)

(where: resistance of the diode is negligible in case of forward connection and infinity in case of reverse connection)

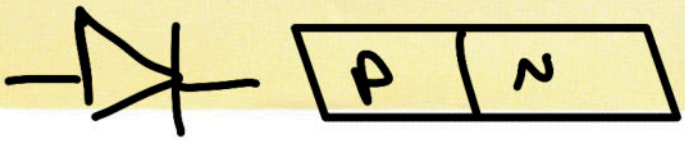
(a) $\frac{5\,(V_b - V_a)}{48}$

(b) $\frac{V_b - V_a}{32}$

(c) $\frac{V_b - V_a}{20}$

(d) $\frac{V_b - V_a}{15}$

Diode



#Chapter⑧ Lesson②

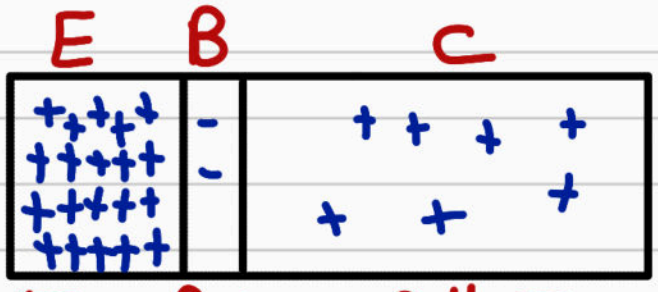
Transistor
Logic gates

on
off

"Triode"

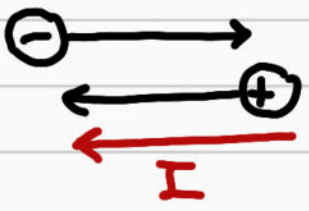
Transistor uses → switch, Inverter, Amplifier

P-Type
$P^+ = N_A^-$
Gallial

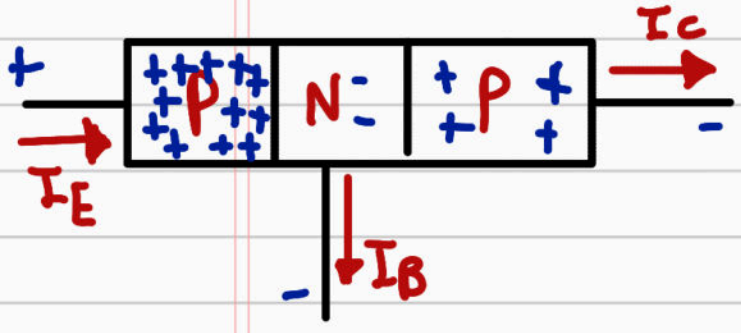


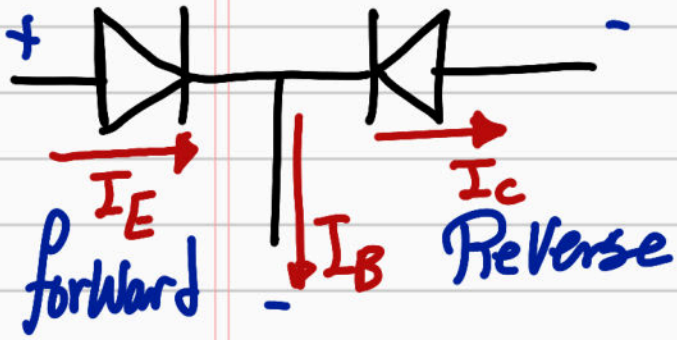
Transistor
PNP
NPN ✓✓✓

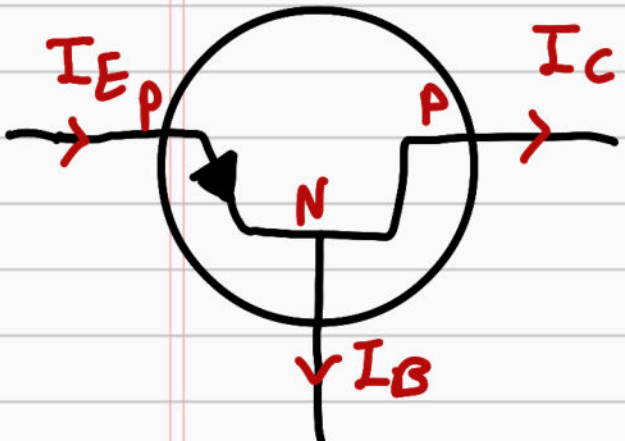
	Emitter	Base	Collector
% Impurities	high	Low	medium
Size	medium	small	Large



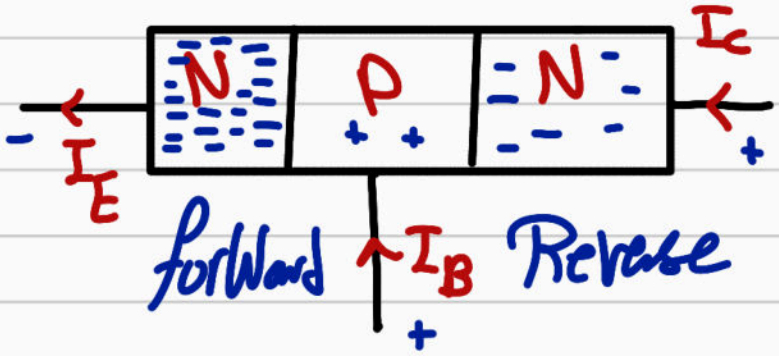
PNP

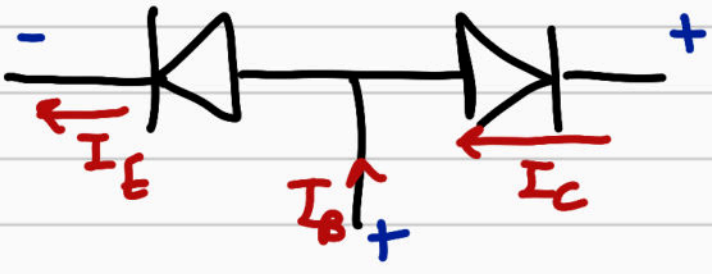


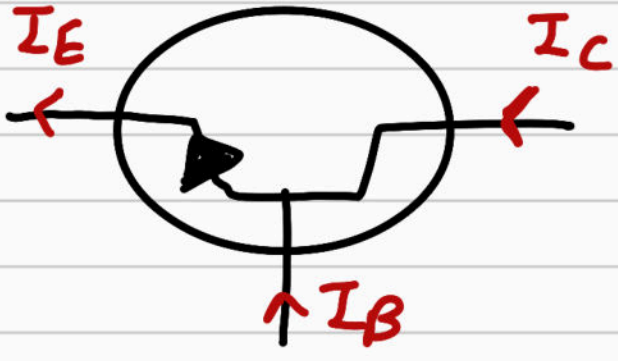


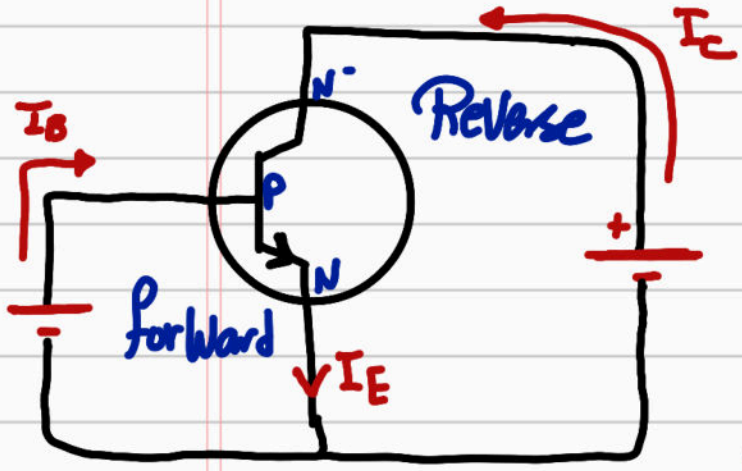


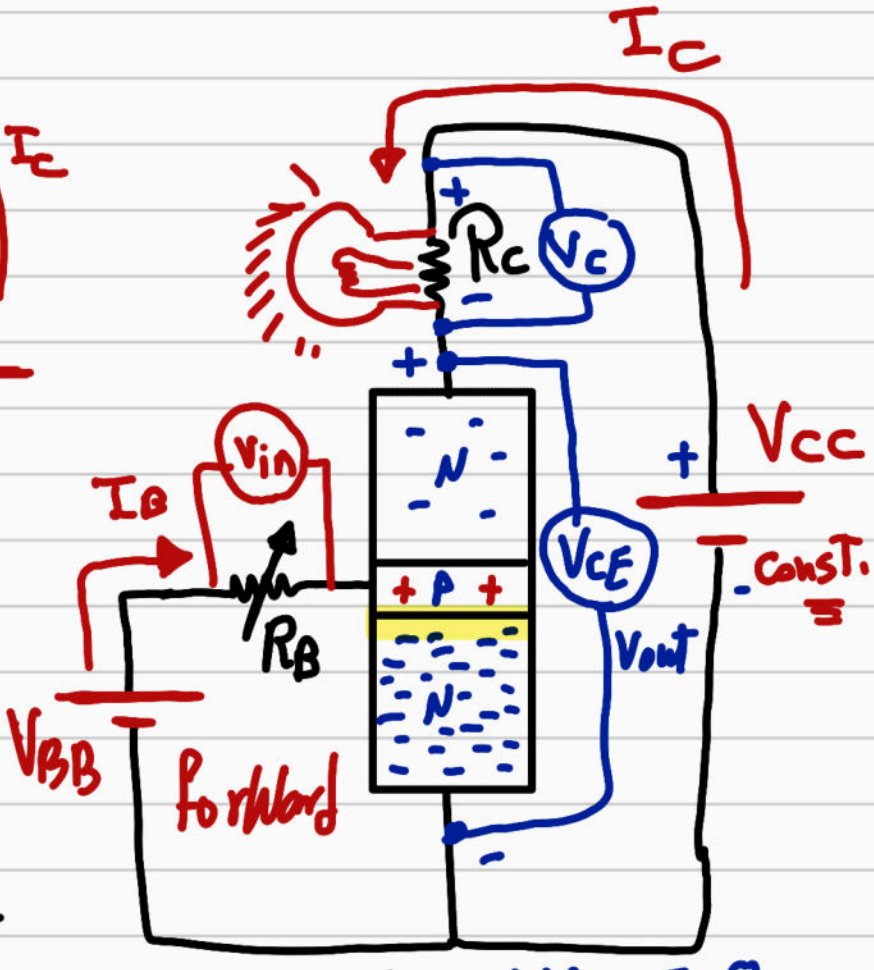
NPN











Grid

$R_B \uparrow$ $I_B \downarrow$ $R_{EB} \uparrow$ $I_E \downarrow$ $I_C \downarrow$ $V_C \downarrow$ $V_{CE} \uparrow$

$R_B \downarrow$ $I_B \uparrow$ $R_{EB} \downarrow$ $I_E \uparrow$ $I_C \uparrow$ $V_C \uparrow$ $V_{CE} \downarrow$

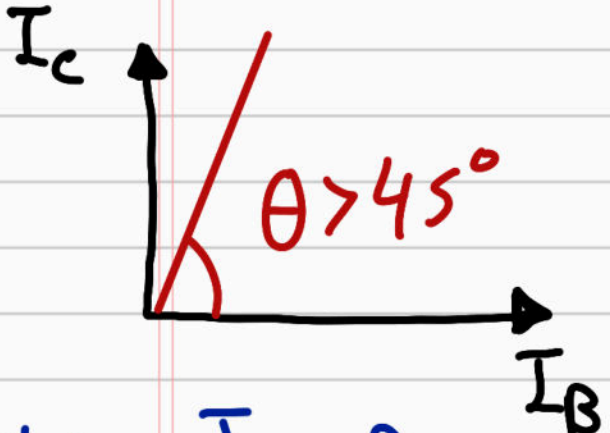
Grid

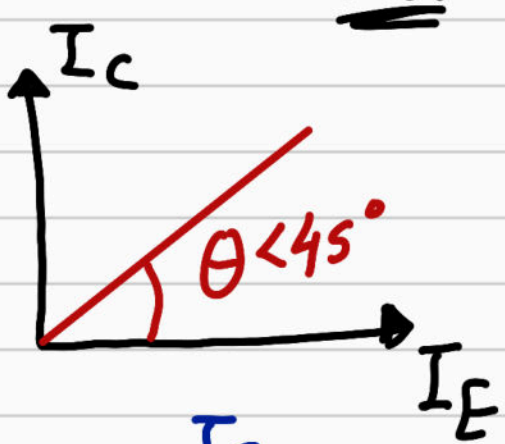
$-V_{CC} + V_{CE} + V_C = 0$

$V_{CE} = V_{CC} - V_C$

$= V_{CC} - I_C R_C$

V_{CC} → const.





$\text{Slope} = \frac{I_C}{I_B} = \beta_e > 1$

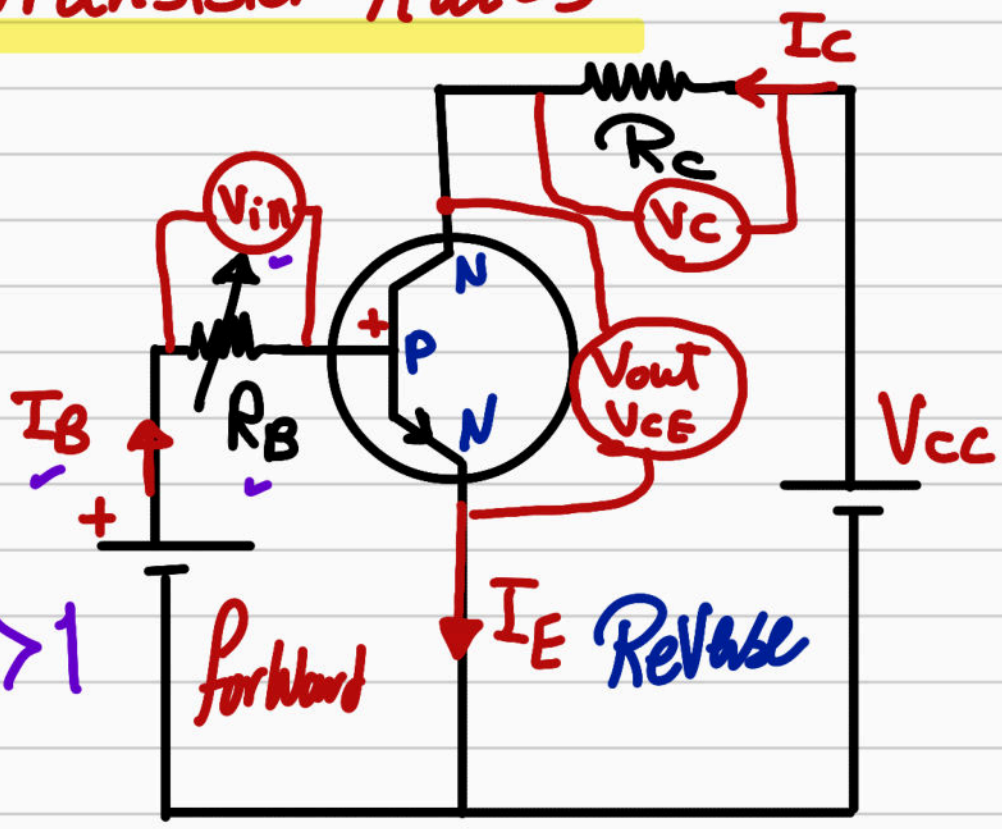
(Current gain Ratio)

$\text{Slope} = \frac{I_C}{I_E} = \alpha_e < 1$

(Current distribution Ratio)

$I_E > I_C > I_B$

"Transistor Rules"



1 $I_E = I_C + I_B$

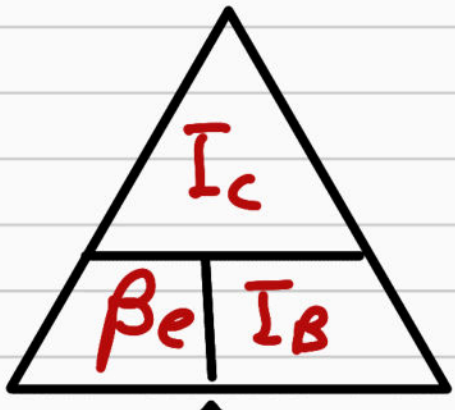
$I_E > I_C > I_B$

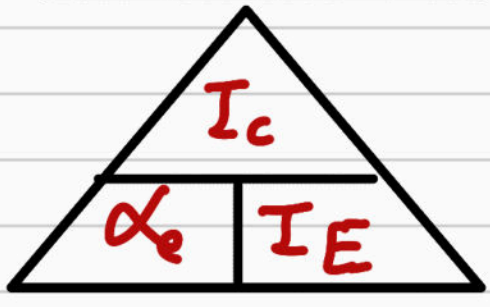
2 $\beta_e = \frac{I_C}{I_B} = \frac{\alpha_e}{1-\alpha_e} > 1$

↳ Gain Ratio

3 $\alpha_e = \frac{I_C}{I_E} = \frac{\beta_e}{1+\beta_e} < 1$

↳ distribution Ratio





4 $V_{in} = I_B R_B$

5 $V_{CC} = V_{CE} + I_C R_C$

$V_{CE} = V_{CC} - I_C R_C$

V_{CC} const.

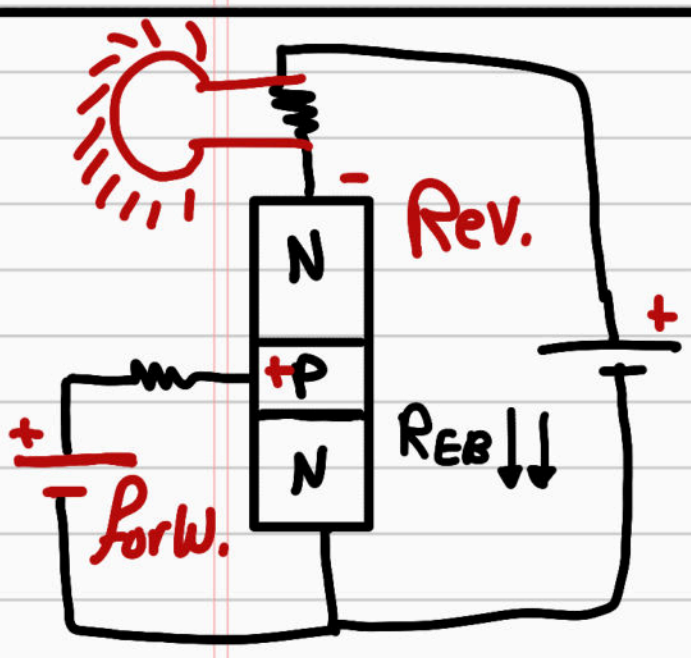
$$\beta_e = \frac{I_C}{I_B} = \frac{I_C \;(\div I_E)}{I_E - I_C \;(\div I_E)}$$

$$I_E = I_B + I_C$$

$$I_B = I_E - I_C$$

$$= \frac{\frac{I_C}{I_E}}{\frac{I_E}{I_E} - \frac{I_C}{I_E}} = \frac{\alpha_e}{1 - \alpha_e}$$

Switch on

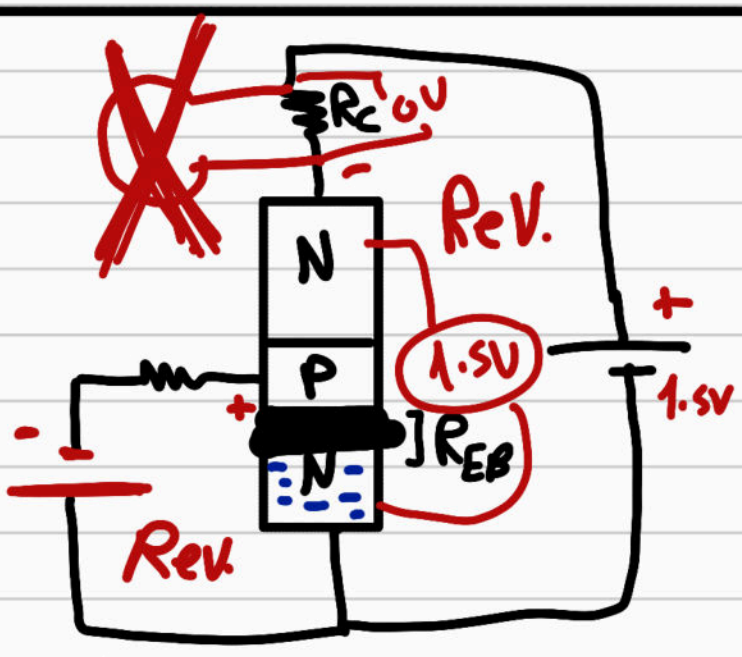


$I_B \uparrow \; I_C \uparrow \; I_E \uparrow$

$R_{EB} \downarrow$

$V_{CE} \downarrow = V_{CC} - I_C \uparrow R_C$

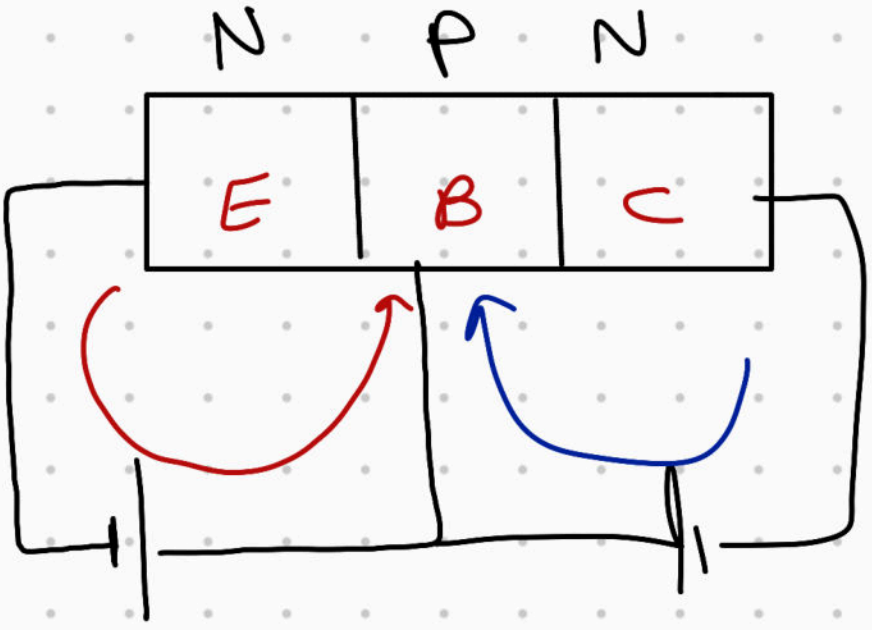
Switch off



$I_B \downarrow = 0 \quad I_C \downarrow = 0 \quad I_E \downarrow = 0$

$R_{EB} \uparrow\uparrow = \infty$

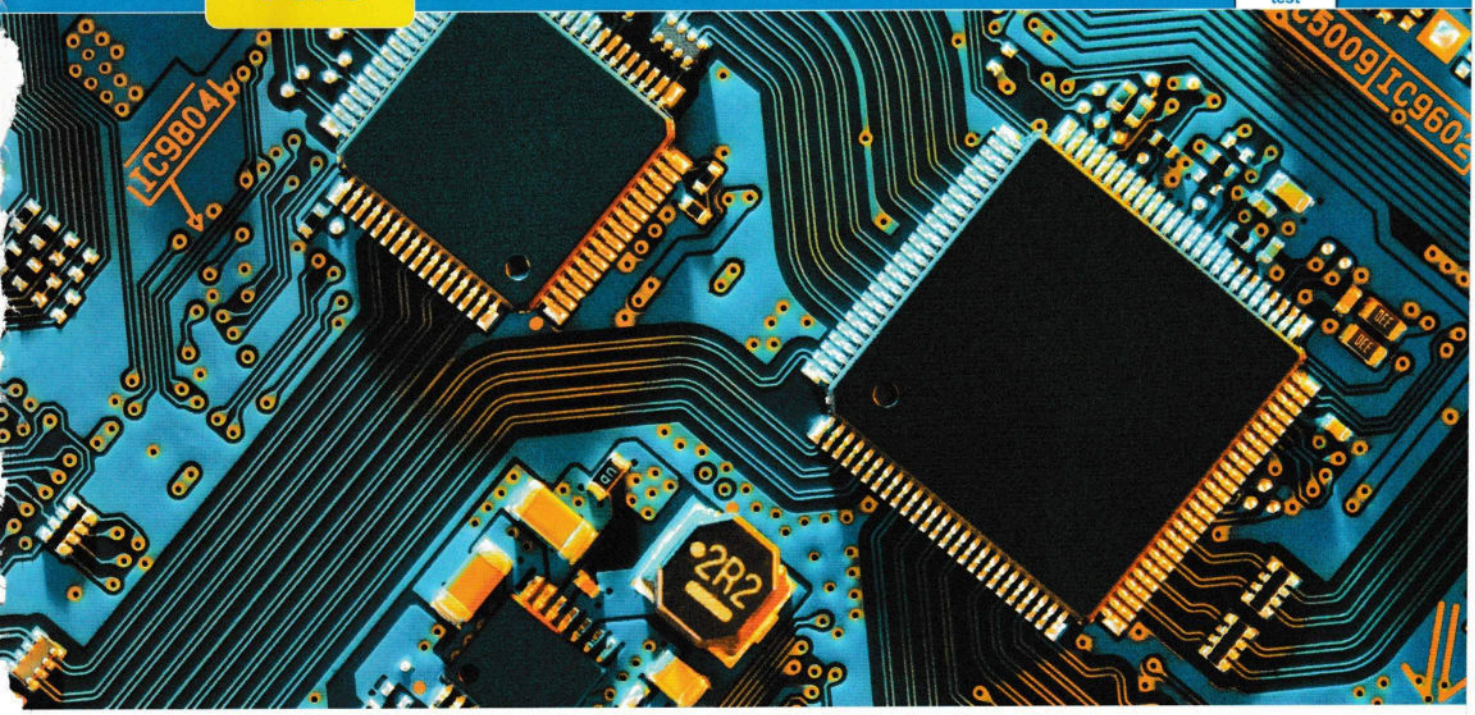
$V_{CE} = V_{CC} - I_C R_C$ ($I_C \to 0$)


N
P
N
E
B
C

Questions on Lesson Two

- The Transistor.
- Digital and Analog Electronics.

Interactive test



The questions signed by ✳ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

Important guidelines

⇒ **Transistor as an amplifier:**

- To determine the emitter current (I_E): $I_E = I_C + I_B$
- To determine the current gain (β_e): $\beta_e = \frac{I_C}{I_B} = \frac{\alpha_e}{1 - \alpha_e}$
- To determine the current transfer ratio (α_e): $\alpha_e = \frac{I_C}{I_E} = \frac{\beta_e}{1 + \beta_e}$

⇒ **Transistor as a switch:**

- To determine the battery voltage (V_{CC}): $V_{CC} = V_{CE} + I_C R_C$

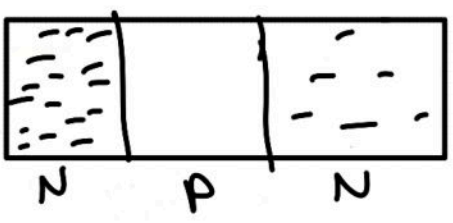
First Multiple choice questions

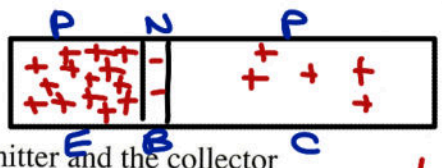
The transistor

1 The base region in the transistor is relatively .....

(a) thick and has higher concentration of impurities
(b) thin and has higher concentration of impurities
(c) thick and has lower concentration of impurities
(d) thin and has lower concentration of impurities ✓

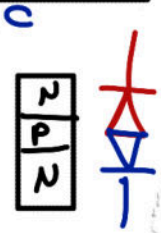
Emitter % ↑↑ Size med.
Base % ↓↓ Size small
Collector % Size large





2 In a pnp transistor, the dominant charge carriers in both of the emitter and the collector are ..... .

(a) acceptor ions (b) donor ions (c) free electrons (d) holes



3 The npn transistor is considered to be consisting of ..... .

(a) one pn junction (b) two pn junctions (c) three pn junctions (d) four pn junctions

4 The two opposite figures (1) and (2) represent transistor connection circuits in a common-emitter configuration. In which of them is the transistor used as a switch in the off state ?

(a) Figure (1). (b) Figure (2).
(c) Both figures. (d) Neither of them.

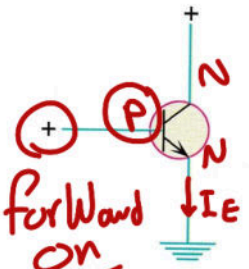


Figure (1)

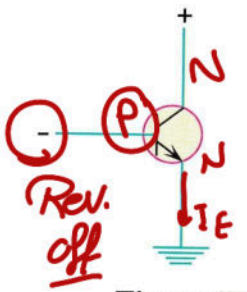


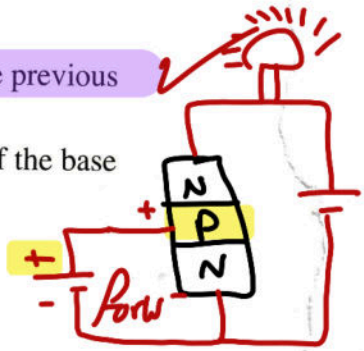
Figure (2)

5 In the transistor, the ratio of the concentration of impurities in the emitter to that in the collector is ..... .

(a) equal to one (b) greater than one (c) less than one (d) indeterminable

6 When connecting a transistor in the common base mode, if the current transfer ratio is α_e and current gain is β_e then ..... .

(a) $\alpha_e < 1$ (b) $\beta_e > 1$ (c) $\alpha_e = \frac{\beta_e}{1+\beta_e}$ (d) all the previous



7 Transistor npn is connected in a circuit in a common emitter configuration. If the base is given a high positive potential, the transistor acts as ..... .

(a) a half wave rectifier for current (b) a closed switch
(c) an opened switch (d) a full wave rectifier

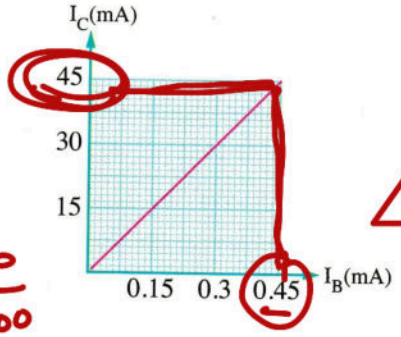
8 If the current through the collector of an npn transistor is 18 mA and that represents 90% of the emitter current, so the emitter current (I_E) and the base current (I_B), respectively, are ..... .

(a) 20 mA, 2 mA (b) 30 mA, 12 mA (c) 19 mA, 1 mA (d) 20 mA, 4 mA

9 The opposite graph represents the relation between the collector current (I_C) and the base current (I_B) in a pnp transistor, so;

(i) The value of α_e equals ..... .

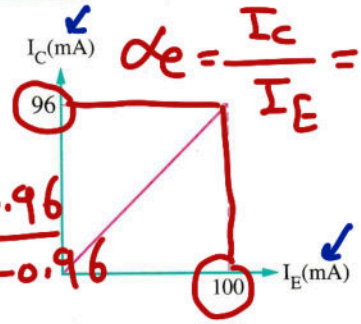
(a) 0.96 (b) 0.98
(c) 0.99 (d) 1



(ii) The intensity of the emitter current, when $I_C = 45$ mA, equals ..... .

(a) 20.12 mA (b) 30.15 mA (c) 40.25 mA (d) 45.45 mA

10 * The opposite graph represents the relation between the collector current (I_C) and the emitter current (I_E) for an npn transistor, so the values of α_e and β_e, respectively, are ..... .



(a) 0.96, 24 (b) 0.96, 48

(c) 0.49, 32 (d) 0.49, 64

11 * If α_e for a transistor is 0.99 and the base current is 100 μA, then the current gain (β_e) and the collector current (I_C), respectively, equal ..... .

(a) 98, 9.8×10^{-3} A (b) 99, 9.9×10^{-3} A

(c) 99, 9.9 A (d) 98, 9.8 A

12 * If the base current in a transistor is 24 μA and its current gain is 24, the current transfer ratio (α_e) and collector current (I_C), respectively, are ..... .

(a) 0.76, 576×10^{-3} A (b) 0.76, 576×10^{-6} A

(c) 0.96, 576×10^{-3} A (d) 0.96, 576×10^{-6} A

13 * When using a transistor as an amplifier, if the electrical signal into the base of the transistor is 200 μA and the collector current becomes 10 mA, then β_e and α_e, respectively, are ..... .

(a) 20, 0.95 (b) 20, 0.98 (c) 50, 0.95 (d) 50, 0.98

14 If the current gain in a transistor of npn type is 99 and the collector current is 10 mA, hence the current transfer ratio (α_e) and emitter current (I_E) respectively are .....

(a) 0.98, 10.1×10^{-3} A (b) 0.98, 0.1×10^{-3} A

(c) 0.99, 0.1×10^{-3} A (d) 0.99, 10.1×10^{-3} A

15 * In a transistor circuit in which ($V_{CC} = 1.5$ V), the potential difference between the collector and the emitter ($V_{CE} = 0.5$ V) and the resistance of the collector circuit ($R_C = 500\ \Omega$), so the collector current (I_C) equals ..... .

(a) 1×10^{-3} A (b) 2×10^{-3} A

(c) 3×10^{-3} A (d) 4×10^{-3} A

$R_1 \uparrow R_B \uparrow R_{EB} \uparrow I_E \downarrow I_B \downarrow I_C \downarrow$ | $R_1 \uparrow \infty$ Reverse

16 In the opposite transistor circuit, when the resistance of R_1 increases, the reading of the voltmeter ..... .

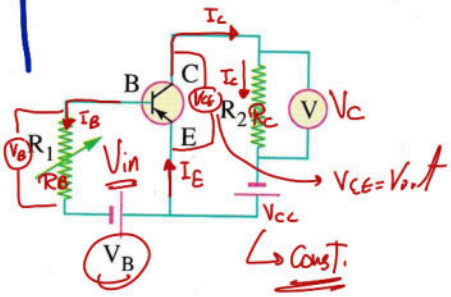
(a) increases

(b) decreases but doesn't reach zero

(c) remains constant

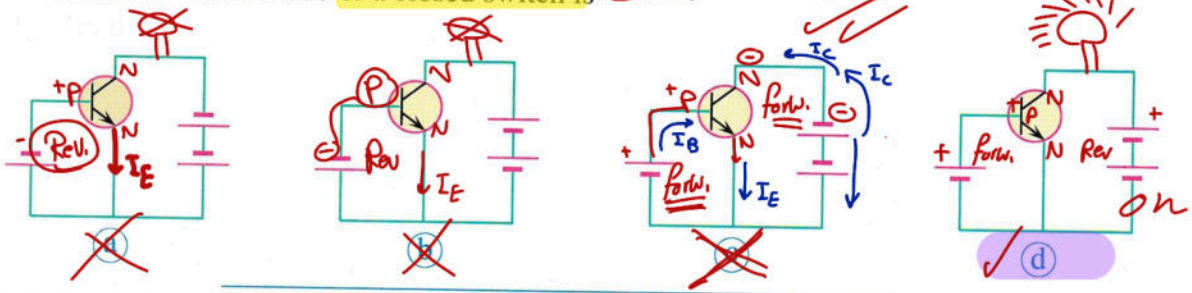
(d) equals zero

$V_C = I_C R_C$



17 The circuit that shows the correct way of connecting the batteries in a transistor circuit in a common emitter mode of a closed switch is ..... .

Switch on



(d)

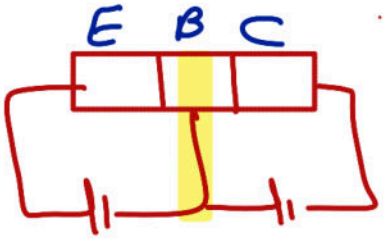
18 In the shown transistor circuit, the transistor is connected as ..... .

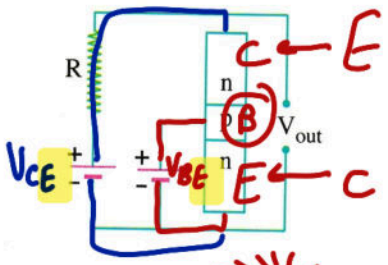
(a) an opened switch

(b) a common emitter

(c) a common base

(d) answer is indeterminable





19 * The opposite circuit has an npn transistor, so from the given data, the two constants β_e and α_e, respectively, equal ..... .

(a) 50, 0.98

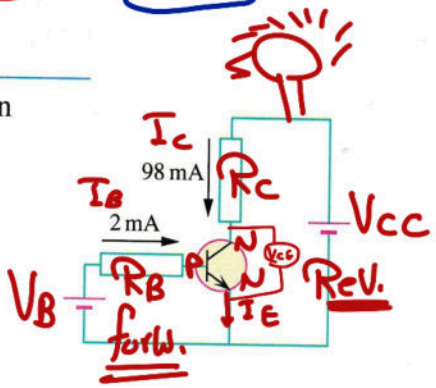
(b) 49, 0.99

(c) 49, 0.98

(d) 50, 0.99

$\beta_e = \frac{I_C}{I_B} = \frac{98}{2} = 49$

$\alpha_e = \frac{\beta_e}{1+\beta_e} = \frac{49}{50}$



20 * The opposite diagram represents a circuit of npn transistor, so α_e equals ..... .

(a) 0.92

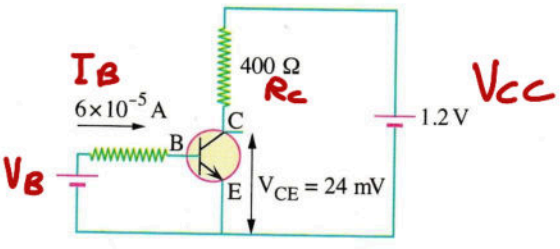
(b) 0.95

(c) 0.96

(d) 0.98

$\alpha_e = \frac{I_C}{I_E} = \frac{I_C}{I_B + I_C}$

$= \frac{2.94 \times 10^{-3}}{6 \times 10^{-5} + 2.94 \times 10^{-3}}$

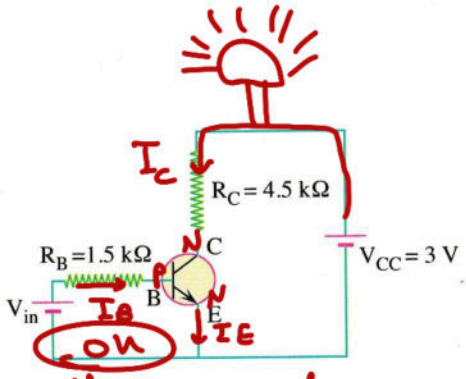


$V_{CC} = I_C R_C + V_{CE}$

$1.2 = I_C \times 400 + 24 \times 10^{-3}$

$I_C = 2.94 \times 10^{-3}$ A

21 The opposite diagram represents a circuit in which the transistor is used as a switch, so if the potential difference across the base resistor R_B equals 0.01 V and $\beta_e = 75$, then ..... .



$V_{in} = V_B$

	The intensity of the base current (I_B)	The output voltage (V_{CE})
(a)	3.33×10^{-6} A	0.55 V
(b)	3.33×10^{-6} A	0.75 V
(c)	6.67×10^{-6} A	0.55 V
(d)	6.67×10^{-6} A	0.75 V

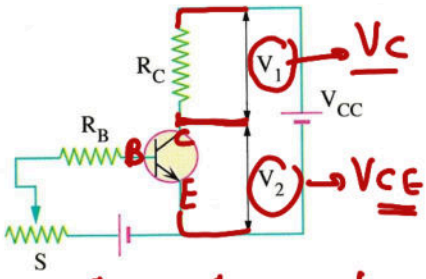
$I_B = \frac{V_B}{R_B} = \frac{0.01}{1500} = 6.67 \times 10^{-6}$ A

$I_C = \beta_e \times I_B = 75 \times \checkmark = 5 \times 10^{-4}$ A

$V_{CE} = V_{CC} - I_C R_C$

$= 3 - 5 \times 10^{-4} \times 4500$

22 The opposite figure represents a transistor (npn) circuit, where the transistor is switched on. When the resistance of the rheostat (S) increases, then the magnitudes of potential differences V_1 and V_2, respectively, will ..... .



$V_1 \rightarrow V_C$

$V_2 \rightarrow V_{CE}$

(a) decrease, decrease

(b) decrease, increase

(c) increase, increase

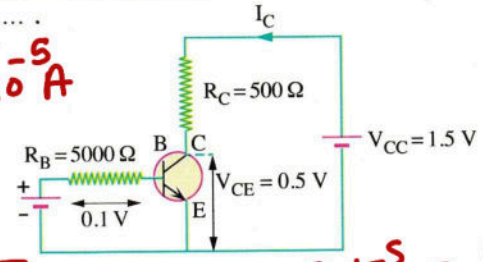
(d) increase, decrease

$R_B \uparrow R_{EB} \uparrow$ I_B I_C I_E

$V_1 = V_C = I_C R_C$

$V_2 = V_{CE} = V_{CC} - I_C R_C$

23 ✱ In the opposite circuit, the value of I_E equals ..... .



(a) 2×10^{-3} A

(b) 2.02×10^{-3} A

(c) 2×10^{-5} A

(d) 2.02×10^{-5} A

$I_B = \frac{V_B}{R_B} = \frac{0.1}{5000} = 2 \times 10^{-5}$ A

$V_{CC} = V_{CE} + I_C R_C$

$1.5 = 0.5 + I_C \times 500$

$I_C = 2 \times 10^{-3}$ A

$I_E = I_B + I_C = 2 \times 10^{-5} + 2 \times 10^{-3}$

24 In npn transistor, if the difference between the intensities of the collector and base currents ($I_C - I_B$) equals 97 mA, and the intensity of the emitter current (I_E) equals 99 mA, so the current gain (β_e) equals ..... .

(a) 97 (b) 98 (c) 99 (d) 100

$I_C - I_B = 97$ → ①

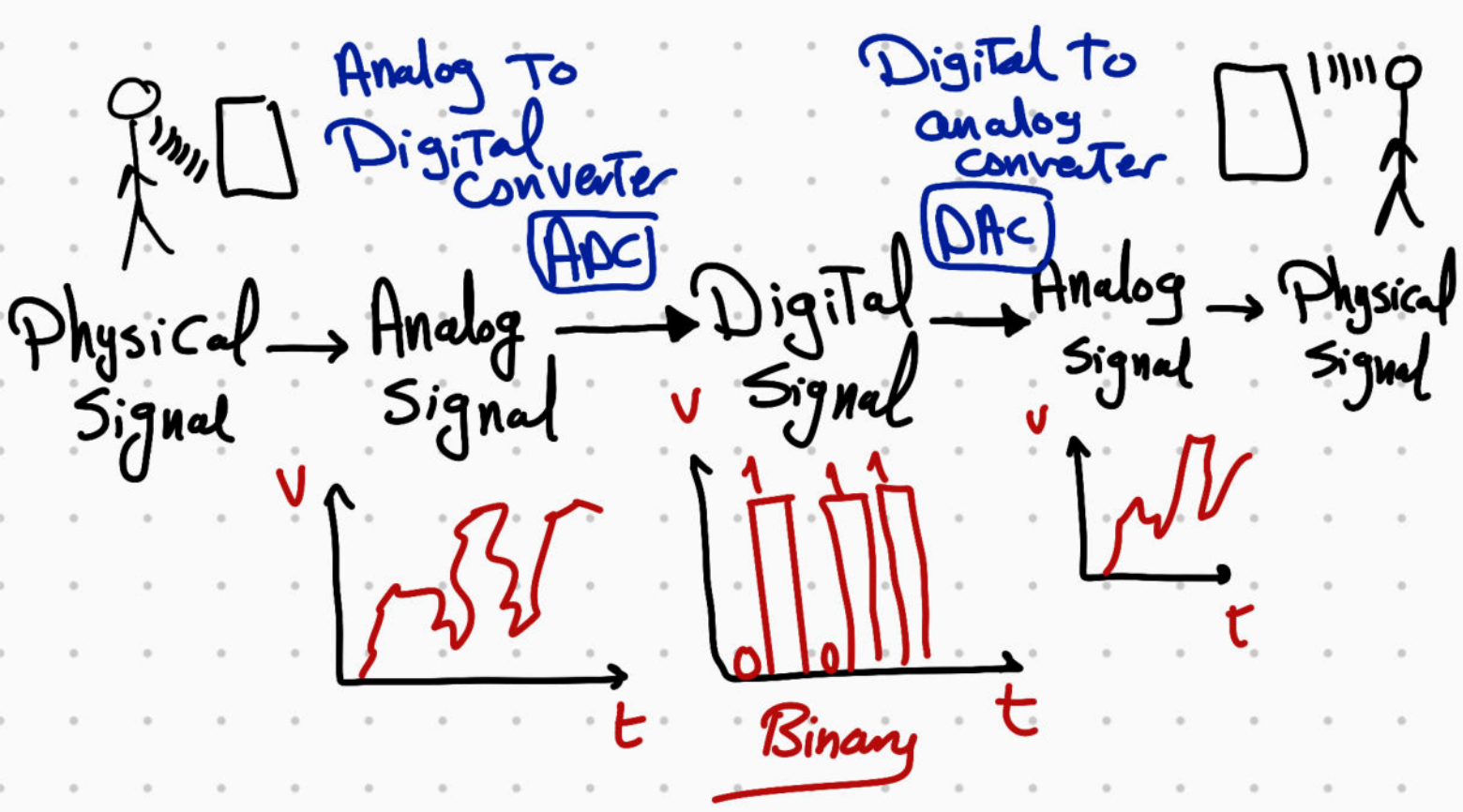
$I_E = I_C + I_B = 99$ → ②

Mode (S) (1)

1	-1	97
1	1	99

$X = I_C = 98$ $Y = I_B = 1$

$\beta_e = \frac{I_C}{I_B} = \frac{98}{1} = 98$



$$[2008]_{10} = [0000\,1111\,1011\,1000]_{2}$$

Decimal — Binary

Mode → Base-N (4)

log BIN — x^2 DEC

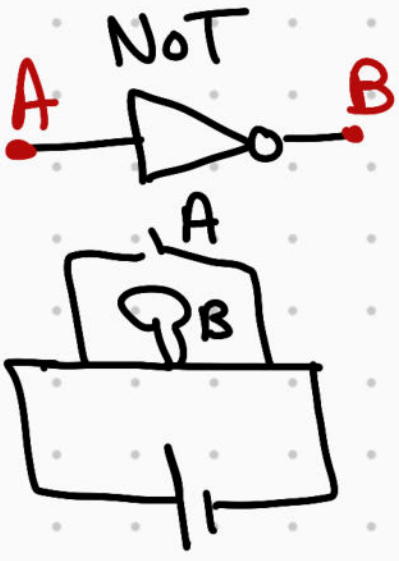
$$(1100\,1100\,11)_{2} = (819)_{10}$$

Binary — Decimal

2008	0	✓
1004	0	✓
502	0	✓
250	251	1 ✓
124	125	1 ✓
62	0	✓
30	31	1 ✓
14	15	1 ✓
6	7	1 ✓
?	3	1 ✓
	1	1 ✓
	0	0

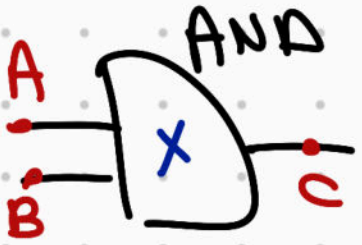
$$(\cancel{XXXX}\,1111\,1011\,1000)_{2}$$

Logic Gates



A	B
0	1
1	0

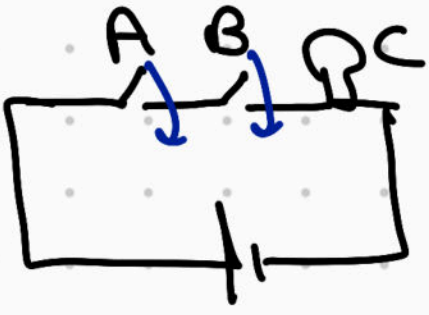
On → 1
off → 0

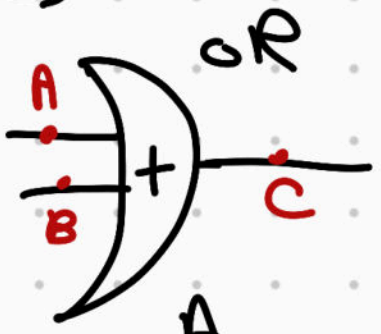


A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

open switch → 0

Closed switch → 1





A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

25 In the shown transistor circuit in figure (1), the relation between the ammeter reading (I_C) and voltameter reading (V_{CE}) is represented graphically as shown in figure (2),

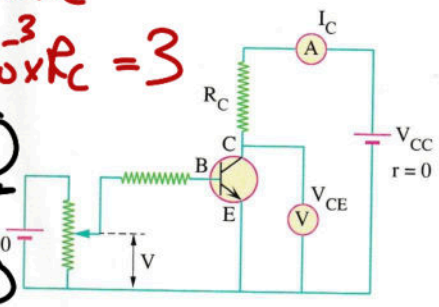


Figure (1)

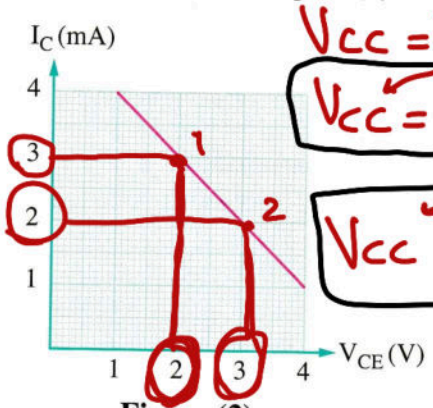


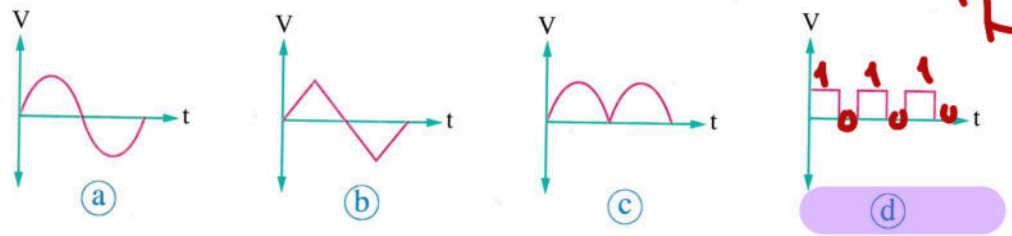
Figure (2)

then the values of R_C and V_{CC}, respectively, are .....

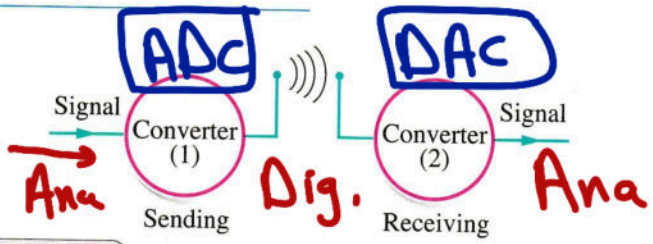
(a) 1 kΩ, 4 V (b) 1 kΩ, 5 V (c) 2 kΩ, 4 V (d) 2 kΩ, 5 V

Digital and analog electronic devices

26 Which of the following graphs represents the change of voltage (V) for a signal in a digital electronic device with time (t)?



27 In digital electronics, to send and receive signals that carry information, converter (1) is used at sending and converter (2) is used at receiving, then ..... .



	Converter (1)	Converter (2)
(a)	analog to digital	analog to digital
(b)	analog to digital	digital to analog
(c)	digital to analog	analog to digital
(d)	digital to analog	digital to analog

28 Which of the following statements correctly represents the order of the stages of digital transmission and reception?

(a) Physical quantity ⟶ Analog signal ⟶ Digital signal ⟶ Analog signal ⟶ Physical quantity

(b) Physical quantity ⟶ Digital signal ⟶ Analog signal ⟶ Digital signal ⟶ Physical quantity

(c) Analog signal ⟶ Physical quantity ⟶ Digital signal ⟶ Physical quantity ⟶ Analog signal

(d) Digital signal ⟶ Analog signal ⟶ Physical quantity ⟶ Analog signal ⟶ Digital signal

29 The code for the decimal number 20 according to the binary system is ..... .

(a) $(10101)_2$ (b) $(10100)_2$ (c) $(11100)_2$ (d) $(00111)_2$

30 The binary number which is equivalent to the summation of the following decimal numbers (1 + 1 + 1 + 1) is .....

(a) $(1111)_2$ (b) $(101)_2$ (c) $(110)_2$ (d) $(100)_2$

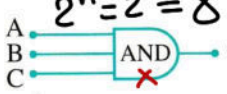
31 The decimal number which is equivalent to the binary number $(1010)_2$ is ..... .

(a) 4 (b) 8 (c) 10 (d) 12

32 The probability of getting an output of (1) in the opposite logic gate is ..... .

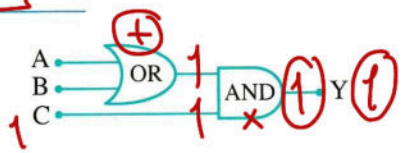
(a) 10 % (b) 20 % (c) 50.9 % (d) 87.5 %

33 The opposite figure shows a logic gate, so the number of probable combinations that gives high value at the output equals ..... .



(a) 0 (b) 1 (c) 2 (d) 3

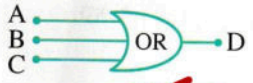
34 In the shown logic gates, to get an output Y = 1, the inputs (A, B, C) have to be .....



	A	B	C
(a)	0	1	0
(b)	0	0	1
(c)	1	0	1
(d)	1	0	0

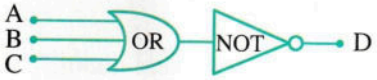
35 Which of the following logic gates matches the opposite truth table?

A	B	C	D
0	0	1	1
0	1	0	1
0	1	1	1

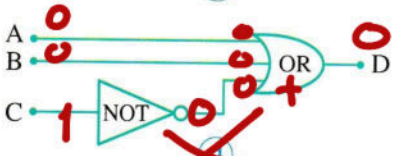


(a) ✓

(b)



(c)

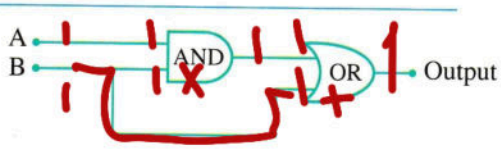


(d)

36 When digital signal $(11001)_2$ is applied at the input of a NOT gate, the output signal will be ..... .

(a) $(00111)_2$ (b) $(00110)_2$ ✓ (c) $(10101)_2$ (d) $(11001)_2$

37 In the opposite logic circuit, if the inputs are as shown in the opposite table, so the outputs will be ..... .



A	B
0	0
1	1

Output
0
1

(a) ✓

Output
1
1

(b)

Output
0
0

(c)

Output
1
0

(d)

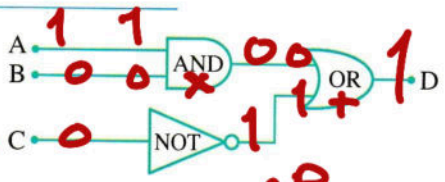
38 In the shown logic circuit, which of the following inputs gives the output D = 1?

(a) A = 0, B = 0, C = 1

(b) A = 1, B = 0, C = 1

(c) A = 1, B = 0, C = 0 ✓

(d) A = 0, B = 1, C = 1



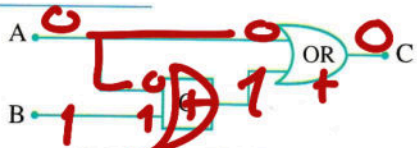
39 The opposite figure shows a logic circuit with its truth table, so the logic gate G is ..... .

(a) AND ✓

(b) OR

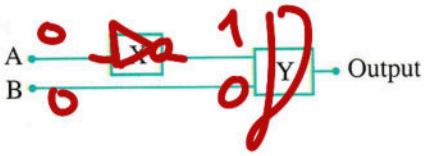
(c) NOT

(d) AND or OR



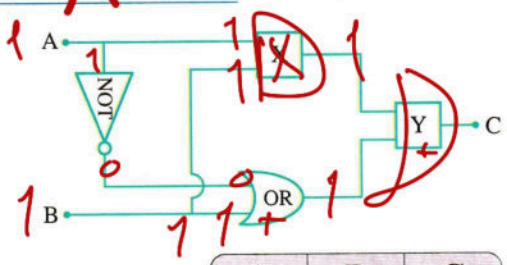
A	B	C
0	0	0
0	1	0
1	0	1
1	1	1

40 The opposite figure shows a logic circuit, if the output of this circuit becomes (1) when its inputs on A and B are (0, 0), the components X and Y, respectively, are ..... .



(a) OR, AND (b) NOT, AND (c) NOT, NOT (d) NOT, OR

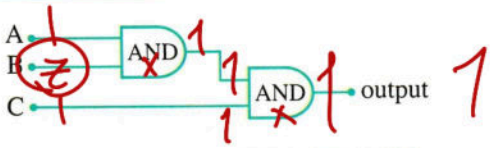
41 The opposite figure shows a logic circuit and its truth table, so the two logic gates X and Y are ..... .



	Gate X	Gate Y
(a)	OR	AND
(b)	AND	AND
(c)	OR	OR
(d)	AND	OR

A	B	C
0	0	1
0	1	1
1	0	0
1	1	1

42 The opposite figure shows a logic circuit, and the truth table that follows the circuit shows some input and output values for this circuit, so the values of x, y, and z in the truth table are ..... .



input			output
A	B	C	
x	1	1	0
1	1	0	y
1	z	1	1

(a)

x	y	z
0	0	0

(b)

x	y	z
1	0	0

(c)

x	y	z
0	0	1

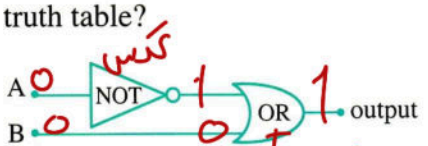
(d)

x	y	z
0	1	1

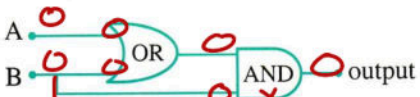
43 Which of the following logic circuits matches the opposite truth table?

A	B	output
0	0	1
0	1	1
1	0	0
1	1	1

(a)

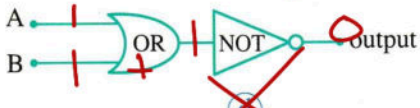


(b)

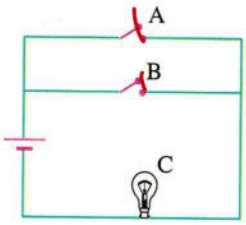


(c)

(d)



44 In the opposite electric circuit, if the two switches (A) and (B) represent the inputs and the lightbulb (C) represents the output, the correct truth table for this circuit is ..... .



A	B	C
0	1	1
1	1	0

(a)

A	B	C
0	1	0
1	1	1

(b)

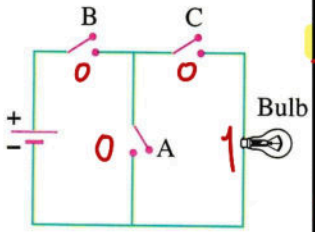
A	B	C
1	0	1
0	1	1

(c)

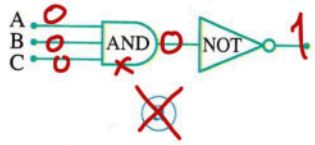
A	B	C
1	0	0
0	1	0

(d)

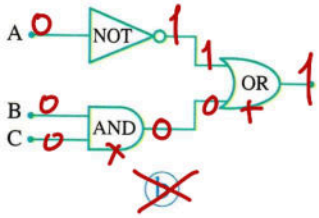
45 Which of the following logic gates represents the opposite electric circuit?



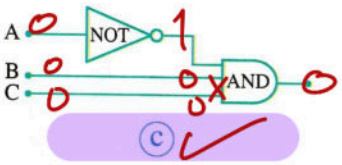
A	B	C	D
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0



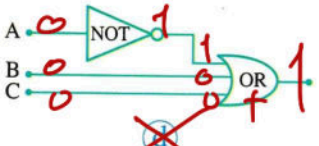
(a)



(b)



(c)

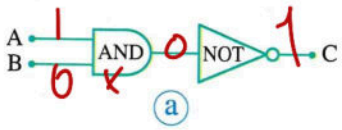


(d)

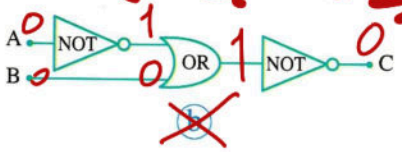
$2^n = 2^3 = 8$

46 In which of the following logic gates will the output be equivalent to the decimal number 11, if the inputs are listed in the opposite table?

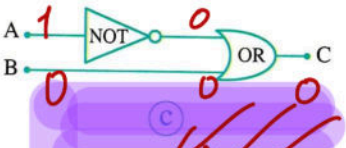
A	B	output
0	0	1
0	1	1
1	0	0
1	1	1



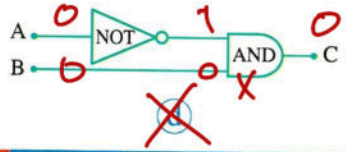
(a)



(b)



(c)



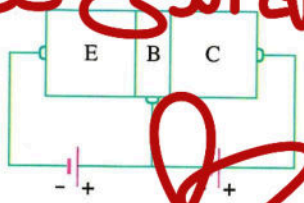
(d)

$(11)_{10} = (1011)_2$

Second Essay questions

The transistor

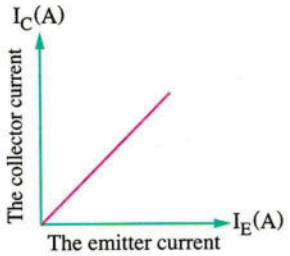
1 **Explain what happens to** the free electrons in each of the regions of an npn transistor when it is connected as shown in the opposite figure.



2 **Give reasons for:** In the transistor, the current transfer ratio (α_e) is close to one while the current gain (β_e) is greater than one.

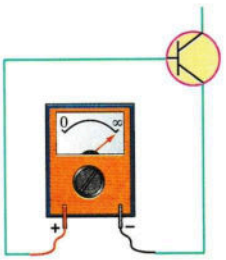
3 **Compare between** the emitter and the collector in the npn transistor in the case of a common base connection (in terms of : valency of impurities - type of connection with the base - the potential barrier with the base).

4 **Write the mathematical equation that relates the two variables in the opposite graph and mention what the slope of the line equals.**



5 **What** are the conditions required for an npn transistor to work as a closed switch (on)?

6 A transistor was connected to an ohmmeter where the reading of the device was as shown in the opposite figure, then **what** is the type of the transistor (npn or pnp)? **With explanation**



7 When using an npn transistor in a common emitter amplifier circuit, the collector current was 99 mA and the product of the current transfer ratio (α_e) and the current gain (β_e) was 98.01, **calculate:**

(a) The base current (I_B).

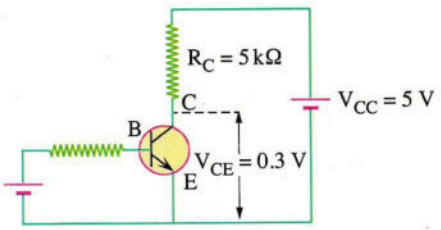
(b) The emitter current (I_E).

8 In a common emitter transistor circuit, **prove that:** $I_B = I_C\left(\frac{1}{\alpha_e} - 1\right)$

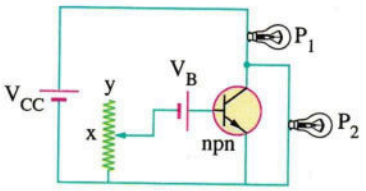
9 **In the opposite transistor circuit, if the current gain is 30, calculate:**

(a) The base current (I_B).

(b) The current transfer ratio (α_e).



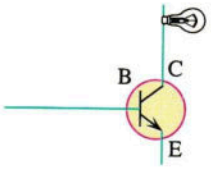
10 The opposite diagram shows an npn transistor circuit working as a closed switch that contains two identical bulbs P_1 and P_2:



(a) **Explain what happens to** the brightness of the two bulbs when moving the slider from point x in the direction of point y.

(b) **Plot the graph** that represents the relation between the voltage (V_1) across the terminals of bulb P_1 and the voltage (V_2) across the terminals of bulb P_2.

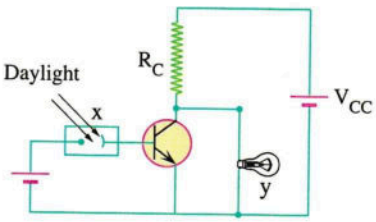
11 The opposite figure represents a transistor connected in series with a small bulb that works on a direct potential difference:



(a) **Complete the diagram** of the electric circuit to light up the lamp.

(b) **What** modification would be made to the circuit in the previous case to prevent the bulb from lighting?

12 In the opposite electronic circuit, a photocell (x) is connected to the base of an npn transistor whose emitter and collector terminals are connected to a lightbulb (y), **explain why** the lightbulb glows when the daylight is gone.



Digital and analog electronic devices

13 **Give reasons for :**

(a) The presence of some defects in sound and image in analog transmission.

(b) Digital electronics is preferred to analog electronics in the electronic devices.

14 **Mention the scientific base** of the digital electronics.

15 **Find the equivalent binary number for each of the following decimal numbers:**

(a) 59 (b) 120

16 **Find the corresponding decimal number for each of the following binary numbers:**

(a) $(100110)_2$ (b) $(10011011)_2$

17 **Find** both of the decimal and binary numbers of the quotient of dividing these two binary numbers $\frac{(11110)_2}{(110)_2}$

18 **Deduce the truth table for:**

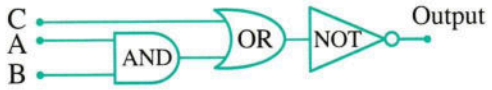
(a) AND gate of two inputs followed by an inverter.

(b) OR gate of two inputs followed by an inverter.

19 **Complete the truth table for each of the following electronic circuits and convert the output result into decimal number:**

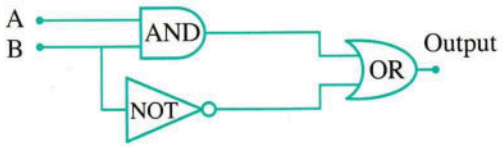
(a)

(A)	(B)	(C)	Output
1	1	0	.....
1	0	1	.....
0	0	1	.....



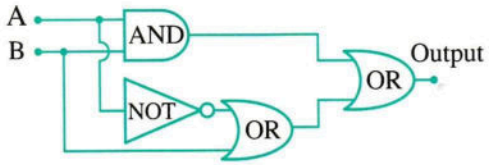
(b)

(A)	(B)	Output
0	0	.....
0	1	.....
1	0	.....
1	1	.....



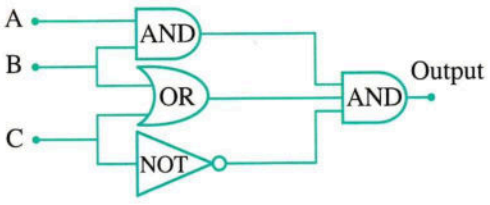
(c)

(A)	(B)	Output
0	0	.....
0	1	.....
1	0	.....
1	1	.....



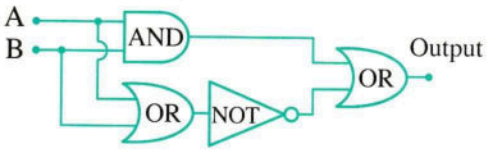
(d)

(A)	(B)	(C)	Output
0	0	0	.....
1	1	0	.....
1	0	1	.....
0	1	1	.....
0	0	1	.....
1	1	1	.....



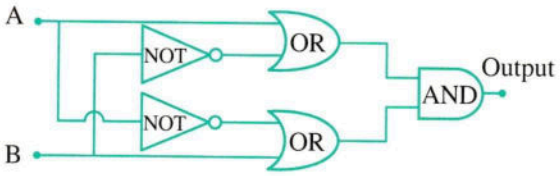
(e)

(A)	(B)	Output
0	0	.....
0	1	.....
1	0	.....
1	1	.....



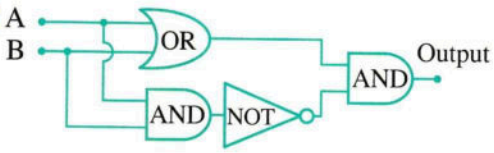
(f)

(A)	(B)	Output
0	0	.....
0	1	.....
1	0	.....
1	1	.....

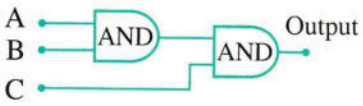


20 **Write the truth table of the following logic circuits :**

(a)

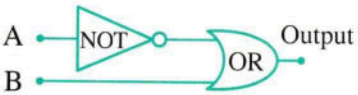


(b)



(c)

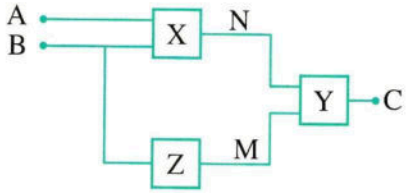
(d)



21 **From the following truth table:**

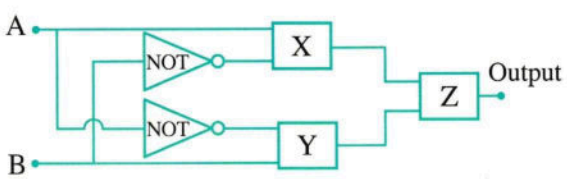
(a) **Conclude** the types of gates X, Y, Z.

(b) **Complete** the table:



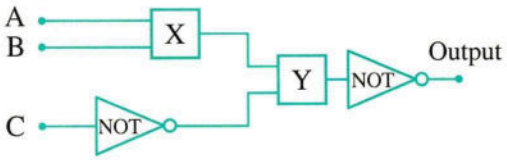
Input				Output
A	B	N	M	C
0	1	1	0	0
1	1	.....	0	.....
1	0	1	.....	1

22 **From the following truth table, conclude the types of gates (X,Y,Z):**



(A)	(B)	Output
0	0	0
0	1	1
1	0	1
1	1	0

23 **The following truth table represents some of input and output values for the logic gates circuit shown in the figure:**



(a) **Identify** the type of gate (X) and gate (Y).

(b) **Find** the output (Z) in the table.

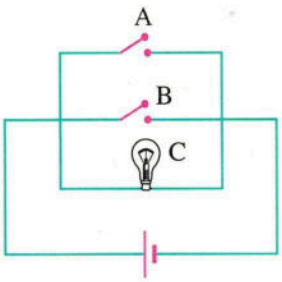
(A)	(B)	(C)	Output
1	1	1	0
0	1	1	1
0	0	0	Z

24 The shown electric circuit is equivalent to a number of logic gates where the switches A and B represent the inputs and the light of the lamp C represents the output.

(a) **Complete** the opposite truth table according to the circuit.

(b) **Draw** the logic gates circuit which fulfill the opposite truth table.

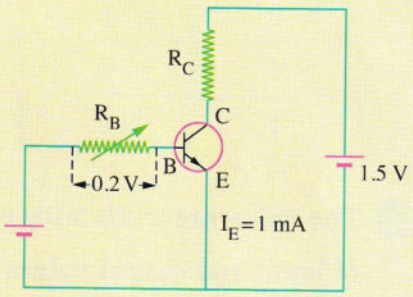
Input		Output (C)
(A)	(B)	
1	1	.....
0	1	.....
1	0	.....
0	0	.....



Questions of the Previous Years Exams on Lesson 2

The transistor

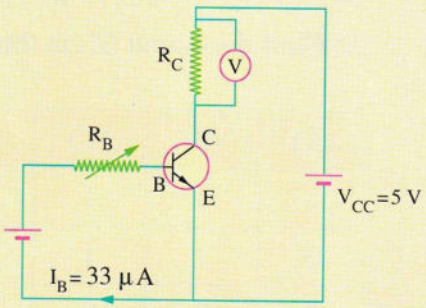
1 The circuit represents a transistor as an inverter gate, if the output potential (V_{CE}) = 0.8 V when the resistance of the base circuit (R_B) = 4000 Ω, the value of the collector circuit resistance (R_C) = ... approximately. *(Exp. June 21)*



(a) 7.37×10^2 Ω (b) 73.7×10^2 Ω

(c) 0.737×10^2 Ω (d) 7370×10^2 Ω

2 The circuit represents a transistor as an amplifier, if the voltmeter reads 4.8 V and the value of R_C = 4.5 kΩ, the values of β_e and α_e respectively are ..... . *(Exp. June 21)*



(a) 32.32, 0.97 (b) 33.67, 0.95

(c) 99, 0.99 (d) 3, 0.75

3 If the base current in npn transistor = 2 mA and α_e = 0.97, the collector current = ...

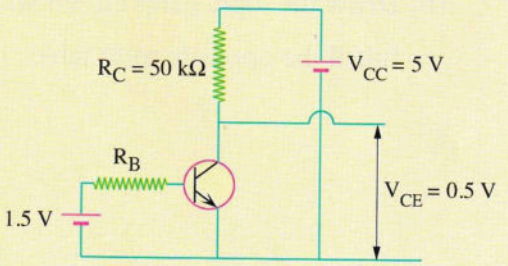
(a) 64.67 mA (b) 1.97 mA (c) 10 mA (d) 50.67 mA

(1^{st} session 21, Exp. 25)

4 An npn transistor is used as a current amplifier, if the base current equals 1 mA and the current gain (β_e) is 200, so the collector current equals ..... . *(1^{st} session 21)*

(a) 0.02 A (b) 0.2 A (c) 2 A (d) 20 A

5 In npn transistor circuit, the resistance connected to its collector R_C = 50 kΩ, if its current gain β_e = 30, from the data on the figure, the intensity of base current (I_B) = ... *(2^{nd} session 21)*

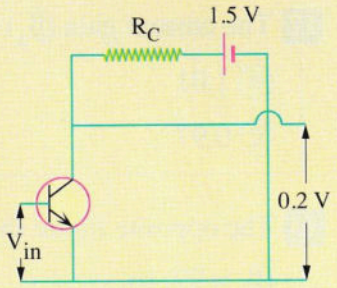


(a) 3×10^{-6} A (b) 9.3×10^{-5} A

(c) 9×10^{-5} A (d) 8.7×10^{-6} A

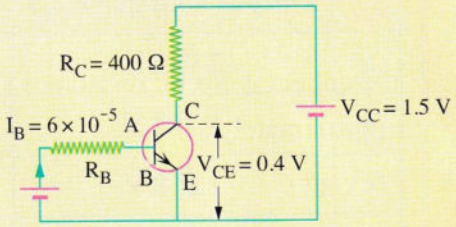
6 A transistor was used as a switch, the output potential (V_{CE}) = 0.2 V and the battery voltage in the collector circuit was 1.5 V, so the potential difference across the resistor in the collector circuit (R_C) = ...

(a) 1.7 V (b) 1.3 V
(c) 0.3 V (d) 7.5 V *(2nd session 21)*



7 The figure shows a circuit of transistor (npn) as amplifier, so the ratio of $\frac{\alpha_e}{\beta_e}$ = ...

(a) 2.7×10^{-3} (b) 2.1×10^{-2}
(c) 1.1×10^{-2} (d) 2.8×10^{-3}
(1st session 22)



8 If the current division (transfer) ratio of a transistor $\alpha_e = 0.99$ then, the ratio :

$$\frac{\text{the current of the emitter } I_E}{\text{the current of the base } I_B} = \text{.....}$$

(2nd session 22)

(a) 100 (b) 99 (c) 200 (d) 198

9 If the base current in an npn transistor is 6 μA and ($\alpha_e = 0.95$), then the currents through the emitter and the collector, respectively, are ..... . *(Exp. 23)*

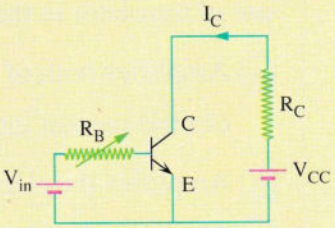
	I_E	I_C
(a)	120 μA	114 μA
(b)	114 μA	120 μA
(c)	11.4 μA	12 μA
(d)	240 μA	242 μA

10 In the transistor circuit, if the current of the emitter is 120 times as that of the base, then α_e equals ..... *(1st session 23)*

(a) 0.96 (b) 120 (c) 119 (d) 0.99

11 The electric circuit shown in the figure represents a transistor used as a switch, so on increasing the value of R_B to double its value. Which of the following choices correctly describes what happens to the value of V_{CC}? *(2nd session 23)*

(a) It remains constant.
(b) It decreases to half its value.
(c) It increases to double its value.
(d) It becomes equals to zero.



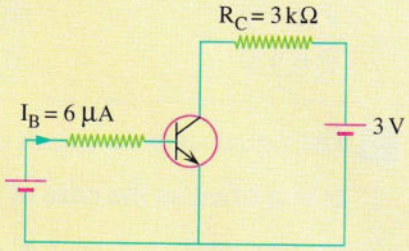
12 The current gain (β_e) for a transistor = 150, so (α_e) = ... *(2nd session 23)*

(a) 1.01 (b) 0.99

(c) 0.97 (d) 1.1

13 The opposite figure shows npn transistor (β_e = 99) so, ... *(1st session 24)*

	Output voltage (V_{CE})	Collector current I_C
(a)	2.982 V	0.06 μA
(b)	1.782 V	16.5 μA
(c)	1.218 V	594 μA
(d)	2.982 V	16.5 μA

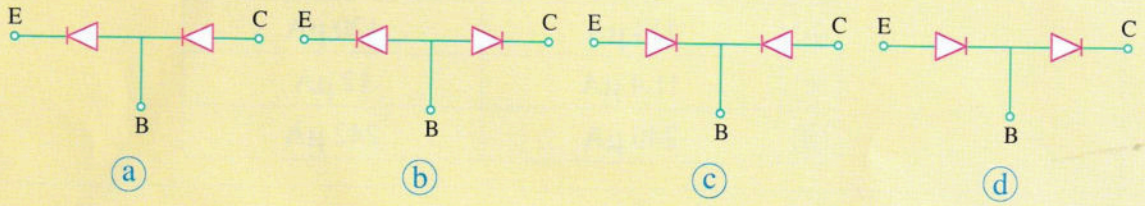


14 If the current gain (β_e) in a transistor = 93.6, then the ratio $\left(\frac{I_E}{I_B}\right)$ = ... *(2nd session 24)*

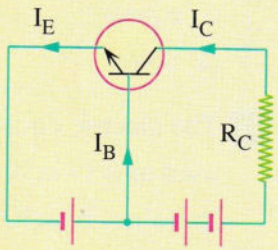
(a) 93.6 (b) 95.6

(c) 94.6 (d) 92.6

15 The npn transistor can be considered as two diodes connected together. Which of the following figures is the correct figure representing the npn transistor? *(Exp. 25)*



16 A transistor is connected as in the opposite figure. Which of the following choices describes correctly the use of transistor in that circuit? *(Exp. 25)*



(a) Amplifies both of the electric power and the electric current intensity.

(b) Amplifies the electric power but does not amplify the electric current intensity.

(c) Amplifies the electric current intensity but does not amplify the electric power.

(d) Neither amplifies the electric power nor the electric current intensity.

17 In a transistor circuit in which $V_{CC} = 1.5$ V, the potential difference between the collector and the emitter (V_{CE}) = 0.5 V and the resistance of the collector circuit (R_C) = 500 Ω, **calculate** the collector current (I_C). *(Exp. 25)*

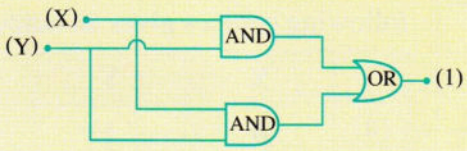
.....

.....

.....

Digital and analog electronic devices

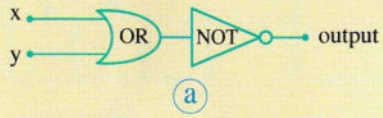
18 A group of logic gates has an output voltage (1), as shown in the figure which of the probabilities shown in the table verifies that? *(Exp. June 21)*



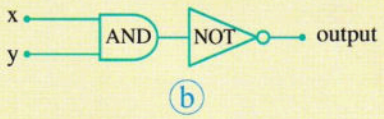
	X	Y
(a)	0	0
(b)	1	0
(c)	1	1
(d)	0	1

19 Which group of the following logic gates verifies the shown truth table?

input		output
x	y	
1	0	1

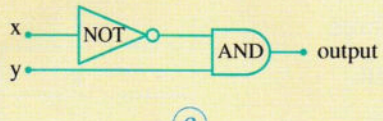


(a)

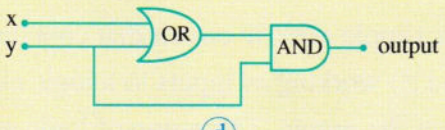


(b)

(1st session 21)

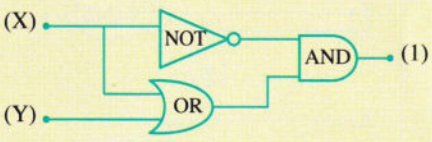


(c)



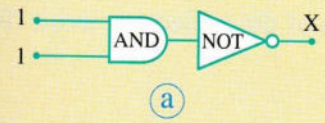
(d)

20 The group of logic gates in the figure has an output voltage (1), which of the following choices in the table for the inputs potentials verifies that? *(2nd session 21, 1st session 22)*

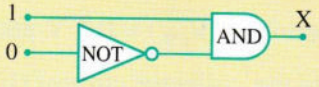


	X	Y
(a)	0	0
(b)	1	0
(c)	1	1
(d)	0	1

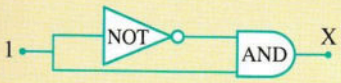
21 Which logic circuit has a high output voltage at point X? *(2nd session 22)*



(a)



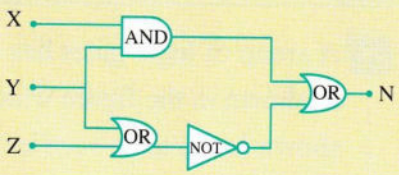
(b)



(c)

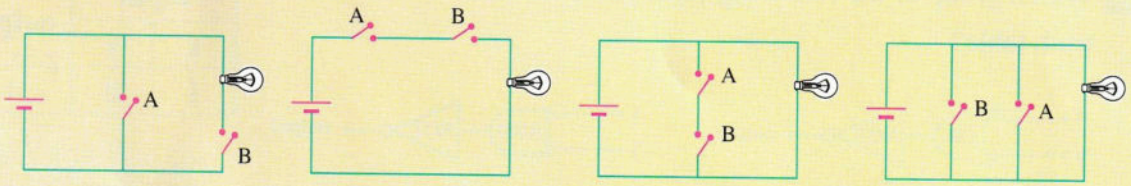
(d)

22 In the illustrated circuit of logic gates, which of the following choices gives an output of N = 0? *(Exp. 23)*



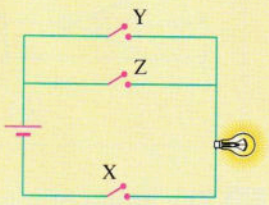
	X	Y	Z
(a)	0	1	0
(b)	1	1	0
(c)	0	0	0
(d)	1	0	0

23 Which of the following simple electric circuits represents the combination of the opposite logic gates? *(1st session 23)*

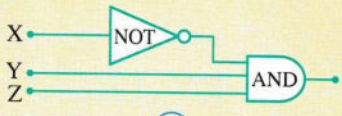


(a) (b) (c) (d)

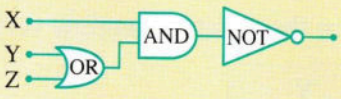
24 The figure shows a simple electric circuit that contains three switches (X , Y and Z) working as inputs to a logic circuit, while the lamp gives the output. The group of logic gates that is represented by this simple electric circuit is ..... .



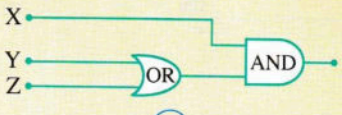
(2nd session 23)



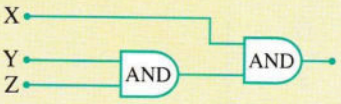
(a)



(b)

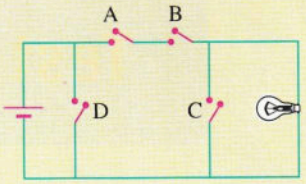


(c)

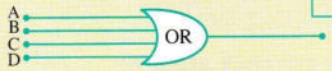


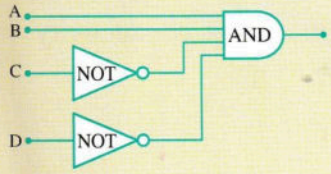
(d)

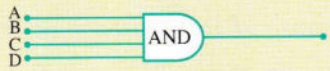
25 The opposite electric circuit is equivalent to the logic gate of shape ..... . *(1st session 24)*



ⓐ

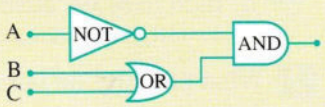
ⓑ 

ⓒ 

ⓓ 

26 From the opposite figure:

Which choice produces high output voltage?



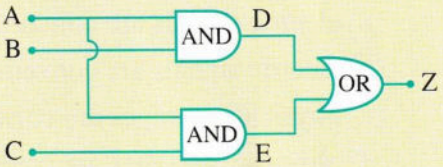
	A	B	C
ⓐ	0	0	0
ⓑ	0	0	1
ⓒ	1	1	0
ⓓ	1	1	1

(2nd session 24)

27 The logic gate whose equivalent electric circuit has two switches connected in parallel is ..... *(Exp. 25)*

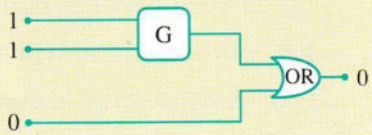
ⓐ NOT ⓑ AND ⓒ OR ⓓ NOT or AND

28 A group of logic gates are connected together as in the opposite figure, which of the following choices can be obtained from that group? *(Exp. 25)*

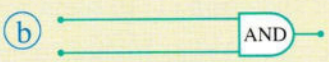


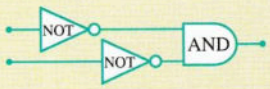
	A	B	C	Z
ⓐ	0	1	1	1
ⓑ	1	0	1	0
ⓒ	1	1	0	0
ⓓ	0	1	1	0

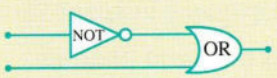
29 The opposite figure shows a circuit containing a group of logic gates where the output is (Low) when the input is as shown, so (G) may be ..... *(Exp. 25)*



ⓐ

ⓑ 

ⓒ 

ⓓ 

Test on Chapters 7 and 8

Answered

First **Choose the correct answer:**

1 When the temperature of pure semiconductor increases, its electric conductivity ..... .

(a) decreases because of the decrease of charge carriers
(b) decreases because of the increase of charge carriers
(c) increases because of the increase of charge carriers
(d) increases because of the decrease of charge carriers

2 What property do laser rays have that makes them usable in guiding missiles and airplanes?

(a) High speed.
(b) Spectral purity.
(c) Having short wavelengths.
(d) Being parallel and concentrated.

3 All laser devices consist of three basic elements which are:

(I) effective medium **(II)** power source **(III)** resonant cavity

And the argon laser differs from the ruby laser in the nature of ..... .

(a) element (I) only
(b) both elements (I), (II) only
(c) both elements (I), (III) only
(d) all elements (I), (II), (III)

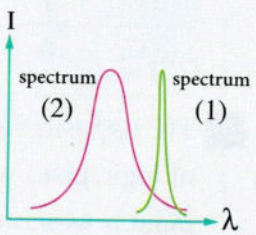
4 In either analog or digital transmitting devices, ...

(a) electric signals are converted into analog electromagnetic waves
(b) electric signals are converted into digital electromagnetic waves
(c) physical messages are converted into digital electric signals
(d) physical messages are converted into analog electric signals

5 Which of the following rays **doesn't** represent electromagnetic radiation?

(a) Laser rays. (b) X-rays. (c) Cathode rays. (d) Blackbody radiation.

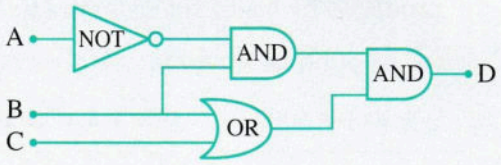
6 Two spectra (1) and (2), each of them is monochromatic and they are analyzed by a spectrometer. The opposite figure represents the relation between radiation intensity (I) and wavelength (λ) for both spectra, so which of the following statements can be considered a suitable conclusion?



(a) The capability of spectrum (1) to scatter is higher than spectrum (2).
(b) A photon of spectrum (1) has higher energy than that of spectrum (2).
(c) Spectrum (1) propagates in space faster than spectrum (2).
(d) Spectrum (1) has coherent photons.

7 In the opposite logic circuit, which of the following inputs gives the output D = 1?

	A	B	C
(a)	0	0	1
(b)	1	0	0
(c)	0	1	0
(d)	1	0	1



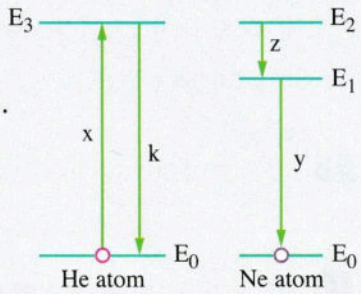
8 The opposite figure illustrates some transitions x, y, z and k in the helium-neon laser. The correct arrangement for these transitions according to the order of their occurrence is ..... .

(a) k ⟶ y ⟶ z ⟶ x

(b) x ⟶ y ⟶ z ⟶ k

(c) x ⟶ k ⟶ z ⟶ y

(d) z ⟶ y ⟶ x ⟶ k



9 Number 40 in the decimal system is equivalent in the binary system to ..... .

(a) $(110000)_2$ (b) $(101000)_2$ (c) $(001010)_2$ (d) $(001100)_2$

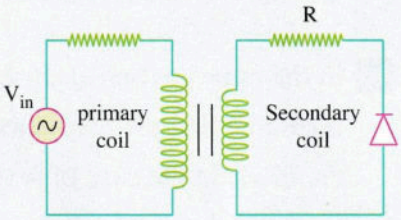
10 In the opposite circuit, what is the type of current through resistor R?

(a) Direct current.

(b) Half-wave rectified current.

(c) Alternating current.

(d) Unidirectional of variable intensity current.



11 The opposite figure illustrates multiple excitation levels for an atom of an element and the lifetime of each level, hence the atoms of this element can reach population inversion when they are excited to level ..... .

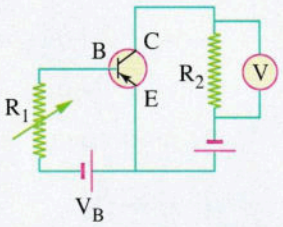
(a) E_1 (b) E_2

(c) E_3 (d) E_4

12 In the opposite transistor circuit, what happens to the voltmeter reading when increasing the resistance of R_1?

(a) Increases. (b) Decreases.

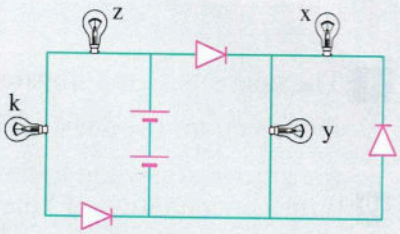
(c) Becomes zero. (d) Doesn't change.



13 The radiation produced by stimulated emission is distinguished from its counterpart produced by spontaneous emission in all the following characteristics **except** that ..... .

(a) its photons travel in one direction
(b) its waves are in phase
(c) its photons have one wavelength
(d) its photons have a greater speed

14 The opposite electric circuit contains a battery, multiple identical lightbulbs and multiple identical diodes, so the lightbulb that glows with the greatest brightness is ..... .

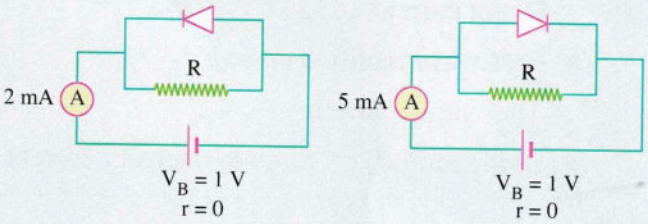


(a) x
(b) y
(c) z
(d) k

15 The drift current in a pn junction is due to the motion of ..... .

(a) positive ions from the n-region to the p-region
(b) free electrons from the p-region to the n-region
(c) holes from the p-region to the n-region
(d) negative ions from the p-region to the n-region

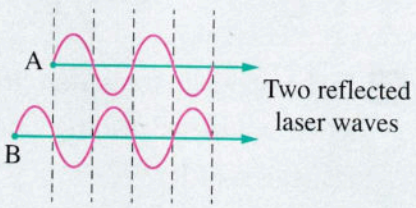
16 In the opposite two electric circuits (A, B), assuming that the resistance of the diode in the case of reverse biasing is infinite, so the resistance of R and the resistance of the diode in the case of forward biasing, respectively, are ..... .



Circuits (A) **Circuits (B)**

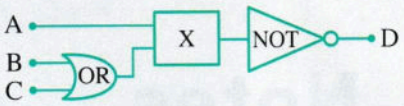
(a) 300 Ω, 333.33 Ω
(b) 300 Ω, 155.33 Ω
(c) 500 Ω, 333.33 Ω
(d) 500 Ω, 155.33 Ω

17 The opposite figure illustrates two laser waves reflected from two points A and B lying on a body during 3D hologram recording, if the path difference between them is 3164 Å, so the wavelength of the laser equals ..... .



(a) 1582 Å
(b) 3164 Å
(c) 6328 Å
(d) 9492 Å

18 The opposite figure illustrates logic gates circuit where an unknown logic gate (x) is one of them, so according to the logic circuit and its truth table, logic gate X is ..... .



A	B	C	D
0	0	0	1
1	0	1	0
0	1	1	1
1	1	1	0

(a) AND

(b) OR

(c) NOT

(d) OR or AND

19 If the concentration of holes or that of free electrons in a pure semiconductor crystal is 2×10^{8} cm^{-3} and when atoms of a certain element are added into it, the concentration of the holes inside it has increased to 4×10^{10} cm^{-3}, then the crystal type and the concentration of free electrons are ..... .

(a) n-type, 10^{6} cm^{-3}

(b) p-type, 2×10^{8} cm^{-3}

(c) n-type, 2×10^{8} cm^{-3}

(d) p-type, 10^{6} cm^{-3}

Second Answer the following question

20 The opposite figure illustrates a transistor circuit, then:

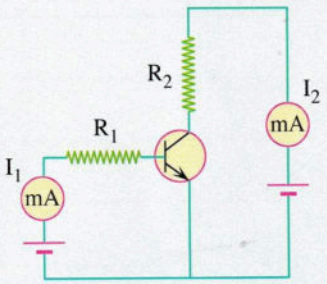
(1) Mention one use of this circuit.

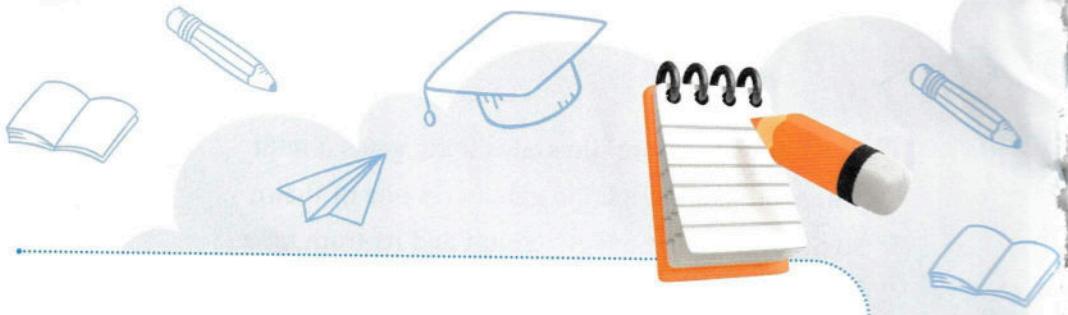
(2) If $\alpha_e = 0.98$, **find the ratio of** $\left(\frac{I_2}{I_1}\right)$.

.....

.....

.....





Notes